PHYSICAL AND MENTAL WELLNESS

6

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PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW I : PHYSICAL FITNESS

"Take care of your body. It's the only place you have to live."



TEACHING INSTRUCTIONS

Period : **One (01)**
Type : **Lecture and Practice**
Year : **1st Year SD/SW**
Conducting Officer : **Associate NCC Officer**
Training Aids : **Script/Lesson Plan; Computer with OHP, Screen, Pointer Staff, Presentation, Chart; Open Training Area or Ground.**

Time Plan

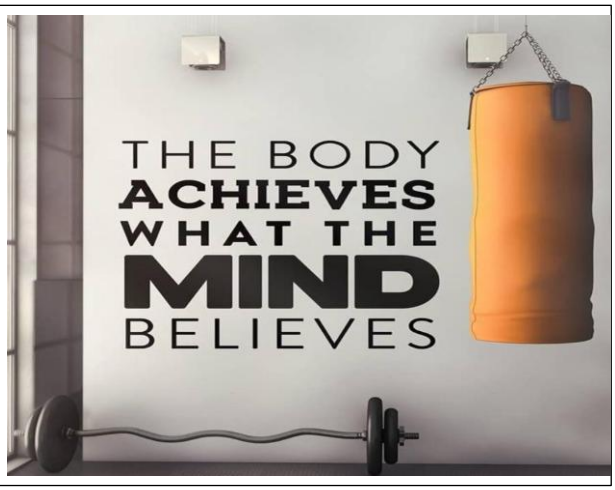
- Introduction : 02 Mins
- Part I : 06 Mins
- Part II : 06 Mins
- Part-III : 06 Mins
- Part-IV : 07 Mins
- Practice : 10 Mins
- Conclusion : 03 Mins



INTRODUCTION

1. Physical fitness is not just about being healthy, but it is also essential for mental well-being, academic performance and future success.

2. Physical fitness enhances cognitive abilities, self-perception, and coping skills. It can improve concentration, focus, and self-esteem. Engaging in sports and exercise cultivates teamwork and leadership. Furthermore, it can reduce anxiety and promote self-reliance to navigate modern social pressures.



LEARNING OBJECTIVES

- Develop good habits
- Imbibe healthy daily routine
- Involve in outdoor activities
- Follow a healthy and nutritious diet
- Develop habit to do regular physical training

PREVIEW

The lecture will be conducted in the following parts:-

(a) Part I : Significance of Physical Fitness

(b) Part II : Importance of Outdoor Activities

(c) Part III : Staying Fit

(d) Part IV : Steps to Physical Fitness

INTERESTING FACTS

- **Life Transforming Morning Habits**
  - Wake up early
  - Do Exercise
  - Sit in Silence and meditate
  - Plan your Day
  - Practise Positive Self-Talk
  - Have Dedicated Off-Screen time

PART I : SIGNIFICANCE OF PHYSICAL FITNESS

3. Regular physical activity is an excellent way to maintain physical health and enhance mental well-being. Studies demonstrate that exercise prompts the release of endorphins, the body's natural feel-good chemicals, in the brain. Even a brief, brisk 10-minute walk can boost mental alertness, energy levels, and mood. This chapter examines the importance of a regular daily routine, physical training, and recreational activities in supporting effective daily living.



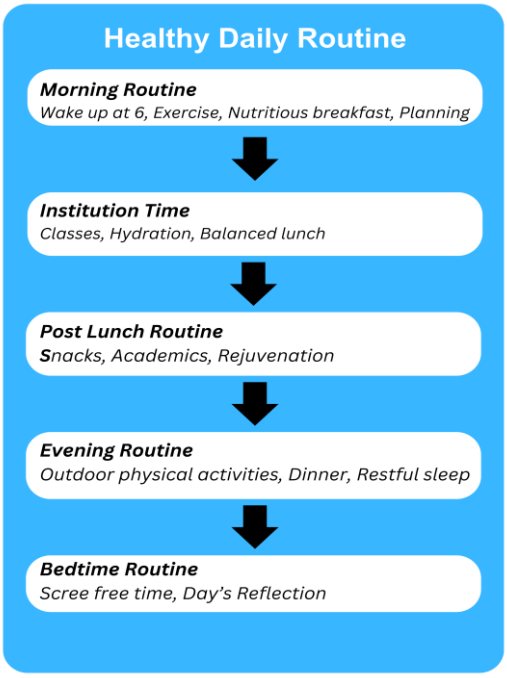
4. **Physical Fitness**. Physical fitness refers to an individual's ability to perform day-to-day tasks without fatigue. It is a measure of the body's efficiency and effectiveness in work and leisure pursuits. To maintain fitness and well-being, one must engage in physical activities and adopt measures to stay physically fit.

5. **Habit**. Habits are regular routines and behaviors that people engage in, consciously or unconsciously. Positive habits help maintain motivation and make tasks easier when motivation is low. Established habits become part of daily life, requiring little effort. For instance, brushing teeth before bed is a habit that requires no extra motivation, as it has been practiced for years and is now seamless in nightly routine.



6. There are numerous ways to classify habits, such as mental, physical, or emotional; motor, intellectual, or character-building. The more beneficial, constructive, and progressive habits individuals cultivate in their life, the greater the boost to their physical, mental, and emotional well-being.

7. **Healthy Daily Routine**. A person's daily routine comprises actions and practices that structure their day. These routines can impact efficiency or cause struggle. Individuals may adopt energizing, time-saving routines or more draining, inefficient ones. The daily routine of a cadet includes morning activities, academic pursuits, evening and bedtime components. A suggested daily healthy routine can be as under :-



(a) **Morning Activities (0600-0800 hrs)**. Wake up, exercise, get ready, have a nutritious breakfast and plan the day's activities by setting goals, prioritizing tasks and creating a to-do list.

(b) **Institution Time (0800-1400 hrs)**. Attend classes with active participation, maintain proper hydration and enjoy a balanced meal for lunch comprising of a mix of carbs, proteins and vegetables.

(c) **Post Lunch Routine (1400-1700 hrs)**. Focus on fulfilling academic responsibilities such that an effective planning can be done for preparation of examination, take nutritious snacks and include periodic breaks in your study routine, allowing for 10–15 minute intervals of physical movement, stretching, or mindful practices to rejuvenate.

(d) **Evening Activities (1700-2100 hrs)**. Undertake outdoor physical activities, eat nutritious dinner, unwind through engaging hobbies, literature, or music.

(e) **Bedtime Routine (2100-2200 hrs)**. Have a **screen-free time** for at least an hour before bedtime, reflect on the day by introspection, brush your teeth, change into comfortable clothes, and aim for **8-10 hours of sleep** to help recharge your body and mind, as is suggested for an age group of around 13 to 16 years.

8. Consistent physical activity and exercise are fundamental strategies for enhancing physical fitness, leading to a variety of benefits that go beyond the musculoskeletal system.

PART II: IMPORTANCE OF OUTDOOR ACTIVITIES

9. **Physical Education**. Physical education aims to nurture the comprehensive development of an individual. Its ultimate goal is to cultivate physical, mental, and emotional well-being in all people, while also providing them with opportunities to foster the personal and social attributes that enable them to coexist harmoniously with others and become responsible citizens.

TIPS

- **For a Healthy and Balanced Life**:-
  - **Stay Organized.** Use a planner, calendar, or app to keep track of your schedule and tasks.
  - **Prioritize Personal Well-Being.** Make time for positive activities that bring you joy and help you relax.
  - **Seek Help When Needed.** Don't hesitate to ask teachers, parents, or friends for help when you're struggling with academics or personal issues.

10. **Significance of Outdoor Activities (Physical Training and Games).** Physical fitness is essential for the physical and mental development of cadets. It boosts the immune system by increasing blood flow and adaptability to exertion. Physical training and games have a positive impact on physical health and mental well-being. The impacts are as under:-

(a) **Physical Health Benefits.** Exercise triggers the release of endorphins, improving mood and reducing stress and anxiety. Regular physical activity enhances self-esteem and can alleviate depression symptoms.

(b) **Academic Performance Benefits.** Physical activity increases blood flow to the brain, boosting concentration, memory, and problem-solving skills, leading to improved academic performance.

(c) **Development of Valuable Life Skills.** Sports and physical training teach teamwork, communication, leadership, perseverance, and goal-setting—skills essential for success in life.

(d) **Social Benefit.** Engaging in physical activities helps individuals form friendships, develop social skills, and learn to respect diversity while working toward shared goals.

Types of Outdoor Activities

11. **Physical Training(PT) and Games**. PT and Games are crucial for physical well-being, promoting teamwork, coordination. Individual sports offer a chance to exert physically; develop strength, skill, stamina and endurance. Team Games like football, kho-kho, Kabaddi, basketball etc help one develop a sense of camaraderie, shared purpose and healthy competitive spirit, facilitating social connection alongside physical exertion. Games channelise our energies in right direction, fostering health, social and personal growth. It is rightly said that 'A healthy mind resides in a healthy body'.



12. **Hobbies and Adventure Activities.** Trekking, Rafting, Rock-climbing, Cycling, Gardening, Nature-photography, star-gazing etc are some of the hobbies cadets can undertake.



13. Outdoor activities give us a break from the classroom, allowing us to explore our creativity more deeply. Classroom learning alone can limit the ability to think critically. When we experience new surroundings, we gain fresh perspectives on our world, which sparks our curiosity and inspires our creativity to address challenging questions.

14. **Benefits**. Various benefits of physical training and games are as under:-

(a) **Improved Physical Health**. Regular outdoor activity enhances fitness and well-being.

(b) **Mental Health Benefits**. Outdoor activities reduce stress, anxiety and depression.

(c) **Social Skills Development.** Team sports and group activities foster communication, teamwork and leadership.

(d) **Environmental Awareness**. Outdoor activities promote appreciation and understanding of nature.

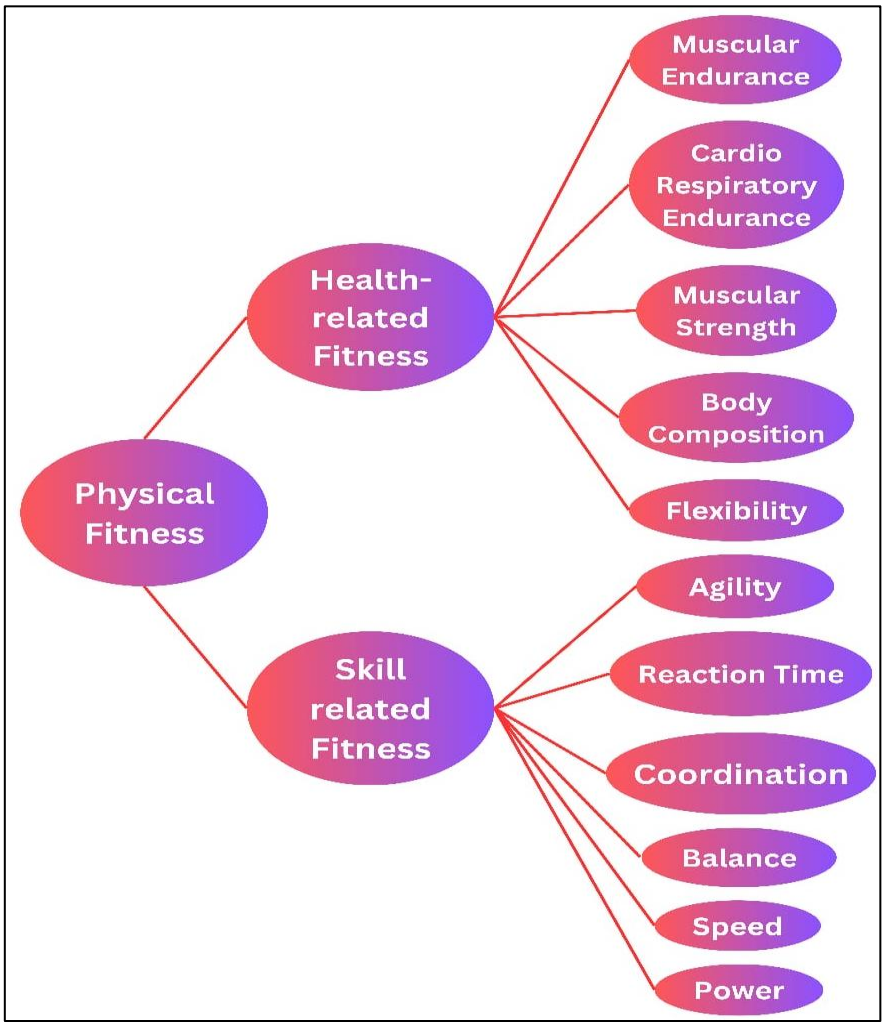
TIPS

- Breathe deeply and naturally while stretching.
- Don't bounce or force your muscles beyond a comfortable stretch.
- Hold each stretch for 15-30 seconds to allow for maximum relaxation of the muscles.
- Stretch regularly to maintain flexibility and range of motion.
- Listen to your body and stop stretching if you experience any pain or discomfort.

(e) **Building Confidence**. Overcoming outdoor challenges and achieving goals boosts self-esteem.

PART III : HOW TO STAY FIT

15. Physical fitness is like carefully tuning an engine. It allows us to reach our full potential, enabling us to look, feel, and perform our best in daily tasks, with energy left for leisure and emergencies. Fitness involves the heart, lungs, and muscles, and also impacts mental and emotional well-being. The need of the hour is that cadets need to maintain both health and skill related fitness for their best performance in any task.



Basic Components of Fitness

16. **Cardio-Respiratory Capability**. It is the ability to deliver oxygen and nutrients to tissues, and to remove wastes, over sustained periods of time. Methods for measuring this component include long runs and swims.

17. **Muscular Strength.** It is the ability of a muscle to exert force for a brief period of time. Upper-body strength can be measured by weight-lifting exercises.

18. **Muscular Endurance.** It is the ability of a muscle, or a group of muscles, to sustain repeated contractions or continue applying force against a fixed object. Tests like push-ups measure endurance of arm and shoulder muscles.

19. **Flexibility.** It is the ability to move joints and use muscles through their full range of motion. The sit-and-reach test is often used to measure flexibility of the lower back and upper legs.



Cooling Down Exercises

20. **Body Composition**. It refers to the makeup of the body in terms of lean mass (muscle, bone, vital tissues, and organs) and fat mass. An optimal ratio of fat to lean mass indicates fitness, and exercise can help reduce body fat while maintaining or increasing muscle mass.

Assessment of Personal Fitness Levels

21. When beginning a fitness program, it is important to assess your current level of physical fitness. Individuals can have varying starting points, often classified as different "beginner" levels. As the saying goes, "Rome wasn't built in a day," and the same principle applies to physical fitness.

22. Professional athletes and sportspeople have dedicated years to training and developing their capabilities. The key is to start slowly and build up gradually, as pushing too hard too soon can lead to burnout and diminish enjoyment of the exercises. Evaluation of your fitness levels, can either be done by Parameters-Based Assessments or Activity-Based Assessments.

23. **Parameters-Based Assessments (PBAs).** Readings of essential parameters can help determine soundness of physical fitness, some of which are as under:-

(a) **Heart Rate.** Measure your resting heart rate by palpating your pulse using your fingertips, either at the carotid artery in the neck (below the windpipe) or the radial artery at the wrist (below thumb-base). If 50-80, then it is considered

to be ranging from very fit to normal and if found greater than 100, then it is below average.

(b) **Body Fat.** Fitness does not solely depend on weight. People can have diverse body types yet still be healthy and physically fit. The recommended body fat range is typically 10-15% for men and 15-20% for women.

(c) **BMI (Body Mass Index).** One is underweight if the BMI is less than 18.5 and overweight when BMI is greater than 25. The method of calculating BMI is:-

(i) Multiply your weight by 10,000. Example: 60 x 10,000 = 600,000

(ii) Square your height in cms(multiply it by itself). Example: 155 x 155 = 24,025

(iii) Divide the result from step 1 by the result from step 2. Example: 600,000 ÷ 24,025 = 24.97 (BMI = 25, rounded off)

(d) **Basal Metabolic Rate**. It is the minimum number of calories needed to sustain your body weight based on typical body composition (required for basic activities like breathing, digestion & blood circulation). In order to determine BMR, multiply your body weight in kgs with 24. This can then be multiplied with the activity level number which is 1.45, 1.60,1.70 & 1.88 for sedentary, light, medium and heavy activity.

(e) **Vmax or VO_2 Test**. It is a measure of the maximum amount of oxygen your body can use during intense exercise, and it is a key indicator of cardiovascular fitness. It reflects your aerobic capacity, which is the ability of your heart, lungs, and muscles to work together efficiently during endurance activities like running, cycling, and swimming. For a healthy adult, the ideal VMax (or VO2max) can vary based on factors such as age, gender, fitness level, and activity type. However, general ranges are:-

(i) **For Men (Age 20-30).** 40-50 mL/kg/min (elite athletes may exceed 60 mL/kg/min)

(ii) **For Women(Age 20-30).** 35-45 mL/kg/min (elite athletes may exceed 55 mL/kg/min)

(iii) As people age, VO2max tends to naturally decrease. But maintaining a higher VO2max (with regular aerobic exercise) is linked to better cardiovascular health and overall fitness.

24. **Activity-Based Assessments(ABAs).** Ability to undertake physical activities/tests with ease can also be a fair yardstick to evaluate fitness levels. Some of the basic tests are as follows:-

(a) **Muscular Strength.** Evaluating muscular strength can be most easily accomplished through the push-up assessment. However, individuals with lower back injuries should refrain from performing this test. For NCC cadets, it is desired that the senior division cadets complete 18-20 and senior wing cadets complete 8-10 push-ups.

(b) **Flexibility.** For this test, use a measuring tape or 36-inch ruler. Warm up your lower back and hamstrings. Sit with legs straight, feet 10 inches apart, and no shoes. Place the tape with the 15 inch mark at your heels. Cross your hands, reach forward, and hold while the measurement is taken. Do three trials and record the highest result. Score below 13 for boys and 10 for girls in 20's is considered to be minimum desired value.

(c) **Abdominal Strength.** The assessment process includes a physical fitness evaluation involving sit-ups. The minimum desired number of sit-ups is 32 for senior division cadets and 25 for senior wing cadets.

Nutrition

25. **Nutrition Basics**. Food serves as the primary energy source for the human body. There are six basic nutrients that are necessary to help support your health, even in small amounts. These are protein, carbohydrates, fats, vitamins, minerals and water. The energy content of food is quantified in calories.

26. **Essential Nutrients**. Essential nutrients are compounds the body requires trusted source to perform basic functions and grow. The six essential nutrients can be broken down into following categories:-

(a) **Macronutrients**. These are nutrients which the body requires in substantial quantities to fuel its activities and maintain its structure. It includes carbohydrates, proteins and fats. Carbohydrates are the primary source of energy.

(b) **Micronutrients**. These include vitamins and minerals that strengthen the immune system, and metabolism

(c) Water is neither a macronutrient nor a micronutrient in the traditional sense. However, it is often categorized alongside macronutrients because the body requires it in large amounts for survival.

27. **Weight Management**. A **caloric deficit/gain** is a key factor in weight management, meaning the correlation between energy (calories) consumed and expended. So, for weight management following aspects bear importance:-

(a) For weight loss, Calories consumed need to be less than Calories expended, creating a **caloric-deficit, essential for weight loss.**

(b) For weight gain, Calories consumed need to be more than Calories expended, achieving a caloric surplus.

(c) For maintaining body weight, Calories consumed and expended need to be approximately the same.

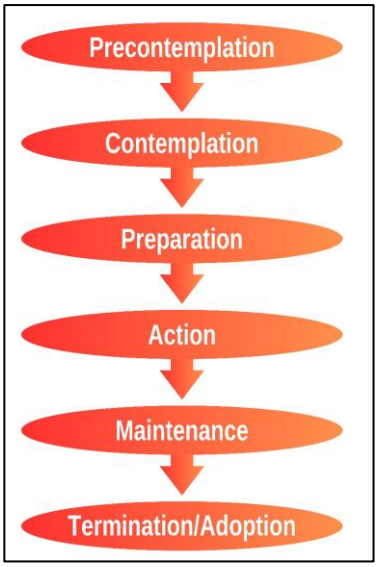
PART IV : STEPS TO PHYSICAL FITNESS

28. Changing habits by modification of one's lifestyle is hard. Many people want to eat better, exercise more, or lose weight, but find it difficult for various reasons. There are five stages to making these lifestyle changes:-

(a) **Pre-contemplation**. The individual is unaware of the problem and doesn't want to change, ignoring warnings. Education and highlighting future consequences may help.

(b) **Contemplation**. The person recognizes the problem and considers change but isn't ready yet. Support from others can an encourage action.

(c) **Preparation**. The individual is ready to change within a month, setting goals and taking small steps, with peer support being important.



(d) **Action**. The person actively works on their goals, but setbacks are common. Continued effort for six months leads to progress.

(e) **Maintenance**. After six months, the individual focuses on maintaining the change, especially in stressful situations, by following specific guidelines.

(f) **Termination/Adoption**. The person has sustained the change for over five years, with no likelihood of relapse for negative behaviors, though relapse is still possible for positive ones.

(g) **Relapse**. A relapse is a return to old behaviors, often during action or maintenance stages. Recovery involves assessing progress, identifying weaknesses, and creating a plan for the future.

29. **Aim & Principles of PT.** The primary aim of physical training is to empower youth physically and mentally by imparting appropriate skills and knowledge, thereby enabling them to engage in a lifetime of healthy physical activities. Three main principles of PT are: -

(a) **Balanced Development.** To give equal exercise to every part of the body.

(b) **Progressive Improvement.** Aim for a steady improvement rather than gains

(c) **Maintain Continuity**. To do PT consistently so that it develops into a habit.

30. **Essentials of PT**. The five golden essentials of PT are:-

(a) A good environment.

(b) Hard work.

(c) Sound sleep.

(d) Recreation.

(e) Patience and satisfaction.

31. An ideal **PT program** includes all activities from warming up to cooling down. . It's a good idea to go for a run to build up stamina and endurance, followed by strengthening and stretching exercises. When it comes to doing exercises, remember to **do the lateral(sideways) ones before the dorsal(forward and back) ones**, to avoid putting of sudden pressure on the core and risk injury. Suggested structure of a 45-minute PT session is as follows:-

(a) **Warming Up.** 03 Mins of slow jogging to warm up the muscles.

(b) **Running.** 10 Mins of running, ending in a sprint for endurance and speed.

(c) **Strength/Endurance Training.**15 Mins of Push-ups, Pull-ups, Skipping, jumping, sprints, sit-ups, rope climbing or competitive games, for gaining strength and power.

(d) **Stretching.** 10 Mins of free stretching exercises to maintain flexibility and mobility.

(e) **Cooling Down**. 07 Mins of Cooling down exercises to restore the metabolism to pre-exercise state.



DID YOU KNOW?

- You need to burn off 3,500 calories more than you take in, to lose just one pound!
- Regular physical activity can increase the size of the 'hippocampus', the part of your brain responsible for memory and learning.
- So, exercising not only improves physical fitness but also cognitive function.

32. **Freehand Exercises.** These exercises boost flexibility, strength, and stamina without the need for any specialized equipment. They activate various muscle groups, improve coordination, and support joint flexibility. They can be performed anywhere and offer an easy & convenient way to maintain overall fitness. Some of the basic ones are: -



(a) **Push-ups.** Strengthens the chest, shoulders, and arms.

(b) **Squats.** Targets the legs

(c) **Sit-ups.** Strengthens the core

(d) **Lunges**. Focuses on legs and glutes.

(e) **Planks**. Engages the core, shoulders, and back muscles.

(f) **Mountain Climbers**. A full-body exercise that enhances cardio too.

(g) **Jumping Jacks**. Improves cardiovascular health.

33. While both freehand and equipment exercises are good for fitness, their impact on the body is different. Freehand exercises aid development and improve overall physical fitness, while equipment exercises work better for developing specific muscle mass. Some salient points of both are as under:-

Free Hand Exercises	Equipment/Gym Exercises
➢ Are better for improving overall movement patterns, enhancing joint mobility, and building strength using just your body. ➢ These exercises can include push-ups, squats, lunges, planks, vertical rope etc. Do not entail expenditure on equipment. ➢ These engage the core and multiple muscle groups, improving coordination, stability, and endurance. ➢ Since exercise is against body's own weight, movements are more natural and less stressful on joints compared to lifting heavy weights. This is ideal for beginners or people with joint issues. ➢ Baseline level of fitness, strength, form and body-shape is maintained even during periods of reduced training ➢ Less chances and severity of injury/muscle damage.	➢ Are more focused on increasing mass and hypertrophy (muscle growth) by adding external resistance, targeting specific muscle groups ➢ Using equipment like dumbbells, barbells, kettlebells or machines. Normally entail considerable expenditure. ➢ These are good for increasing resistance & strength beyond body-weight. ➢ Muscle adaptations from equipment-heavy training are more specific, dependent on higher-intensity loading, and prone to faster loss during inactivity ➢ Form, mass and shape is gained quicker but is also lost more rapidly as compared to body-weight training in periods of reduced training ➢ More chances and severity of injury/muscle damage

34. **Activities as Per Age, Health, Location and Weather.**

(a) **Age-appropriate Adjustments.** Junior cadets should focus on engaging, simple exercises that build basic strength and coordination, like bodyweight squats and jumping jacks. Senior cadets, however, should increase intensity, incorporating strength training like weightlifting and resistance exercises.



(b) **Health Adjustments.** Fitness routines should be adapted to accommodate cadets with injuries or chronic conditions by offering low-impact alternatives, such as swimming, cycling etc

(c) **Weather and Location Adjustments.** Areas with extreme climates or high-altitude locations, demand separate considerations and allowances in performance. For places experiencing heavy rainfall, indoor sessions in covered spaces can ensure training continues safely.

35. Following are the desired **minimum** physical standards for SD/SW cadets (18-25 yrs) **on completion of their three years training**:-

	Boys	Girls
Weight (BMI)	18.5-21	18.5-22
Walking (30 mins)	4-5 km/hr	3-4 km/hr
1.6 km Running	8-11 mins	10-13 mins
2.4 km Running	12-14 mins	14-16 mins
Sit-ups	32	25
Push-ups	18-20	8-10
Plank Hold	60-90 sec	45-60 sec
Yoga	Proficiency in at least 10 Asanas	
Swimming	50 metres (Any style/stroke)	

Note: These desired standards are only meant to serve as a reference guide for cadets to achieve minimum levels of physical fitness. Cadets will NOT be assessed or tested on the same.

36. In addition to above physical standards, to ensure overall physical and mental wellness, following steps can be taken on routinely:-

(a) **Regular Exercise.** Engage in at least an hour of physical activity daily to improve strength, endurance and cardiovascular health.

(b) **Balanced Diet**. Eat a balanced diet rich in fruits, vegetables, lean proteins, and whole grains while limiting processed foods.

(c) **Oral Health.** Brush and floss daily, visit the dentist regularly, and avoid excessive sugar to maintain strong teeth and gums.

(d) **Adequate Sleep**. Ensure 8-10 hrs of quality sleep each night for proper body and brain function.

(e) **Yoga and Meditation**. Practice yoga for flexibility, and relaxation, and meditate to reduce stress and enhance focus.

(f) **Hydration**. Drink enough water (at least 8 glasses) throughout the day to maintain energy levels.

(g) **Stress Management**. Engage in hobbies and mindfulness techniques to reduce anxiety and mental fatigue.

(h) **Limit Screen Time**. Reduce excessive phone and computer use to prevent eye-strain, stress and sleep disturbance.

ACTIVITY

- ANO to divide the cadets into groups of 10-15 and take them outdoors for carrying out Warm up, Strengthening and Cooling down exercises as given in the chapter.

CONCLUSION

37. This lesson equips cadets with practical knowledge and techniques to adopt a healthy lifestyle, including structured routines, physical training and balanced nutrition. Adopting and implementing these practices can lead to long-term health benefits and personal growth.

38. Practising a healthy lifestyle enhances an individual's quality of life by boosting physical fitness. It improves heart and lung function, increases oxygen supply to body tissues, enhances muscle tone, promotes good posture and a balanced figure, fosters a positive body image, speeds up recovery from illness or injury, and reduces the risk of cardiovascular disease.

SUMMARY

- A healthy daily routine combines adequate sleep, regular physical activity and nutritious meals.
- Physical training improves physical and mental health, discipline and endurance.
- Outdoor activities promote holistic development and reduce stress.
- Following a structured exercise routine with proper techniques prevents injuries and maximizes benefits.
- A balanced diet is essential for fuelling physical activities and supporting recovery.
- Regular PT empowers the youth physically and mentally with proper skills and knowledge to enjoy a lifetime of healthy activities physically.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. Which of the following is a key component of a balanced diet?

(a) Only processed foods

(b) Rich in fruits, vegetables, lean proteins, and whole grains

(c) Only sugary snacks

(d) High-fat foods

Q2. Bedtime Routine should be devoid of:-

(a) Sleep. (b) Relaxation.

(c) Screen Time. (d) All of the above.

Q3. Physical training and games are essential for:-

(a) Physical growth.

(b) Mental growth.

(c) Both physical and mental growth of the cadets.

(d) None of the above.

Q4. Being part of the Physical Training and Games/ sports team or group activities lets cadets make:-

(a) Money.

(b) Friends and imbibe social skills.

(c) Good Communication Skills.

(d) None of the above.

Q5. What is the primary component of physical fitness that involves the ability of the body to transport oxygen and nutrients?

(a) Muscular strength. (b) Cardiovascular endurance.

(c) Flexibility. (d) Body composition.

Q6. Which of the following exercises is an example of aerobic exercise?

(a) Weightlifting. (b) Sprinting.

(c) Jumping jacks. (d) Yoga.

Q7. What do you call it when an individual is unaware of the problem and doesn't want to change, ignoring warnings?

(a) Contemplation (b) Pre-contemplation

(c) Action (d) Preparation

Q8. Proper warm-up and cool-down exercises to prevent:-

(a) Headache. (b) Digestion.

(c) Injuries. (d) None of the above.

Q9. What is the term for the ability to move quickly and easily?

(a) Agility. (b) Speed.

(c) Power. (d) Endurance.

Q10. Macronutrients consist of:-

(a) Carbohydrates. (b) Proteins.

(c) Fats. (d) All of the above.

Q11. Cadets must limit consumption of which food item?

(a) Sugary Drinks.

(b) Fast food and processed snacks.

(c) Saturated and trans fats.

(d) All of the above.

Q12. Essentials of PT involves:-

(a) A good environment. (b) Lethargy.

(c) More sleep. (d) Watch movie.

Q13. Which of the following is True in respect of PT?

(a) More people can participate.

(b) Only select, persons can participate.

(c) Every part of the body gets exercise.

(d) (a) and (c) above.

Q14. Strength training includes activities to:-

(a) Improve strength.

(b) Improve Sleep.

(c) Improve Appetite.

(d) None of the above.

Q15. The purpose of Cooling down exercises is to:-

(a) Restore metabolic rate to pre-exercise levels.

(b) Remove stress.

(c) Not sweat anymore.

(d) None of the above.

Short Answer Questions

Q1. What is Healthy Daily Routine?

Q2. What are Habits?

Q3. What are Macronutrients?

Q4. What is the aim of PT?

Q5. What is muscle endurance? Give two exercises which help in measuring muscle endurance.

Long Answer Questions

Q1. Discuss the significance of physical fitness.

Q2. Briefly describe the importance of outdoor activities.

Q3. Explain in detail how does a person stay fit.

Q4. Briefly discuss the term PT.

Q5. Explain the Parameters Based Assessment (PBA) method of fitness assessment.

PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW II : IMPORTANCE OF HYGIENE AND SANITATION

"Sanitation and cleanliness are among the humblest of the civic virtues, and it is easy to underestimate their significance."

Former President, Shri Ram Nath Kovind



TEACHING INSTRUCTIONS

Period : **One (01)**
Type : **Lecture**
Year : **1st Year SD/SW**
Conducting Officer : **Associate NCC Officer**
Training Aids : **Script/Lesson Plan: Computer with OHP, Screen, Pointer Staff, Presentation, Charts/posters.**

Time Plan

- Introduction : 03 Mins
- Part I : 04 Mins
- Part ll : 15 Mins
- Part-III : 10 Mins
- Part-IV : 05 Mins
- Conclusion : 03 Mins



INTRODUCTION

1. Maintaining hygiene and sanitation is essential for overall health and well-being. This chapter explores the importance of personal, oral, and food hygiene, as well as camp sanitation. Personal hygiene includes practices such as bathing, handwashing, and grooming. Oral hygiene, which involves caring for teeth and the mouth, prevents dental issues and supports overall health. Food hygiene ensures food safety and prevents contamination, protecting against foodborne diseases.

2. Camp sanitation is crucial for groups in close quarters, such as NCC camps, to maintain cleanliness and prevent the spread of infectious diseases. By adopting proper hygiene and sanitation practices, individuals can reduce the risk of disease, promote healthier lifestyles, and create a safer environment for themselves and others.

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I : Importance of Hygiene and Sanitation

(b) Part II : Personal, Oral and Food Hygiene

(c) Part III : Camp Sanitation

(d) Part IV : Dos and Don'ts

INTERESTING FACTS

- Soap comes from the Roman word "sapo," meaning "fat" or "oil."
- The first toothbrush with nylon bristles was introduced in 1938
- The term Hygiene is derived from the word Hygeia, representing the Greek Goddess of health, cleanliness, and sanitation.

LEARNING OBJECTIVES

- Comprehend the basics of Hygiene and Sanitation
- Comprehend the basics of Personal, Oral and Food Hygiene
- Learn about common problems caused by poor hygiene
- Understand the importance of food safety from production to consumption
- Learn about foodborne diseases and prevention methods
- Cater for aspects of sanitation of camp during recce of camp site
- Ensure sanitation of cook house, water-point and living area

PART I : IMPORTANCE OF HYGIENE AND SANITATION

3. As NCC cadets, maintaining hygiene and sanitation is vital for your health and performance. These practices prevent disease, promote cleanliness, and create a positive environment.

4. **Hygiene**. Hygiene encompasses practices like bathing, handwashing, and grooming (trimming hair and nails) to promote health. According to WHO, hygiene refers to conditions and practices that help maintain health and prevent the spread of diseases.

5. **Sanitation**. Sanitation involves water quality, sewage treatment, and public health. It includes tools and actions that maintain a clean environment and helps prevent diseases like roundworm infections caused by infected faeces. Stopping open defecation eliminates such infections and more serious ones like E. coli.

6. **Importance of Hygiene and Sanitation**.

(a) Crucial for personal, social, psychological, and health reasons.

(b) Proper hygiene and sanitation prevent diseases and infections.

(c) Maintaining good personal and environmental hygiene significantly reduces many diseases.

(d) It involves not just personal hygiene but also avoiding littering and open defecation to promote better health and quality of life.

7. **Hygiene of the Society**. Societal hygiene is as important as individual hygiene. The cleanliness of our environment and the hygiene practices of those around us influence our hygiene standards.

8. Overcrowded areas lacking sanitation facilities and with accumulating garbage create unhygienic environments, impacting residents, particularly slum dwellers. Individual hygiene practices alone are insufficient in such circumstances; the entire community must adopt hygienic practices to maintain a clean environment.

PART II : PERSONAL, ORAL AND FOOD HYGIENE

9. **Personal Hygiene**. Personal hygiene is essential for maintaining health and well-being. It involves keeping the body, clothing, and surroundings clean to prevent illnesses and promote confidence.

10. **Key Aspects of Personal Hygiene**.

(a) **Body Cleanliness**. Bathe regularly, use soap and shampoo, and focus on areas prone to sweat and bacteria.

(b) **Oral Hygiene**. Brush twice daily, floss, and use mouthwash.

(c) **Hand Hygiene**. Wash hands with soap, especially after using the restroom and before meals.

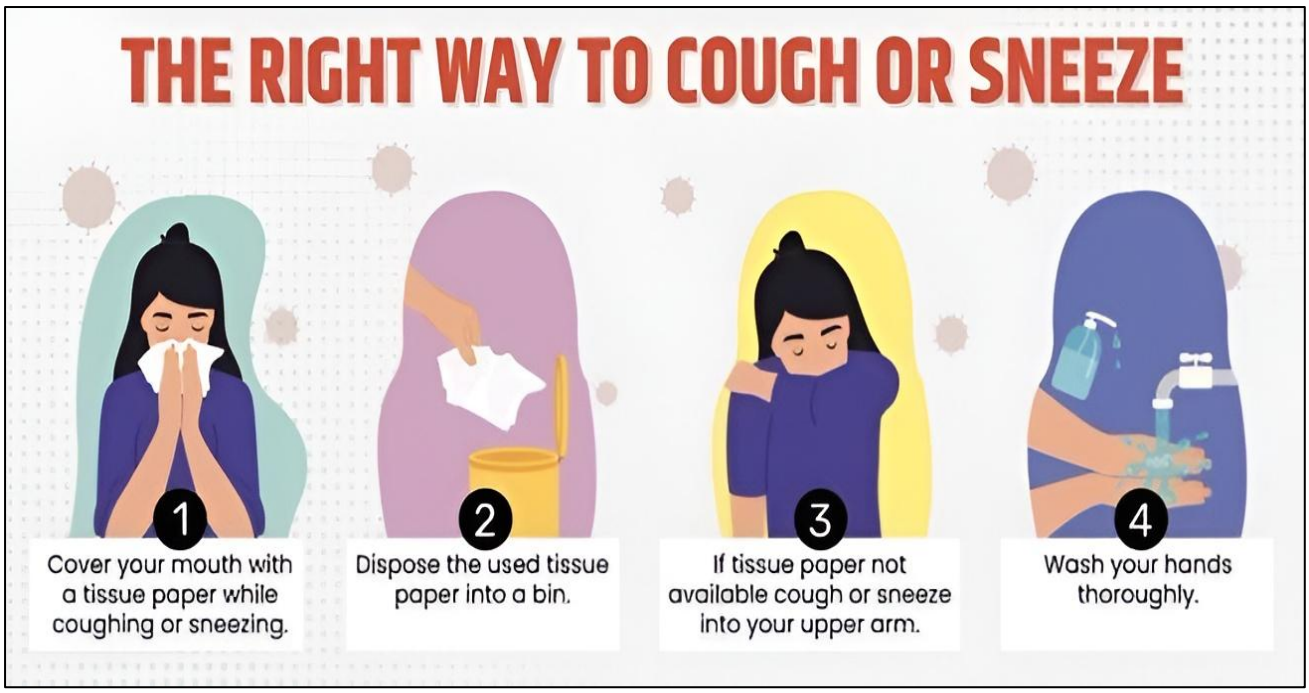
(d) **Hair and Scalp Care**. Wash hair regularly to maintain cleanliness.

(e) **Nail Care**. Keep nails clean and trimmed.

(f) **Clothing and Laundry**. Wear clean clothes and wash them frequently.

(g) **Foot Care**. Wash feet daily, dry them well, and trim toenails.

(h) **Genital Hygiene**. Maintain proper hygiene, particularly during menstruation and urination.



11. **Importance of Personal Hygiene**.

(a) **Prevents Illnesses**. Reduces disease transmission.

(b) **Boosts Confidence**. Enhances self-esteem and confidence.

(c) **Maintains Physical Health**. Prevents body odour, acne, and fungal infections.

(d) **Promotes Mental Well-being**. Reduces stress and anxiety.

(e) **Respects Others**. Shows consideration for those around you.

12. **Practices Performed by Individuals to Maintain Good Personal Hygiene.**

Some daily habits to maintain personal hygiene and prevent diseases are as under:-

Morning Routine

- **Wake-up Wash**. Wash your face, hands, and mouth when you wake up.
- **Brush and Floss**. Brush your teeth at least twice a day and floss once a day to maintain good oral hygiene.
- **Wash Hands**. Wash your hands with soap and water after using the toilet, before eating, and after blowing your nose, coughing, or sneezing.

Bathing and Grooming

- **Shower or Bathe**. Shower or bathe at least once a day, using soap and shampoo to keep your body clean.
- **Wash Hair**. Wash your hair regularly, depending on your hair type and lifestyle.
- **Trim Nails**. Trim your fingernails and toenails regularly to keep them clean and tidy.

Hand Hygiene

Wash Hands Frequently. Wash your hands with soap and water frequently throughout the day.

Foot Hygiene

Wash Feet. Wash your feet daily, paying special attention to the areas between your toes.
Dry Feet. Dry your feet thoroughly, especially between the toes.

Additional Habits

- **Wear Clean Clothes**. Wear clean clothes every day, and wash clothes regularly.
- **Use Clean Towels**. Use clean towels to dry yourself after bathing or showering.
- **Cover Mouth and Nose**. Cover your mouth and nose when coughing or sneezing to prevent the spread of germs.
- **Avoid Touching Your Face**. Avoid touching your face, especially your eyes, nose, and mouth, to prevent the spread of germs.

Oral Hygiene

13. **Oral hygiene**. Oral hygiene ensures a healthy mouth, teeth and gums, preventing dental issues and infections.

14. **Key Aspects of Oral Hygiene.**

(a) Brush teeth at least twice daily using a fluoride toothpaste. It helps remove bacteria and freshens the breath.

(b) Floss daily to remove plaque and food particles.

(c) Rinse with water or mouthwash to keep the mouth fresh.

(d) Use a tongue scraper to remove bacteria.

(e) Visit the dentist regularly for check-ups and cleanings.

15. **Consequences of Poor Oral Hygiene.**

(a) Tooth decay and cavities

(b) Gum disease and tooth loss

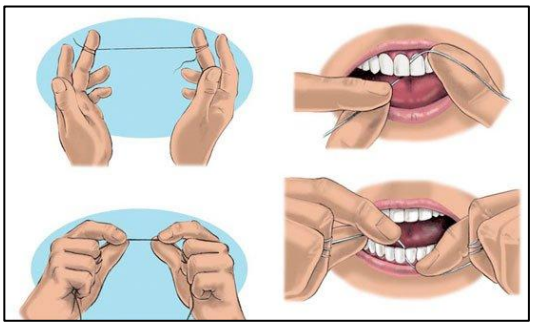
(c) Bad breath (halitosis)

(d) Gingivitis and Periodontitis

(e) Increased risk of systemic diseases like heart disease and diabetes.



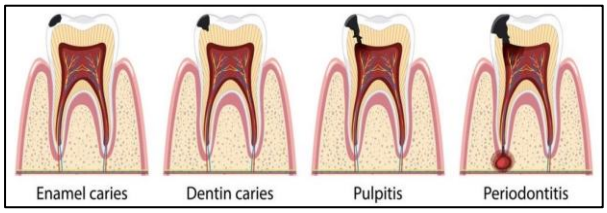
Brushing



Flossing

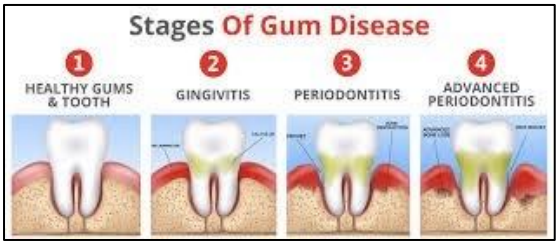
16. **Some Common Oral Health Challenges**. Oral health, If neglected, may lead to following issues/problems :-

Tooth Decay and Cavities



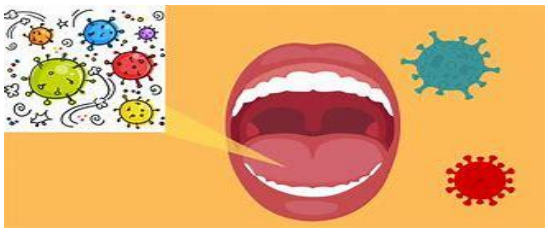
- **Tooth Sensitivity. Pain or discomfort when consuming hot or cold foods and drinks.**
- **Toothaches. Pain or discomfort in the tooth or surrounding area.**
- **Cavities. Bacterial infections that cause de-mineralisation of tooth enamel.**

Gum Disease and Periodontal Issues



- **Gingivitis. Inflammation of the gums (gingiva) caused by plaque build-up.**
- **Periodontitis. Infection of the gums and bone that support the teeth.**
- **Bad breath (Halitosis). Persistent bad breath due to poor oral hygiene, gum disease, or other oral health issues.**

Oral Infections



- **Oral Thrush. Fungal infection that causes white patches on the tongue and inside the mouth.**
- **Cold Sores. Viral infection that causes painful blisters on the lips, mouth, or gums.**
- **Mouth Ulcers. Painful sores on the tongue, lips, or inside the mouth.**

17. **Common Problems Caused by Poor Hygiene.** Are as under :-

(a) **Skin Infections**. Bacterial or fungal infections like acne, ringworm, and scabies.

(b) **Dental Issues**. Cavities, gum disease, and bad breath due to poor oral hygiene.

(c) **Body Odour.** Caused by the build-up of sweat and bacteria on the skin.

(d) **Gastrointestinal Infections**. From consuming contaminated food or water due to unclean hands.

(e) **Respiratory Illnesses**. Spreading of colds, flu, and other airborne infections.

(f) **Parasitic Infections**. Lice, scabies, and intestinal worms due to poor cleanliness.

(g) **Urinary Tract Infections (UTIs**). Can occur due to improper hygiene, both males and females, especially the latter.

(h) **Fungal Infections**. Athlete's foot, yeast infections, and nail fungus

Food Safety

18. **Food Safety**. Food safety involves proper handling, storage, and preparation to prevent contamination and foodborne illnesses.

19. **Key Aspects of Food Safety.**

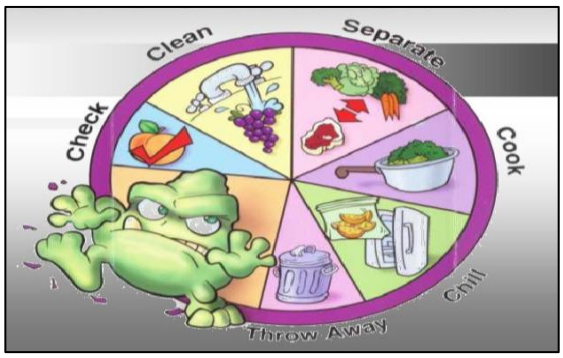
(a) **Handling & Storage.** Store food properly to prevent contamination.

(b) **Temperature Control**. Maintain safe cooking and storage temperatures.

(c) **Personal Hygiene**. Wash hands before handling food.

(d) **Cleaning & Sanitation.** Regularly clean utensils and surfaces.

(e) **Food Inspection.** Check for spoilage before consumption.





20. **Prevention of Foodborne Diseases.**

(a) **Cook Food Thoroughly**. Kill bacteria with proper cooking temperatures.

(b) **Refrigerate Promptly**. Prevent bacterial growth by storing food correctly.

(c) **Avoid Cross-Contamination**. Keep raw and cooked foods separate.

(d) **Avoid Raw or Undercooked Foods**. Avoid consuming raw or undercooked foods, especially for vulnerable populations such as the elderly, pregnant women, and young children.

(e) **Stay Informed**. Be aware of food recalls and safety alerts.

21. **Common Foodborne Diseases/Conditions**. Bacteria, virus, parasites or toxins may cause the following:-

(a) Diarrhoea

(b) Abdominal and Stomach Cramps

(c) Urinary Tract Infections

(d) Pneumonia

(e) Vomiting and Fever

Food Safety and Hygiene in Daily Life

22. Food safety and hygiene are vital for anyone handling food in the kitchen.

(a) **Avoiding Cross-contamination**. Use separate cutting boards for raw meat and vegetables.

(b) **Safe Storage Practices**. Refrigerate perishables within two hours.

(c) **Mindful Ingredient Sourcing**. Source ingredients from reliable suppliers.

(d) **Regular Hand Hygiene**. Wash hands thoroughly before food preparation.

Personal Hygiene

Wash Hands. Wash hands with soap and water before eating, after using the toilet, and after handling pets or animals.
Keep Fingernails Clean. Keep fingernails short and clean to prevent dirt accumulation.
Bathe Regularly. Bathe or shower regularly to maintain personal hygiene.

Food Handling and Preparation

Separate Raw and Cooked Foods. Separate raw and cooked foods to prevent cross-contamination.
Use Clean Utensils and Equipment. Use clean utensils and equipment when handling food.
Cook Food Thoroughly. Cook food thoroughly, especially meat, poultry, and eggs.
Reheat Food Safely. Reheat food to a minimum internal temperature of 165°F (74°C) to prevent bacterial growth.

Food Storage

Store Food at Proper Temperatures. Store food at proper temperatures (below 40°F or 4°C for refrigerated foods and 0°F or -18°C for frozen foods).
Use Airtight Containers. Use airtight containers to store food and prevent contamination.
Label and Date Leftovers. Label and date leftovers to ensure they are consumed within a safe time frame.

Safe Water and Food Consumption

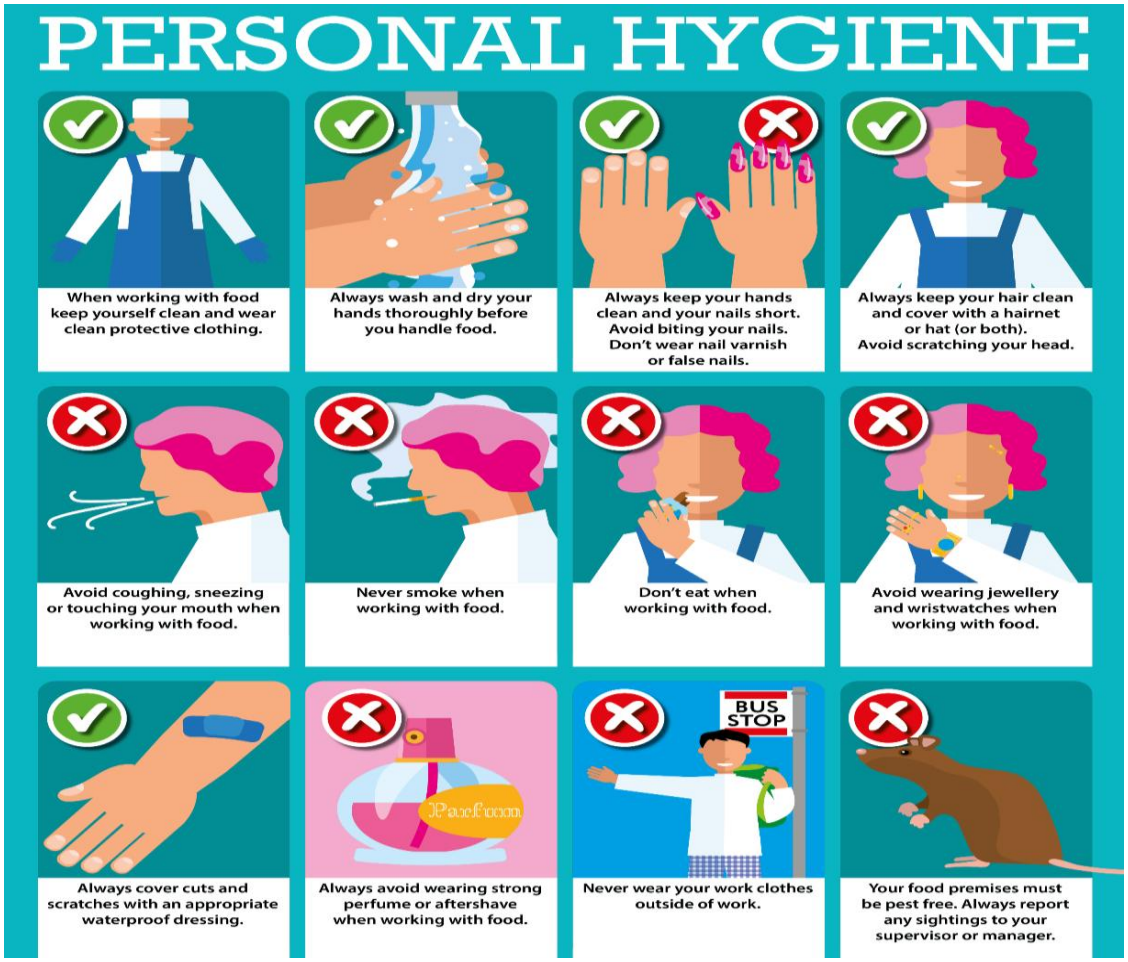
Drink Safe Water. Drink water from a safe source, such as a municipal water supply or a filtered water system.
Avoid Raw or Undercooked Foods. Avoid consuming raw or undercooked foods, especially meat, poultry, eggs, and seafood.
Avoid Expired or Spoilt Food. Avoid consuming expired or spoilt food, as it can cause foodborne illnesses.

Additional Tips

Wash fruits and Vegetables. Wash fruits and vegetables thoroughly before consuming them.
Avoid Cross-Contamination. Avoid cross-contamination by separating raw and cooked foods and using separate utensils and equipment.
Report food Safety Concerns. Report any food safety concerns to a teacher, parent, or school administrator.

23. By maintaining personal hygiene, oral care, and food safety, we can significantly reduce health risks, promote well-being, and create a safer environment for all:-





PART III : CAMP SANITATION

24. Hygiene and sanitation are critical for maintaining the health and well-being of cadets during NCC training camps. Proper planning and discipline in these aspects ensure a safe and efficient training environment while preventing diseases and health hazards.

Preliminary Recce for Selection of Camp Sites

25. Before a camp begins, a reconnaissance (recce) is conducted about four months in advance to ensure a suitable site is chosen. The ideal camp location should:-

(a) Be accessible by road or rail.

(b) Be away from populated areas to prevent disturbances.

(c) Have level ground with adequate built-up structures for accommodation and storage.

(d) Provide open space for training activities.

(e) Be free from water hazards, pests, and epidemic diseases.

(f) Have a reliable water source and proper drainage.

26. **Move to Camp and Setup.** An advance party arrives four days before the camp starts to set up essential facilities, including:-

(a) **Sanitation and Hygiene.** Cleaning the area, marking camp layout, and establishing water points and bathing facilities.

(b) **Living Arrangements.** Setting up accommodation, cookhouses, offices, storage areas, and medical rooms.

(c) **Security and Coordination.** Collaborating with local authorities for necessary services and ensuring camp safety.

Camp Hygiene and Sanitation

27. Sanitation means keeping the living area and its surroundings neat and clean. This involves removal of waste products and refuse. Maintaining hygiene during camps is crucial for the health and efficiency of cadets. Certain important aspects are given in succeeding paragraphs.

28. **Cook House.** The quality of food in the Camp should be of high standard and purchased from authorised or reputed dealers. All precautions should be taken to ensure that no case of Food Poisoning occur in the Camps. A sample of food provided to cadets will be preserved for 36 hours after the meal is served.

29. **Cooking and Serving of Food**. The following points need attention:-

(a) All preliminaries, such as, cutting, peeling and washing of food items will be

done in a separate room. Proper preparation tables, cutting and chopping blocks will be used for the purpose

(b) The cooked food should be served hot and so timed that food is ready just a few minutes before distribution.

(c) Regular inspection of the hygiene, sanitation and cleanliness of the cook house staff and utensils should be carried out.

(d) Avoid over peeling, use of excessive oil, condiments, salt and sugar.

(e) If food is being out-sourced, it should be from approved vendors, stored hygienically, cooked properly, and served hot.

(f) Waste food disposal must be systematic to avoid pests and contamination.

30. **Soakage Pits**. Each cook house will have soakage pit, 6 feet deep and 4 1⁄2 feet wide. These pits will be fly-proofed, filled with lime stone/brick and covered with a top layer of earth. Waste food will be buried in refuse pits which will be located away from the cook houses.

31. **Water Points.** Water used in the Camp must be from authorised source and must be checked by the local medical authority and certified as potable. Chemical for water treatment should be procured depending on requirement. Emphasis be laid on the following:-

(a) Cleanliness of the water containers and the surroundings be ensured.

(b) Proper drainage facilities/soakage pits to avoid stagnation of water.

(c) Wherever possible, provision of RO plant should be ensured for drinking water.

32. **Personal and Camp Hygiene.**

(a) **Personal Hygiene.** Cadets must maintain personal hygiene by bathing regularly, washing hands before meals, and wearing clean clothes.

(b) **Clean Living Spaces.** Living spaces should be kept dry, ventilated, and free from clutter.

(c) **Proper Waste Disposal.** Waste must be disposed of in designated areas using covered bins.

(d) **Clean Toilets.** Toilets and bathing areas should be cleaned regularly, with proper drainage systems in place.

(e) **Cookhouse Sanitation.** Cookhouse sanitation must follow strict rules, including handwashing stations, food inspections, and waste management.

33. **Health and First Aid.**

(a) **First aid.** First aid measures should be in place to handle minor injuries, with serious cases reported immediately.

(b) **Hydration.** Hydration must be maintained to prevent dehydration and heat exhaustion.

(c) **Nutrition.** Proper nutrition is necessary to ensure cadets stay fit and active during training.

34. **Discipline and Responsibility of Cadets.**

(a) Every cadet is responsible for keeping their surroundings clean.

(b) Camp hygiene activities, including waste disposal and area maintenance, should be followed diligently.

(c) Reporting hygiene or sanitation concerns to authorities ensures the well-being of all cadets.

35. Hygiene and sanitation are essential components of NCC training camps, ensuring cadets remain healthy, disciplined, and focused on their training. A well-maintained camp fosters teamwork, morale, and efficiency, contributing to a successful and enriching NCC experience.

36. **Camp Latrines.** Built-up accommodation may not always be available and camps may have to be conducted in open/field conditions, necessitating digging of latrines. Camp latrines are of following types:-

(a) **Deep Trench Latrines(DTLs).** A pit three feet wide, at least eight feet deep and of a length suitable to the requirement is constructed and wooden seats placed over it with proper partitions and curtains. Soil may necessitate reverting of sides with sand bags, bamboos or wire netting. On vacation of camp, these are filled up with soil to assist in disintegration and prevent breeding of flies.

(b) **Shallow Trench Latrines**. For camps of less than a week's duration, dig a row of trenches in parallel, each trench being 3 feet long, 1-foot-wide and 2 feet deep. Each trench should be 2 feet apart. The ratio is 5 trenches for the first hundred users and three for each subsequent hundred. After defecation, the excreta are covered with loose earth with a shovel or a scoop. These trenches are filled up after 24 hours and new trenches are dug up.

(c) **Urinals.** The most common urinals used for camps are the Funnel Urinals which are constructed over a simple soakage pit.

37. DTLs of proper specifications will be constructed and covered by proper latrine frames. Latrines will be smoked every morning. The wooden frames will also be washed

twice a day and treated with lime. Partition screens will be erected for each latrine seat. Night Latrines will be earmarked and lighting provided. Sufficient urinals should be provided within the camp area. These will be treated daily with creosote and lime quick. proper security ensured.

HIGHER ORDER THINKING SKILLS (HOTS)

- **What are the possible consequences of not washing hands regularly? Analyse the situation and explain your answer.**
- **Compare and contrast the importance of personal hygiene in rural and urban areas, with examples.**
- **How does poor hygiene contribute to the spread of diseases? Explain the chain of events that leads to the spread of diseases.**

PART IV : DOs AND DON'Ts IN CAMPS

38. **Dos**.

(a) Wash hands or use hand-sanitiser, especially after coughing/sneezing.

(b) Brush and floss teeth daily.

(c) Sanitise food preparation areas.

(d) Clean surroundings regularly.

(e) Wash hands before and after having food.

(f) Take only that much quantity of food which you can eat, avoid wastage.

(g) Wear clean clothes and undergarments, wash them daily.

39. **Don'ts**.

(a) Avoid sharing personal hygiene items (e.g., towels, toothbrushes).

(b) Don't leave food uncovered or improperly stored.

(c) Avoid littering camp areas, make use of dustbins/refuse pits.

(d) Never use untreated water for drinking or cooking.

(e) Don't clog the washbasin/utensil-washing area with waste food.

(f) Don't spit casually, respect your surroundings.

(g) Don't go to non-designated areas for ablutions.

CONCLUSION

40. Adequate sanitation, together with good hygiene and safe drinking water, are fundamental to good health and to social and economic development. Maintaining personal & oral hygiene and ensuring food safety leads to healthy outcomes and promotes environmental sustainability. Additionally, maintaining high hygiene standards fosters a sense of inter-personal respect, creating a cohesive and supportive community. Recognising the link between cleanliness and physical & mental well-being underscores the importance of promoting hygiene and sanitation as a favoured approach towards life.

41. By instilling a culture of hygiene consciousness through training and reinforcement of proper hygiene protocols, cadets can better protect themselves and fellow cadets from preventable illnesses. Prioritising personal hygiene in camps not only boosts individual health but also plays a crucial role in maintaining good physical, mental health and high morale for undergoing training.

SUMMARY

- Hygiene and sanitation practices help prevent the spread of diseases, promote cleanliness and foster a positive and respectful environment.
- Personal hygiene prevents diseases and improves quality of life.
- Good oral hygiene practices like brushing, flossing, and rinsing are essential to avoid dental problems.
- Food safety measures ensure the prevention of food-borne illnesses.
- Adherence to hygiene practices contributes to better physical and social well-being.
- The mission of an effective food safety system is to protect and improve public health.
- Recce of camp site should be carried out diligently to meet the hygiene and sanitation requirements of the camp.
- Various appointments of the camp must ensure the measures to be undertaken to ensure proper camp sanitation and implemented in time.
- Living area, water point, and cookhouse form major aspects to be focused while ensuring maintaining good camp hygiene.
- Good personal and camp hygiene are morale boosters for cadets.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the primary purpose of personal hygiene?

(a) To look attractive

(b) To prevent diseases and infections

(c) To follow societal norms

(d) To reduce the need for medical check-ups

Q2. Which of the following is NOT a key aspect of oral hygiene?

(a) Brushing twice daily

(b) Flossing regularly

(c) Using a tongue scraper

(d) Drinking sugary beverages frequently

Q3. Which of these diseases is caused by poor sanitation?

(a) Diabetes (b) Tuberculosis

(c) Typhoid (d) Hypertension

Q4. Which of the following practices prevents foodborne diseases?

(a) Eating expired food

(b) Washing hands before handling food

(c) Cooking food at a low temperature

(d) Mixing raw and cooked food

Q5. Which organization defines hygiene as "conditions and practices that help maintain health and prevent the spread of diseases"?

(a) UNICEF (b) WHO

(c) FDA (d) FAO

Q6. What is the first step in maintaining proper hand hygiene?

(a) Drying hands with a towel

(b) Using hand sanitizer

(c) Washing hands with soap and water

(d) Rinsing hands with plain water

Q7. Which of the following is a direct benefit of oral hygiene?

(a) Reducing body odour

(b) Preventing tooth decay and gum disease

(c) Preventing acne

(d) Improving digestion

Q8. What is a major cause of foodborne diseases?

(a) Proper food storage (b) Washing hands before eating

(c) Cross-contamination of food (d) Cooking food thoroughly

Q9. Which of the following is NOT a proper waste disposal practice in camp sanitation?

(a) Using covered bins for waste

(b) Burning plastic waste in open areas

(c) Proper sewage treatment

d) Disposing of food waste systematically

Q10. What is the purpose of conducting a recce for selecting a camp site?

(a) To ensure accessibility, sanitation, and safety

(b) To provide a place for entertainment

(c) To avoid interactions with locals

(d) To establish trade with nearby areas

Q11. What is the recommended method to clean drinking water in a camp?

(a) Storing it in an open container

(b) Boiling or treating with chlorine

(c) Keeping it near waste disposal sites

(d) Using only river water

Q12. What is a critical practice to ensure good camp hygiene?

(a) Cooking food only once a day

(b) Not cleaning the cookhouse regularly

(c) Keeping living spaces dry and ventilated

(d) Ignoring proper waste disposal

Q13. What should be done after using the toilet to maintain hygiene?

(a) Spray perfume (b) Wash hands with soap and water

(c) Avoid drinking water (d) Use a cloth to wipe hands

Q14. Which of the following best describes sanitation?

(a) A set of habits for personal grooming

(b) Ensuring safe disposal of waste and clean surroundings

(c) Only washing hands regularly

(d) Brushing teeth twice a day

Q15. Which practice should be avoided to maintain food hygiene?

a) Storing food at safe temperatures

b) Using separate cutting boards for raw meat and vegetables

c) Eating food past its expiration date

d) Washing hands before preparing food

Short Answer Questions

Q1. What are three key aspects of personal hygiene, and why are they important?

Q2. List three consequences of poor oral hygiene and explain their impact on health.

Q3. Explain the role of handwashing in preventing disease transmission.

Q4. Why is camp sanitation crucial in NCC training camps?

Q5. What are two major causes of food contamination, and how can they be prevented?

Long Answer Questions

Q1. Discuss the importance of maintaining hygiene and sanitation in daily life. How does it contribute to individual and community well-being?

Q2. Describe the essential practices for maintaining oral hygiene. Why is regular dental care important?

Q3. Explain the importance of food safety and hygiene in preventing foodborne illnesses. Provide examples of preventive measures.

Q4. What are the key factors to consider while selecting a camp site to ensure proper hygiene and sanitation?

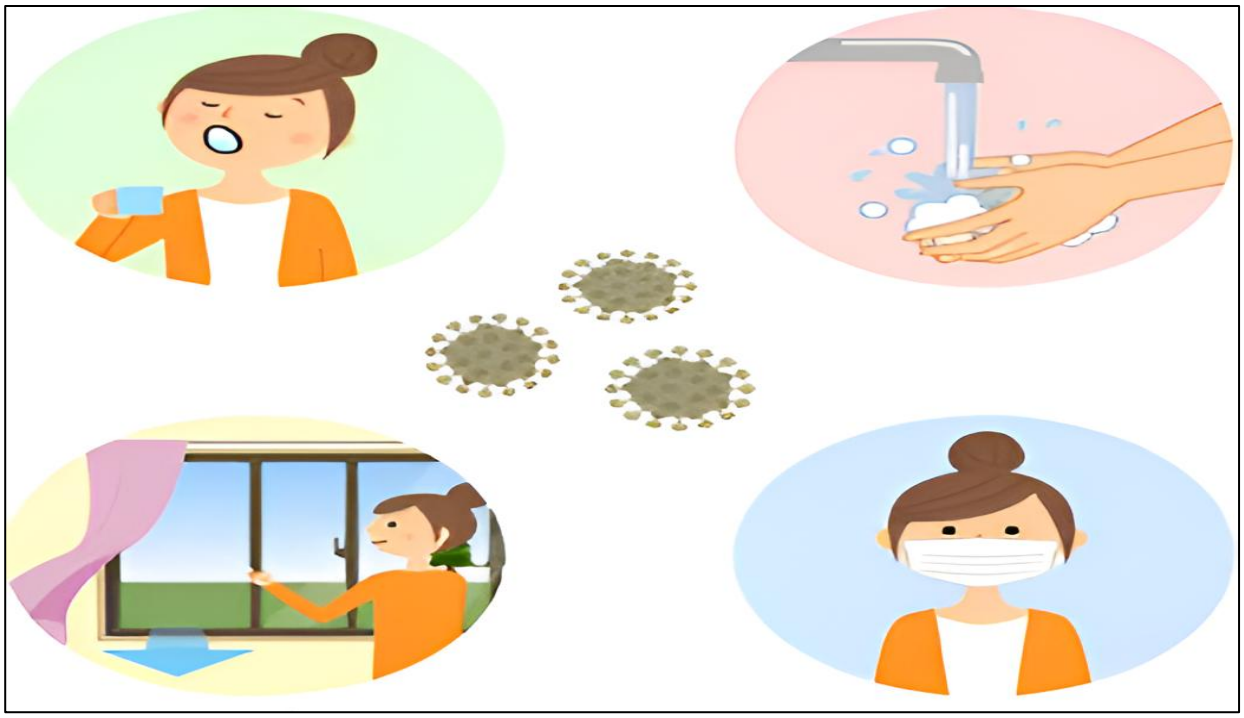
Q5. Describe the dos and don’ts of maintaining hygiene and sanitation in an NCC camp. How do these practices contribute to cadet health and discipline?

PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW III: INFECTIOUS DISEASES AND PREVENTION

"The greatest medicine of all is teaching people how not to need it."

Hippocrates



TEACHING INSTRUCTIONS

Period	**:**	**One (01)**
Type	**:**	**Lecture**
Year	**:**	**2nd Year (SD/SW)**
Conducting Officer	**:**	**Associate NCC Officer**
Training Aids	**:**	**Script/Lesson Plan: Computer with OHP, Screen, Pointer Staff, Presentation**

Time Plan

- Introduction : 02 Mins
- Part I : 15 Mins
- Part II : 12 Mins
- Part III : 08 Mins
- Conclusion : 03 Mins



INTRODUCTION

1. A disease is an abnormal condition affecting part or all of the body. Pathology, the study of disease, investigates its cause. Infectious diseases are medical conditions caused by microorganisms, including bacteria, viruses, fungi, and parasites. These illnesses can be transmitted directly between individuals or indirectly through contaminated food, water, air, and vectors like insects.

2. Understanding how infectious diseases spread and the best practice for prevention is essential for maintaining public health. Effective measures such as proper hygiene, quarantine, vaccination, and timely medical intervention can help reduce the transmission of infectious diseases. Vaccinations, in particular, have been instrumental in controlling and even eradicating deadly diseases such as smallpox and polio. This chapter aims to provide an overview of infectious diseases, their transmission, prevention strategies, and the role of important vaccinations in safeguarding health.

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I: Infectious Diseases

(b) Part II: Prevention of Infectious Diseases

(c) Part III: Important Vaccinations

INTERESTING FACTS

- Mosquitoes are responsible for more human deaths annually than any other insect, due to malaria and other vector-borne diseases.
- Vaccines have eradicated smallpox, a disease that killed millions in history.

LEARNING OBJECTIVES

- Learn about various infectious diseases and identify the pathogens that cause them.
- Study the modes of transmission for infectious diseases.
- Understand the importance of hygiene practices in preventing infectious diseases.
- Learning about quarantine and isolation in controlling outbreaks.
- Understand the mechanism of vaccines and list essential vaccines for common diseases.

3. Infectious Means Spreading or Causing Infection. “Infection” is having a disease, and “to infect” means contaminating with germs. Infectious diseases spread germs.



4. **Infectious Diseases.** Pathogenic microbial agents, such as viruses, bacteria, or other microorganisms, cause infectious diseases. Pathogenic means “disease- producing.” Microbial refers to microbes—or microorganisms— that can produce disease. Viruses are sometimes referred to as microbes because they are microscopic and can cause disease. An infectious disease occurs when harmful germs enter the body, spread, and disrupt normal bodily functions.

5. Examples of infectious diseases include chickenpox, cholera, common cold, COVID-19, Ebola, flu, hantavirus, hepatitis A & B, HIV/AIDS, Lyme disease, malaria, measles, meningitis, polio, pneumonia, smallpox, STDs (sexually transmitted diseases), tuberculosis, West Nile virus, and Zika.

6. Some diseases are common, like colds and flu, while others are rare or largely eliminated.

7. **Pathogens**. Pathogens are microorganisms, such as bacteria, viruses, fungi, or parasites, that can cause disease in their host organisms. They have the ability to infect and multiply within the host, leading to illness or harm. The types of pathogens and their roles are as follows:-

(a) **Bacteria.** Cause illnesses like tuberculosis.

(b) **Viruses.** Lead to diseases such as influenza and HIV/AIDS.

(c) **Fungi.** Cause infections like athlete’s foot.

(d) **Parasites.** Responsible for diseases such as malaria.

8. **Contagious Disease**. An infectious disease is contagious if it spreads through direct contact with an infected person, their discharges, or contaminated objects or surfaces.

(a) **Malaria.** It is an infectious disease caused by a parasite. However, **it is not considered contagious** because it cannot be contracted merely by being near or coming into contact with an infected individual. The primary mode of transmission is through the bite of a mosquito, which introduces the parasite into the bloodstream.

(b) **COVID-19.** It is a disease caused by a new type of coronavirus. It spreads easily through contact, droplets, and infected surfaces, making it highly **contagious.**

9. **Causes of Infectious Diseases**. Infectious diseases are caused by pathogens like viruses, bacteria, fungi, and parasites entering the body. They spread through person-to-person contact, contaminated food or water, and bug bites. Symptoms arise from the pathogens damaging cells and the immune system’s response. While some infections are minor, others can be serious.

10. **Modes of Transmission of Infectious Diseases.** Infectious diseases spread in various ways, sometimes through multiple methods. Major ways include:-

(a) **Direct Contact**. Touching, sexual contact, or contact with infected bodily fluids.

(b) **Airborne Spread**. Inhaling droplets containing pathogens (e.g., COVID-19, tuberculosis).

(c) **Vector-Borne Transmission**. Spread via insects like mosquitoes (e.g., malaria).

(d) **Contaminated Food/Water**. Ingesting contaminated substances (e.g., cholera, hepatitis A).

Overview of Common Infectious Diseases

11. Infectious diseases are widespread globally, though some occur more frequently than others. These infections occur when an organism enters body cells, releases toxins, and triggers reactions in host tissues. Anyone can contract an infectious disease. Factors such as a weakened immune system or travel to regions with certain highly transmissible diseases may increase the risk. Individuals at higher risk of infectious disease include:-

(a) Individuals with weakened immune systems, such as those undergoing cancer treatment, living with HIV, or on certain medicines.

(b) Young children, pregnant women, and senior citizens.

(c) Unvaccinated individuals against common infectious diseases.

(d) Medical professionals.

(e) People travelling to areas with mosquitoes that carry malaria, dengue, or Zika virus.

12. **Common Infectious Diseases.** Understanding the symptoms, complications, and global impact of common infectious diseases can aid in their identification and the timely implementation of precautions to prevent or reduce the spread of these diseases:-

Influenza

- **Symptoms. Fever, chills, sore throat, and body aches.**
- **Complications. Pneumonia and worsening of chronic conditions.**
- **Global Impact. Seasonal outbreaks causing significant morbidity.**

Tuberculosis

- **Symptoms. Persistent cough, weight loss, night sweats.**
- **Complications. Lung damage.**
- **Global Impact. A leading cause of death worldwide.**

HIV/AIDS

- **Symptoms. Weak immune system, frequent infections.**
- **Complications. Opportunistic infections, cancer.**
- **Global Impact. A pandemic affecting millions.**

Malaria

- **Symptoms. Fever, chills, headache.**
- **Complications. Severe anaemia, cerebral malaria.**
- **Global Impact. Endemic in tropical regions.**

Covid-19

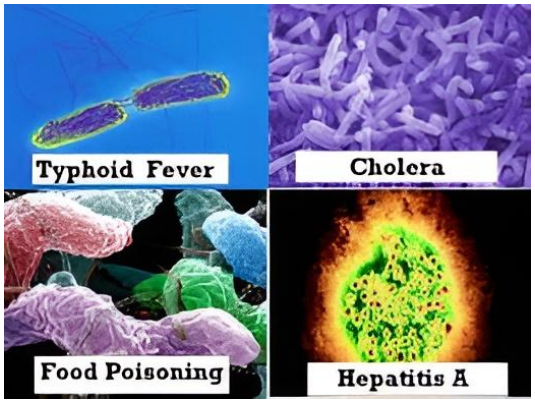
- **Symptoms. Fever, cough, loss of taste/smell.**
- **Complications. Pneumonia, long COVID.**
- **Global Impact. Worldwide pandemic with economic and social effects.**

Hepatitis (A, B, C)

- **Symptoms. Jaundice, fatigue, abdominal pain.**
- **Complications. Liver damage, cirrhosis.**
- **Global Impact. Millions affected annually.**

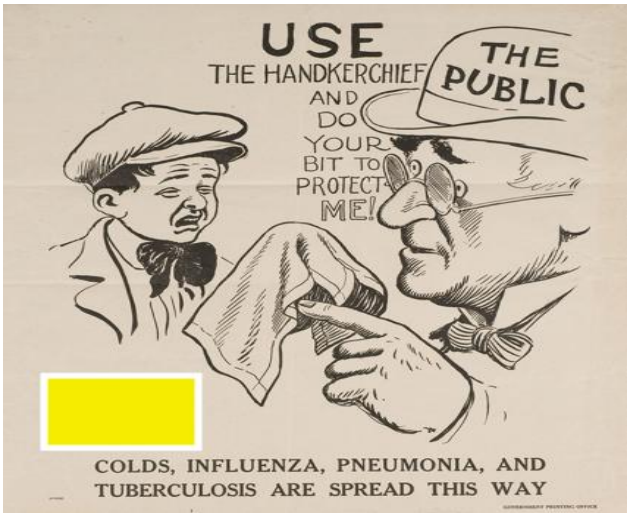
13. Contagious diseases (communicable diseases) are classified as follows:-

(a) **Excremental Diseases.** Diseases spread through human excreta (urine and faeces) can contaminate food, water, or hands of cooks, leading to infections like typhus fever, dysentery, diarrhoea, jaundice, and intestinal worms.



(b) **Droplet Infection.** These diseases are transmitted through pathogens expelled from the nose, throat, or lungs into the air in small droplets of saliva during coughing, sneezing, or even speaking. Common illnesses within this category include the common cold, influenza, diphtheria, meningitis (inflammation of the brain), and tuberculosis.





DID YOU KNOW?

- The first vaccine was developed by Edward Jenner in 1796 for smallpox.
- HIV/AIDS affects over 38 million people globally.
- Quarantine as a public health measure dates back to the 14th century, imposed to control the 'Black Death' Plague.

(c) **Contact Diseases.** These diseases spread through direct contact with an infected person. Examples include venereal diseases like syphilis, gonorrhoea, and skin infections.





(d) **Insect Borne Diseases.**
These diseases spread when germs are transferred from a sick person to a healthy person by blood-sucking insects, known as 'carriers.' These insects bite an infected person and then a healthy one, transmitting the germs. These germs multiply in the new host's blood during incubation, leading to symptoms.

- **Mosquito. Malaria, Dengue, and Filaria.**
- **Sand fly. Sand-fly fever, Kala Azar, Oriental Sore.**
- **Lice. Typhus, Relapsing Fever.**
- **Flies. Diarrhoea, Dysentery, Cholera, Typhoid.**
- **Fleas. Plague, Typhus.**
- **Ticks. Relapsing Fever, Typhus.**

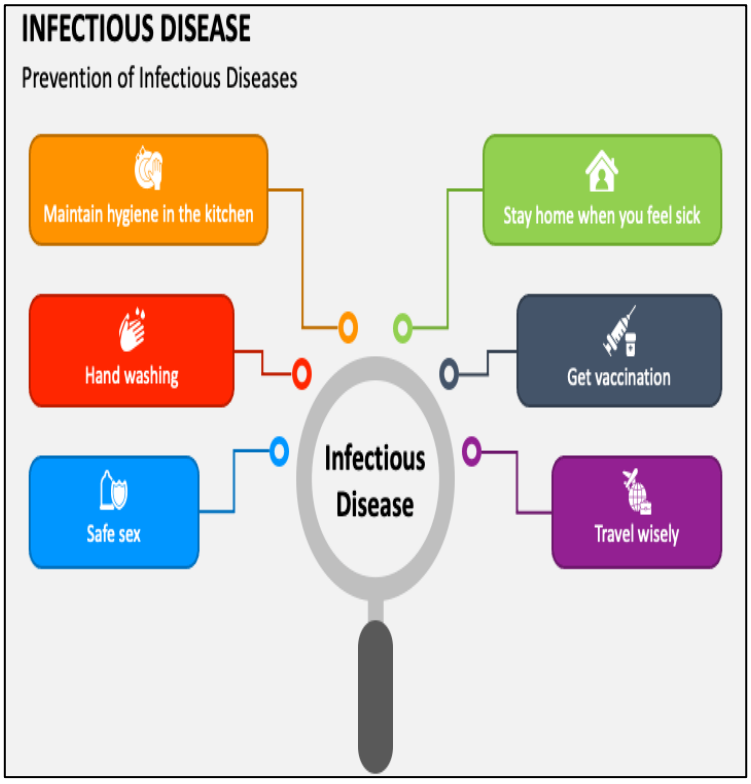
(e) **Waterborne Diseases.** Diseases such as cholera, dysentery, diarrhoea, and jaundice spread through water. Contamination occurs through vomit or faeces entering the water. To prevent epidemics, it is essential to disinfect water and properly dispose of waste. Eliminating all disease sources is crucial.

(f) **Animal-Borne Diseases.** Animals may transmit pathogens through the consumption of milk or via insect vectors. Some common zoonotic diseases include rabies, plague, anthrax, and tuberculosis.

PART II : PREVENTION OF INFECTIOUS DISEASES

14. Hygiene Practices to Prevent Infectious Diseases.

(a) Wash your hands with soap and water. It is particularly important to ensure meticulous handwashing before meal preparation or eating, using the restroom, and following gardening or contact this soil. Peel or thoroughly wash all fruits and vegetables.



(b) Clean food preparation surfaces and utensils thoroughly with soap and water after each use.

(c) Freeze meats at -18ºC until thawing, then cook to a safe temperature before eating.

(d) Avoid consuming raw or partially cooked seafood.

(e) Avoid drinking untreated water.

(f) Consumption of unpasteurised milk is not recommended.

(g) Ensure to cover your nose and mouth when sneezing or coughing.

(h) Disinfect surfaces that are frequently touched in both your home and workplace.

(j) Do not touch or share personal items with infected persons.

(k) Avoid others while sick with an infectious disease. When sick, use a mask.

(l) Refrain from drinking or swimming in water that may be contaminated.

(m) Use tick and mosquito repellent, wear protective clothing, and check for ticks after being in wooded or grassy areas.

(n) Dispose of waste properly.

(o) Maintain a clean environment.

15. **Quarantine and Isolation.**

(a) **Quarantine**. Limits movement of those exposed to a contagious disease but not yet symptomatic.

(b) **Isolation**. Isolates individuals with infections to prevent disease transmission.

(c) **Significance**. Interrupts the transmission chain during outbreaks.

Prevention of Contagious Diseases

16. **Prevention of Excremental and Water-Borne Diseases.** Measures to prevent to prevent excremental and water-borne diseases include:-

(a) Control of water routes can be effectively managed by disinfecting the water supply or providing an alternative source of safe water.

(b) The control of the milk distribution process can be effectively managed by subjecting the milk to boiling or pasteurisation.

(c) Foodborne infection can be managed through food hygiene standards and by preventing sick individuals from handling food.

(d) Proper excreta disposal prevents disease transmission through faecal matter.

17. **Prevention of Droplet Infections.** This can be accomplished through the use of masks, adequate spacing of beds, screening, dust control, preventing overcrowding, proper ventilation, refraining from spitting in public areas, ensuring sufficient sunlight exposure, and appropriate air disinfection.

18. **Prevention of Contact Diseases.** Detailed measures to prevent contact diseases include:-

(a) Implement comprehensive patient segregation.

(b) No direct contact between the patient and relatives or staff.

(c) Early diagnosis prevents patient spread.

(d) Proper disposal of all excreta and disinfection of all patient articles.

19. **Prevention of Insect-Borne Diseases.** Measures to prevent insect-borne diseases:-

(a) Filling, levelling, and drainage breeding places, along with proper water management, can eliminate larvae. Effective sewage and wastewater disposal are crucial for preventing breeding and spread of Culex mosquito.

(b) Utilise kerosene oil, fuel oil, or specialised oil to prevent larvae.

(c) Application of residual sprays such as malathion.

(d) Use mosquito nets, screen doors and windows, apply repellent, and wear long sleeves after sundown.

(e) Control rodents, fleas and mosquitos in and around the house.

(f) Refrain from contact with any wild rodents, particularly those that appear sick or are deceased.

(g) Avoid contact with sick or dead animals or their waste.

Preventive Measures to Inhibit the Spread of Contagious Diseases

20. **Segregation of the Patient.** The following preventive measures are necessary to control the spread of contagious diseases:-

(a) Preferably shift the patient to an isolated room.

(b) Ensure the room conforms to hygiene and sanitation standards, i.e., adequate ventilation, sunlight, and cleanliness.

(c) Nominate one healthy person to undertake nursing and care of the patient.

(d) Nominated person to take preventive measures like the use of masks, gowns, and gloves and avoid direct contact and hand washing before and after every visit.

(e) Clothing and utensils used by the patient to be cleaned/washed separately.

(f) Safe disposal of patients' excreta – Urine, Stool, Sputum, Refuse, e.g., discarded dressings, garbage etc. by burning.

(g) Disinfect frequently touched surfaces like doorknobs, light switches, and furniture with appropriate disinfectants.

(h) Limit visitors to only essential caregivers to minimize the risk of infection spread.

(j) Monitor the patient's symptoms regularly and seek medical attention if their condition worsens.

21. **Destroy Agents (Germs) Causing Infection in Surrounding Areas or Premises**. Immediately upon detecting a communicable disease, take the following actions to destroy the source of agents/germs causing infection:-

(a) Removal or destruction of garbage.

(b) Cleaning the drains & keeping them covered.

(c) Remove/dry out wastewater.

(d) Spray Malathion mixed in water (1 mL of Malathion and 1000 mL of water) in and around the premises. (Caution - Malathion is highly poisonous. Direct breathing/touching should be avoided.) Wash hands after use.

(e) Keep premises free of rodents (rats) and stray dogs.

(f) Use mosquito nets, long sleeves, and screening of doors and windows.

Disinfection Means

22. **Natural Means.** Sunlight and air can be used to disinfect articles like blankets, mattresses, pillows, and also the rooms. Microorganisms thrive in darkness and need moisture for their survival; on being exposed to sunlight and air, they die.

23. **Physical Means.** Ways to physically destroy germs are as under:-

(a) **Heat.** Heat can be used in two ways for sterilising. Moist Heat in the form of boiling kills germs very rapidly. Dry heat sterilization, using methods like flaming or hot air ovens, involves exposing materials to high temperatures without moisture to kill microorganisms. 'Autoclaving' or Steam-under-Pressure is the most effective method used to sterilise all hospital equipment, which can be boiled, like linen, bandage, dressing material, gloves etc.

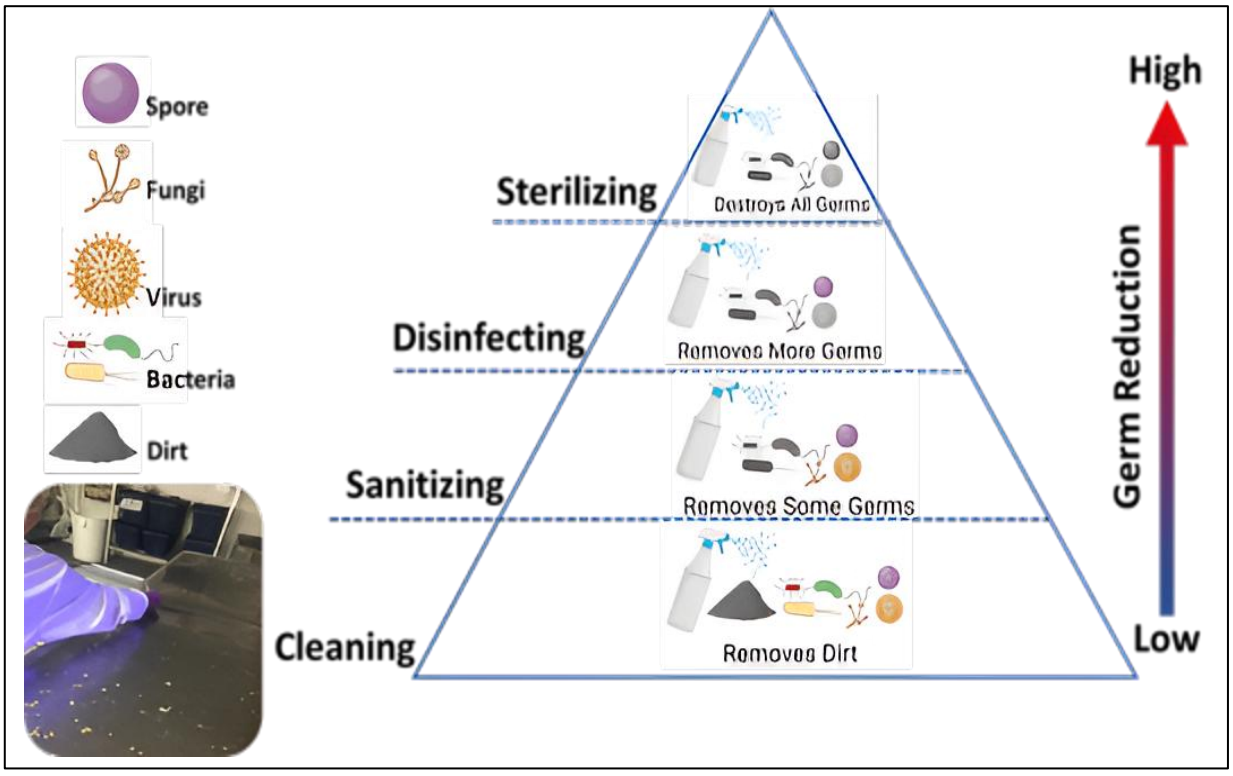
(b) **Cold.** Freezing or freeze-drying can inactivate bacteria. Some of them can, however, survive even at 0° C.

(c) **Radiation.** It includes ionising radiations like X-rays, gamma, beta, and ultraviolet radiation. These are expensive methods and are not suitable for small-scale procedures.

(d) **Chemical Agents.** Agents like phenol, Savlon, potassium permanganate, hydrogen peroxide, etc. are commonly used for disinfection and sterilisation. The strength of the agent depends upon the concentration being used.

(e) **Other Methods.** Disinfection and sterilisation can also be done by using other methods like infrared rays, filtration, etc.





24. **<u>Control of Food and Drink</u>.**

(a) Clean water, preferably boiled, must be served to the patient. Water containers must be kept covered.

(b) A balanced diet, well-cooked, hygienically prepared food using less oil and condiments, should be served hot. Stale, cold, and food exposed to flies & insects should not be served.

(c) Food items sourced from restaurants/dhabas must not be served.

PART III : IMPORTANT VACCINATIONS

Vaccines



25. A vaccine is any substance used to stimulate the production of antibodies, which can provide immunity against one or several diseases.

26. **Mechanism**. A harmless form of a pathogen is introduced to stimulate the immune system to produce antibodies.

27. **Examples of Essential Vaccines**.



(a) MMR (Measles, Mumps, Rubella)

(b) DPT (Diphtheria, Pertussis, Tetanus)

(c) Polio (IPV/OPV)

(d) HPV (Human Papillomavirus)

28. **Inoculation and Vaccinations.** Free inoculations and vaccinations are provided under government programs at Primary Health Centres or Hospitals. If unavailable, they should be obtained from the market. Key inoculations and vaccinations include:-

Name of Vaccine	Disease Prevented
Injection Rabipur	**Rabies**
Injection Tab	**Typhoid**
Injection Hepatitis 'B'	**Hepatitis 'B'**
Injection T T	**Tetanus**
Oral Polio drops	**Polio**
MMR	**Measles, Mumps, Rubella**
DPT	**Diphtheria, Pertussis, Tetanus**
HPV	**Human Papillomavirus**

HIGHER ORDER THINKING SKILLS (HOTS)

- **Why are hygiene practices considered the first line of defence against infectious diseases?**
- **How does the transmission mode of a disease affect its prevention measures?**
- **In what ways do vaccines contribute to controlling global outbreaks of infectious diseases?**

CONCLUSION

29. Infectious diseases pose a major global health challenge, necessitating prevention, treatment, and management. Thus, studying these diseases is both a biological and social effort.

30. Since the beginning of the 21st century, diseases have posed complex challenges for human beings. Effective control of communicable diseases necessitates identifying the causative organism and taking measures to prevent its transmission. It is essential to treat the infected individuals and protect others from exposure to the pathogen. Additionally, legal regulations mandate the reporting of numerous communicable diseases.

SUMMARY

- Infectious diseases stem from bacteria, viruses, fungi, and parasites.
- They spread via contact, airborne particles, vectors, and contaminated food or water.
- Diseases like influenza, tuberculosis, HIV/AIDS, malaria, and hepatitis heavily impact global health.
- Hygiene, quarantine, isolation, and vaccines are essential for disease prevention and control.
- Vaccines against measles, polio, and HPV ensure lower morbidity/mortality rates.

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ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the study of diseases called?

(a) Immunology (b) Pathology

(c) Virology (d) Bacteriology

Q2. Which of the following is NOT a type of pathogen?

(a) Bacteria (b) Viruses

(c) Minerals (d) Fungi

Q3. How are infectious diseases primarily transmitted?

(a) Direct contact (b) Airborne spread

(c) Contaminated food or water (d) All of the above

Q4. Which of the following diseases is caused by a virus?

(a) Tuberculosis (b) Malaria

(c) Influenza (d) Athlete's foot

Q5. What is the primary mode of transmission for malaria?

(a) Airborne (b) Direct contact

(c) Mosquito bites (d) Contaminated Water

Q6. Which of the following diseases has been eradicated due to vaccinations?

(a) HIV/AIDS (b) Smallpox

(c) Tuberculosis (d) Malaria

Q7. What is an excremental disease?

(a) A disease spread by air droplets

(b) A disease spread by contaminated food, water, or human waste

(c) A disease spread through insect bites

(d) A disease that spreads via direct skin contact

Q8. Which of the following is NOT a preventive measure for infectious diseases?

(a) Frequent handwashing

(b) Consuming unpasteurized milk

(c) Using mosquito repellents

(d) Getting vaccinated

Q9. How does quarantine help control infectious diseases?

(a) It isolates infected individuals

(b) It limits movement of those exposed to a disease

(c) It provides medical treatment

(d) It strengthens the immune system

Q10. Which of the following vaccines prevents cervical cancer?

(a) DPT (b) MMR

(c) HPV (d) Polio

Q11. Which of the following is an example of a waterborne disease?

(a) Tuberculosis (b) Cholera

(c) Influenza (d) Syphilis

Q12. What is the main function of a vaccine?

(a) To cure a disease

(b) To kill all bacteria in the body

(c) To stimulate the immune system to produce antibodies

(d) To act as a pain reliever

Q13. What type of pathogen causes athlete's foot?

(a) Bacteria (b) Virus

(c) Fungi (d) Parasite

Q14. Which measure is effective in preventing droplet infections?

(a) Wearing masks (b) Washing fruits and vegetables

(b) Using mosquito nets (d) Avoiding dairy products

Q15. Which of the following statements is True?

(a) All infectious diseases are contagious

(b) Vaccinations can help prevent certain infectious diseases

(c) Insect-borne diseases are transmitted through direct contact

(d) Quarantine is only needed for bacterial infections

Short Answer Questions

Q1. What are the four main types of pathogens that cause infectious diseases?

Q2. How is malaria transmitted?

Q3. What is the difference between quarantine and isolation?

Q4. Name three hygiene practices that help prevent infectious diseases.

Q5. What is an example of an excremental disease, and how does it spread?

Long Answer Questions

Q1. Describe the different modes of transmission for infectious diseases with examples.

Q2. What are the key preventive measures for insect-borne diseases?

Q3. Explain the role of vaccinations in preventing infectious diseases. Provide examples.

Q4. Discuss the significance of hygiene in controlling the spread of infectious diseases.

Q5. What are contagious diseases, and how do they differ from non-contagious diseases? Give suitable examples.

PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW IV : ESSENTIAL FIRST AID AND LIFE-THREATENING EMERGENCIES

"First aid is not just about knowing what to do, but about doing what you know when every second counts."



TEACHING INSTRUCTIONS

Period	:	**Two (02)**
Type	:	**Lecture and Practice**
Year	:	**1st Year & 2nd Year SD/SW**
Conducting Officer	:	**Associate NCC Officer**
Training Aids	:	**Script/Lesson Plan; Presentation on Computer/ Chart; Mannequins.**

Time Plan

- Introduction : 02 Mins (1st Year)
- Part I : 08 Mins
- Part ll : 20 Mins
- Part III : 10 Mins
- Part IV : 35 Mins (2nd Year)
- Conclusion : 05 Mins



INTRODUCTION

1. First Aid is the immediate care given to a person who has been injured or suddenly taken ill. It is a vital skill that can help save lives, prevent further harm, and promote recovery. As NCC cadets, it is essential to learn the basics of First Aid.

2. In Camps accidents may happen and it is crucial to know how to respond. Through this training, you will learn the fundamental principles of First Aid including wound care, bleeding control, and basic life support.

3. It will be good to remember that while First Aid is not a substitute for medical treatment, but it can save lives in emergency situations. It can also prevent infections and reduce pain and discomfort.



LEARNING OBJECTIVES

- Understand the importance of First Aid in daily life.
- Identify common injuries and emergencies.
- Learn basic First Aid principles and how to provide immediate care.
- Be able to perform CPR and assist choking victims.

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I : First Aid

(b) Part II : Life-threatening Emergencies & CPR

(c) Part III : Importance of First Aid Knowledge

(d) Part IV : Practice on CPR

INTERESTING FACTS

- The earliest recorded evidence of First Aid practices comes from ancient Egypt, Greece, and Rome, where soldiers and caregivers used various techniques to treat wounds and injuries.
- The first First Aid manual was written in 1870 by Dr. Peter Shepherd, a British physician. It outlined basic First Aid techniques for treating common injuries.

PART I : FIRST AID



4. **First Aid**. First Aid is the **immediate care** provided to an injured or ill person before professional medical help arrives. Its objectives are to **preserve life, prevent further harm, and promote recovery**. **NCC cadets** should have basic First Aid knowledge to handle common injuries and emergencies.

5. Knowledge of First Aid provides NCC cadets with essential skills to handle injuries and emergencies, contributing to a safer environment during training and in daily life. By learning basic principles, cadets can offer immediate care and reduce complications until professional help is available.

Prevention of Further Harm	**First aid reduces the severity of injuries, prevents further harm, and promotes recovery.**
Preservation of Life	➢ **First Aid can save lives in emergency situations, such as cardiac arrest or severe bleeding.** ➢ **Timely intervention can prevent fatalities.**
Promotion of Recovery	➢ **First Aid promotes recovery by providing temporary treatment and support.** ➢ **Proper care can reduce the risk of complications and speed up the healing process.**
Building Confidence	➢ **Learning First Aid builds confidence and responsibility among NCC cadets.** ➢ **Cadets can respond effectively in emergency situations, making a positive impact.**
Enhancing NCC Training	➢ **First Aid training is an essential component of NCC training.** ➢ **Cadets learn valuable skills that complement their NCC training and prepare them for real-life situations.**
Providing Service	➢ **NCC cadets can apply their First Aid skills in community service initiatives.** ➢ **Cadets can provide assistance and support in emergency situations, demonstrating their commitment to social responsibility.**

6. **Principles of First Aid**.

(a) Assess the situation for safety.

(b) Call for help if necessary.

(c) Provide immediate care while ensuring personal safety.

(d) Stop bleeding, manage shock, and prevent infection.

(e) Keep the injured person comfortable until medical help arrives.

7. **Additional Principles of First Aid**.

(a) Keep the area clean and safe.

(b) Know how to use First Aid equipment correctly.

(c) Stay calm to provide effective First Aid.

8. **Activation of Emergency Medical Services.**

(a) Call emergency services if needed.

(b) Share details of injury or illness with medical staff.

9. **Points to be kept in mind before administering First Aid**. In the event of an emergency or when First Aid is required, the following should be verified before administering First Aid:-

(a) **History**. Ask casualties and witnesses: How did it happen?

(b) **Symptoms**. What does the casualty experience? (e.g., pain, immobility, or numbness)

(c) **Signs**. Observe level of consciousness, bleeding, weak pulse, limb deformities, and skin colour changes. Check for Medic-Alert bracelets or health cards.

(d) Ensure that approaching any incident or casualty does not put you at risk.

(e) Never touch someone in contact with electricity to avoid electrocution.

(f) Do not allow anyone to smoke near a crashed car; as there could be petrol, oil, or chemicals on the road.

10. **Priorities and Principles**. In incidents involving multiple casualties, prioritise treatment for the most severely injured individuals. Be aware that the noisiest casualty may not necessarily be the most seriously injured. Establish ABC (Airway, Breathing, Circulation) within three minutes if the casualty is unconscious to prevent permanent injury to the brain and other vital organs. Administer necessary care if shock is present.

11. **Steps of First Aid**. In an accident or incident, follow these to administer First Aid to the injured person:-

(a) Approach the casualty safely and easily.

(b) Assess the scene.

(c) Check if the person is conscious, semiconscious, or unconscious.

(d) Provide prompt and suitable treatment based on priority.

(e) Quickly arrange to move the casualty.

(f) First aid should be continued, once started.

(g) Minimise handling of the casualty.

(h) Comfort the casualty.

(j) Act swiftly and calmly.

(k) Facilitate the secure extraction of casualties.

PART II : LIFE-THREATENING EMERGENCIES & CPR

General Care of Injured Persons

12. **Principles of First Aid for Injuries**. Despite the wide variety of possible injuries, several principles of first aid generally apply to all emergencies:-

(a) Remain calm and avoid rushing.

(b) Unless the victim's location places them at risk of additional harm, the victim should remain unmoved until the extent of their injury has been assessed.

(c) Check the victim for wounds, burns, and fractures.

(d) Reassure the conscious victim.

(e) The victim's head should be kept at body level unless the person complains of difficulties in breathing.

(f) In the absence of a skull or spine injury, the victim's head and shoulders may be raised slightly to make the person more comfortable.

(g) If vomiting occurs, the head may be turned to one side.

(h) Do not give liquids to an unconscious person.

(j) Immediate relief is crucial for asphyxia, severe bleeding, and poisoning to prevent loss of life.

(k) The victim should be safeguarded against shock.

13. **Asphyxia**. Respiration supplies oxygen to the body's tissues and organs, essential for their function. It includes inspiration (air intake and lung expansion), expiration (air release and lung contraction), and a pause. The respiratory system comprises the respiratory tract & lungs, with muscle actions controlled by the brain's respiratory centre.

14. **Causes**. Asphyxia is a condition characterised by irregular breathing caused by any interference with respiration. The main causes of asphyxia are:-

(a) **Drowning**. Caused by water entering the air passage.

(b) **Hanging and Strangulation.** Cause air obstruction.

(c) Excessive intake of drugs such as sleeping pills, morphine, and pethidine.

(d) Electric Shock.

(e) Medical conditions, such as tetanus, epilepsy, and rabies.

(f) **Suffocation**.

(i) Obstruction of air entry through the air passage.

(ii) Foreign body in airway.

(iii) Inhalation of poisonous gases, e.g., carbon monoxide.

15. **Signs and Symptoms of Asphyxia**.

(a) Dizziness and weakness.

(b) Shortness of breath or rapid breathing.

(c) Rapid pulse.

(d) Partial loss of consciousness.

(e) Neck veins swell.

(f) Blue discolouration of face, lips, nails, fingers, and toes.

16. **General Rules for Treatment of Asphyxia**.

(a) Remove the cause or the casualty from the danger.

(b) Ensure there is a clear airway.

(c) Lay the person on their back. Press the head backward while supporting the neck with your palm. Ensure the tongue is clear of the airway. Administer mouth-to-mouth breathing.

First Aid in Special Cases

17. **First Aid for Drowning Cases**.

(a) Remove the person from water safely.

(b) Loosen wet clothes.

(c) Clear mouth, throat, and nostrils of mucus and foreign objects.

(d) Patient should be made to lie down over his belly, face down, head turned to one side, arms stretched beyond the head, tongue pulled out.

(e) Check for breathing and pulse.

(f) If not breathing, begin CPR immediately. Continue artificial respiration until the patient starts breathing.

(g) Keep the person warm and seek medical help.





18. **First Aid in Case of Strangulation**.

(a) Cut the constriction.

(b) Clear the air passage.

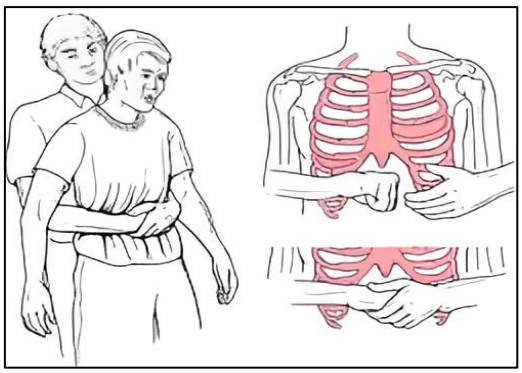
(c) Start artificial respiration.

(d) Administer inhalations if possible.

(e) Keep the patient warm and comfortable.

19. **First Aid in Case of Choking (Heimlich Manoeuvre).**

(a) Bend the casualty's head and shoulders forward to dislodge the obstructions.



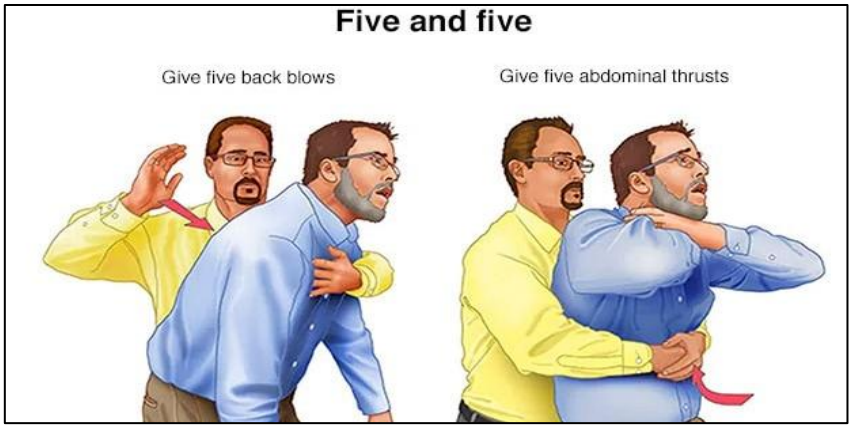
(b) In the case of a small child, hold the child upside down and thump the child's

back hard between the shoulder blades or encourage vomiting by passing two fingers to the back of the throat.

(c) Encourage the victim to cough forcefully.

(d) If choking continues, give five back blows followed by five abdominal thrusts.

(e) If unconscious, begin CPR immediately.



20. **First Aid in Case of Suffocation by Poisonous Gas**. Ensure your own safety and Dial emergency services informing them of the situation. Promptly move the affected individual to an area of fresh air, away from the gas. Position the victim comfortably, preferably on their left side, and monitor their breathing. Administer CPR if required.

21. **First Aid in Case of Electric Shock**.



(a) **Turn off the power source** before touching the person.

(b) Remove the person from contact with the current, if possible. Stand on insulating material, such as rubber-soled shoes or boots, or piles of newspapers. Use a stick or a rod of insulating material.

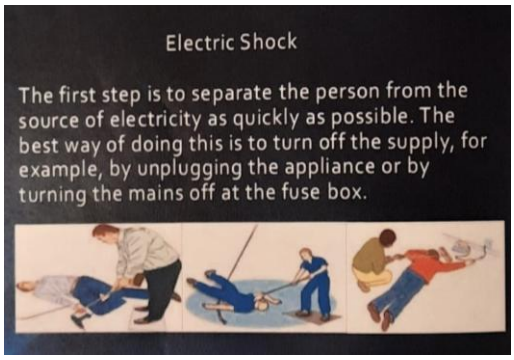
(c) Check for breathing and pulse.

(d) Start CPR if necessary.

(e) Provide treatment for shock and burns.

(f) Seek medical attention immediately.

22. **Insensibility or Unconsciousness**. Insensibility or unconsciousness happens when the brain's activity is disrupted, affecting the central nervous system's functions.



23. **Stages of Unconsciousness**.

(a) **1st Stage(Delirium)**. Restlessness of body and mind is present.

(b) **2nd Stage**. The patient responds to loud commands and provides the maximum response to minimum stimuli.

(c) **3rd Stage(Semi-Coma)**. Patient responds only to painful stimuli (minimal response to maximum stimulus).

(d) **4th Stage**. This stage is characterised by the patient's lack of response to any stimuli.

24. **Causes of Unconsciousness or Coma**.

(a) Head injury and brain haemorrhage.

(b) Hypoglycaemia.

(c) Heart failure.

(d) Physical agents such as heat strokes and electric shocks.

(e) Epilepsy.

25. **General Rules of First Aid in Casualties of Unconsciousness**.

a) Make the patient lie down with their head turned to one side. Pull out their tongue.

b) Loosen the clothing and provide fresh air.

c) If breathing is stopped or is irregular, start artificial respiration.

d) Keep the air passage clear.

e) Do not administer anything orally. Remove false teeth, if present.

26. **Artificial Respiration**. If breathing is irregular or there is cardiac arrest, provide artificial respiration promptly and correctly to save the patient's life. We will discuss the most common method.

27. **Mouth to Mouth Respiration(MMR)**. The simplest and most common method, effective when given promptly, involves the following:-

(a) Place the person on their back.

(b) Clear the mouth and throat.

(c) Extend the neck to straighten air passages.

(d) Cover the patient's mouth with clean gauze and blow directly and slowly into it (10 – 12 times per minute).

Cardiopulmonary Resuscitation (CPR)

28. **CPR is for people experiencing a cardiac arrest**. A heart attack occurs when blood flow to the heart is blocked, but the person is still conscious and not yet in cardiac arrest. **While both heart attack and cardiac arrest are serious heart-related emergencies, they are distinct conditions. A heart attack is a circulation problem where blood flow to the heart is blocked, while cardiac arrest is an electrical problem where the heart stops beating or beats too erratically.** CPR is an emergency procedure aimed to keep blood and oxygen flowing through the body when a person's heart and breathing have stopped.

29. CPR includes performing chest compressions and providing rescue breaths. Providing CPR immediately following a cardiac arrest can significantly increase the likelihood of survival.

30. The old model was ABC (Airway, Breathing, Compressions). **This has been replaced by CAB (Compressions, Airway, Breathing).**The new guidelines say that a person near someone who may be experiencing cardiac arrest should initiate CPR immediately without waiting. **This is because, in the first few minutes of cardiac arrest there's still oxygen in the person's lungs and bloodstream. Starting chest**

compressions first on someone who's unresponsive or not breathing normally can help send this critical oxygen to the brain and heart without delay.

31. **CPR for Adults**. The general procedure is as under:-

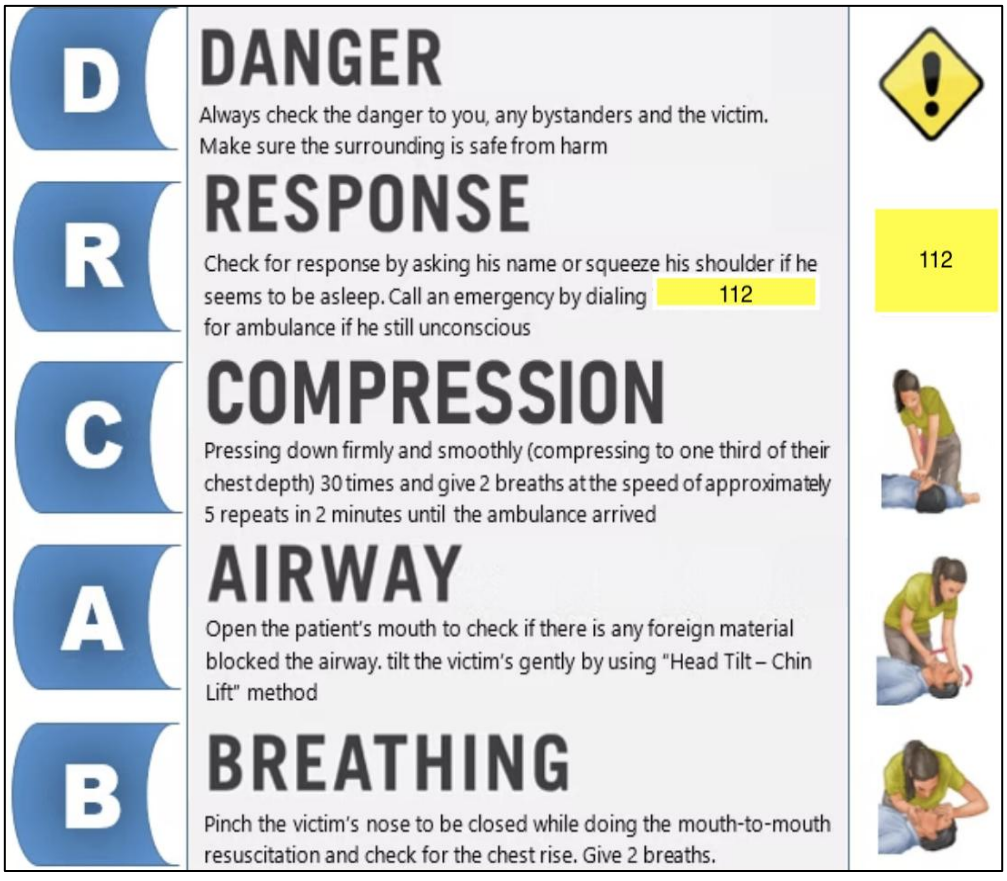
(a) **Call your Local Emergency Number**. If alone, call first before starting CPR. If with someone else, one calls while the other starts CPR.

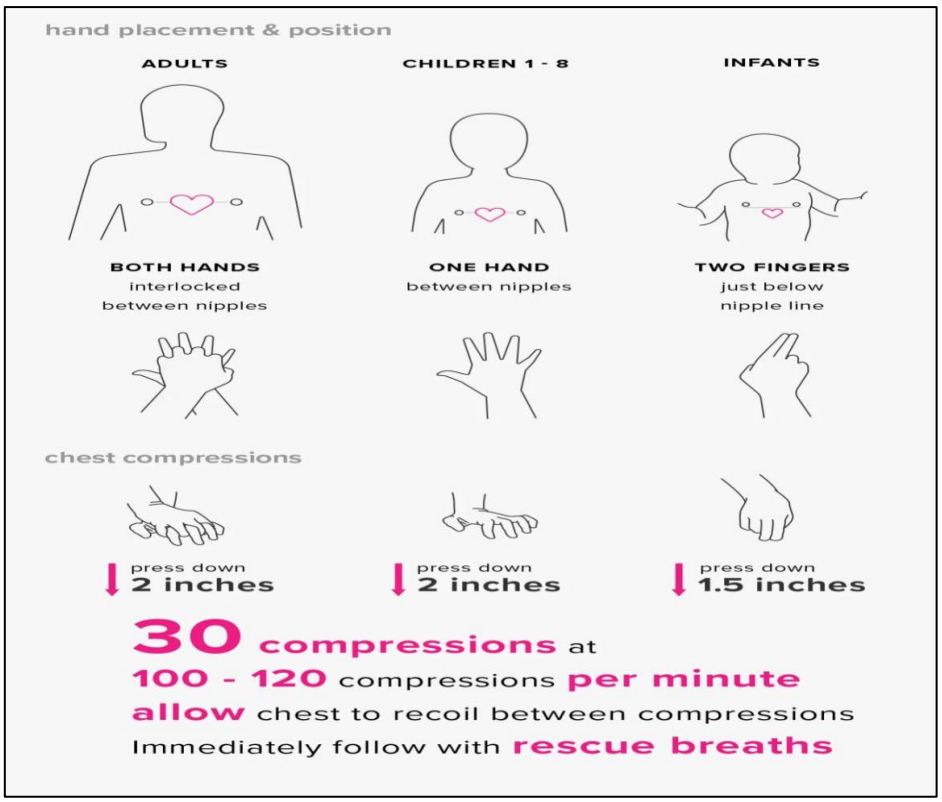
(b) **Start Compressions**. If the person is unresponsive or not breathing normally, begin chest compressions. Place the heel of one hand at the centre of their chest above the nipple line. Place your other hand on top and interlace your fingers. Push down 2-3 inches and release. Repeat.

(c) **Check the Person's Circulation, Breathing, and Airway(CABs)**. Position the person on their back on a firm, flat surface. Tilt their head back and lift their chin to open their airway. Listen for breath sounds and feel for air on the cheek. Check for a pulse in the neck.

(d) **Give Rescue Breaths**. After 30 compressions, give two breaths by MMR. Pinch the nose shut, breathe into their mouth for one second to make the chest rise, then give a second breath.

(e) **Continue CPR**. Keep performing 30 compressions and 2 breaths until medical help arrives, the person shows consciousness, or you can't continue.





32. **CPR for Child(1-8 years) and Infants(0-12) Months**. Mouth-to-mouth resuscitation, or CPR with breaths, is appropriate to be given to an adult, teen, child, or infant. The chest compression technique though is different for each age group:-

(a) **Compressions for a Child/Infant.** Start compressions if the child is unresponsive or not breathing normally. Place one hand's heel at the centre of the chest above the nipple line, interlace fingers with the other hand on top. Push down to about 1/3 of the chest depth, release, and repeat. In case of infants, place your thumbs on the centre of the chest above the nipple line, push down to a depth of 1/4 of the chest cavity, release and repeat.

(b) For children, use **one hand** for compressions; for infants, use **two fingers**.

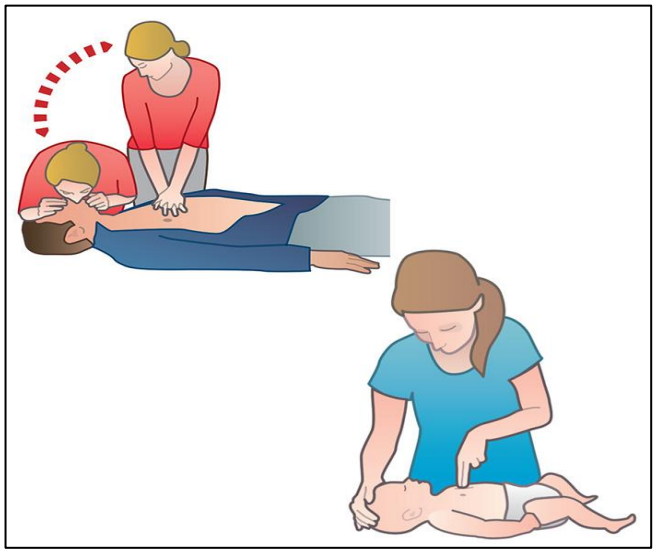
(c) **Check the Child's/Infant's CABs.** Lay the child on their back on a firm surface. Infants need to be placed supine on a firm, flat surface. Tilt their head back, lift their chin to open the airway, and listen for breathing. Feel for air and check the pulse in the neck.

(d) **Give Rescue Breaths.** After 30 compressions, give two breaths. Need to be gentle ones for infants. Pinch the child's/infant's nose and breathe into their mouth for about one second to raise their chest. Repeat for the second breath.

(e) **Continue CPR**. Keep providing compressions and breaths in a 30:2 cycle until emergency medical personnel arrive, the child/infant shows signs of regaining consciousness, or you are physically unable to continue.

Comparison of CPR Technique for Adults, Children and Infants

	Adult	Child	Infant
			
	About age 12 years or older	Between the ages of 1 and 12 years	Younger than 1 year
Hand Position			
	Two hands in center of chest	Two hands in center of chest	Two fingers on center of chest, just below the nipple line
Chest Compressions			
	Compress **at least 2** inches Rate: 100–120 compressions/min	Compress **about 2** inches Rate: 100–120 compressions/min	Compress **about 1½** inches Rate: 100–120 compressions/min
Rescue Breaths			
	Tilt head to **past-neutral position**; pinch nose shut and form seal over mouth 	Tilt head to **slightly past-neutral position**; pinch nose shut and form seal over mouth 	Tilt head to **neutral position**; form seal over mouth and nose 
Sets			
	30 chest compressions and 2 rescue breaths	30 chest compressions and 2 rescue breaths	30 chest compressions and 2 rescue breaths





DID YOU KNOW?

- **The "Golden Hour" is Crucial**. The first hour after an injury or illness is critical, and prompt First Aid can significantly improve outcomes.

PART III : IMPORTANCE OF FIRST AID KNOWLEDGE

33. **Importance of First Aid Knowledge for Cadets**. A few key points to be borne in mind by cadets are as follows:-

(a) **Prompt Treatment**. Knowing first aid techniques enables cadets to provide prompt treatment, reducing the risk of complications and promoting faster recovery.

(b) **Enhanced Safety**. First aid knowledge enhances safety during training, camping, and other activities, reducing the risk of accidents and injuries.

(c) **Leadership Skills**. Cadets who learn first aid and wound care develop leadership skills, as they can take charge in emergency situations.

(d) **Community Service.** Cadets can apply their first aid knowledge to help others in their community, promoting a sense of social responsibility.

(e) **Prevention of Infections**. Proper wound care and first aid can prevent infections, which can be debilitating and even life-threatening.

34. **Spreading Awareness about First Aid**. The awareness about First Aid administration can be spread as follows :-

(a) **Training Sessions**. Organise training sessions to educate cadets and SSCD activities with cadets to educate others about wound care and first aid.

(b) **Social Media Campaigns.** Utilise social media platforms to share information, videos, and infographics about wound care and first aid as part of the SSCD activities.

(c) **Community Events**. Encourage NCC cadets to participate in community events, eg health fairs, to promote awareness about wound care and first aid.

(d) **Collaboration with Medical Professionals**. Collaborate with medical professionals to provide expert advice and guidance on wound care and first aid during SSCD activities.

(e) **Developing Educational Materials.** Develop educational materials, such as brochures, posters, and videos, to educate cadets and others about wound care and first aid.

HIGHER ORDER THINKING SKILLS (HOTS)

- **In a situation with multiple casualties, how would you assess and prioritise administration of First Aid among several patients?**
- **Design a simple first aid kit for a typical outdoor NCC training exercise. What items would you include in the kit?**
- **Imagine a scenario where an unconscious person is lying on the road. What key observations and actions would you take before administering First Aid?**
- **How can First Aid training be integrated into community life beyond NCC camps?**

PART IV: PRACTICE ON CPR

ACTIVITY

- ANO to conduct demonstration of CPR on a mannequin with the help of Institution's Medical Staff.
- ANO to ensure practice by all cadets.

CONCLUSION

35. First Aid is a **lifesaving skill** that all NCC cadets must possess and practice. Whether dealing with a life-threatening emergency or a minor injury, the ability to act promptly and correctly makes a tremendous difference.

36. Mastery of First Aid principles not only ensures **individual safety** but also enhances **community resilience**.

SUMMARY

- First Aid helps preserve life, prevent further harm, and promote recovery.
- The Golden Hour is critical in emergencies—prompt action can save lives.
- CPR is essential for treating unconscious patients and those without a pulse.
- NCC cadets must stay calm, assess danger, and follow structured steps (CAB: Circulation, Breathing and Airway).
- First Aid knowledge builds leadership and community service skills in cadets.

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ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What does CAB stand for in First Aid?

(a) Control, Alert, Breathe (b) Circulation, Airway, Breathing

(c) Check, Apply, Bandage (d) CPR, Assessment, Balance

Q2. What is the correct compression-to-breath ratio for adult CPR?

(a) 15:1 (b) 20:2

(c) 30:2 (d) 10:1

Q3. What is the primary objective of First Aid?

a) To cure the illness or injury

b) To preserve life, prevent further harm, and promote recovery

c) To provide a complete diagnosis of the injury

d) To replace medical professionals

Q4. Which of the following is NOT a sign of shock?

(a) Cold, clammy skin (b) Rapid pulse

(c) Heavy sweating (d) Slow, strong pulse

Q5. What is the first action to take for a choking adult who cannot cough or speak?

(a) Give water (b) Back blows and abdominal thrusts

(c) Check their pulse (d) Have them lie down

Q6. In case of an electric shock, what should you do first?

(a) Pour water on the person (b) Pull them away immediately

(c) Turn off the power supply (d) Cover them with a blanket

Q7. Which of the following is a cause of asphyxia?

a) Heart failure b) Inhalation of poisonous gases

c) Skin burns d) Dizziness

Q8. What is the best position for a person who has fainted?

(a) Sitting with arms raised (b) Standing and moving

(c) Lying down with legs elevated (d) Seated upright against a wall

Q9. Which of the following is a life-threatening emergency?

(a) Small cut on the finger (b) Nosebleed

(c) Cardiac arrest (d) Minor burn

Q10. How many chest compressions should be given per cycle during adult CPR?

(a) 15 (b) 20

(c) 25 (d) 30

Q11. Which method of artificial respiration is commonly used in First Aid?

a) Mouth-to-mouth respiration b) Chest compressions

c) Inhalation of oxygen d) Abdominal thrusts

Q12. Which of the following is a common symptom of shock?

(a) Increased alertness (b) Rapid, weak pulse

(c) High blood pressure (d) Excessive thirst

Q13. A victim with bluish lips and nails may be suffering from:-

(a) Dehydration (b) Fracture

(c) Asphyxia (d) Minor bleeding

Q14. Which of the following best describes a symptom of choking?

(a) Swollen limbs (b) Difficulty speaking or breathing

(c) Numbness in the legs (d) Sharp back pain

Q15. When performing CPR on an adult, how deep should the chest compressions be?

(a) 1-2 inches (b) 2-3 inches

(c) 4-5 inches (d) 6-7 inches

Short Answer Questions

Q1. Define First Aid and list its three main objectives.

Q2. Describe the steps to take when encountering an unconscious person.

Q3. What precautions should be taken before administering First Aid?

Q4. Explain the significance of the “Golden Hour” in emergencies.

Q5. List three symptoms of asphyxia and how you would treat the condition.

Long Answer Questions

Q1. Describe the full CPR procedure for adults, children, and infants.

Q2. Discuss the role of NCC cadets in emergency First Aid and community service.

Q3. Explain the treatment steps for choking, electric shock, and drowning.

Q4. Why is it important to assess danger before administering First Aid? Give examples.

Q5. Elaborate on the causes, symptoms, and First Aid for unconsciousness.

PHYSICAL AND MENTAL WELLNESS (JD/JW)

CHAPTER PMW V : TYPES AND TREATMENTS OF WOUNDS

"Healing is not just the mending of a wound, but the restoration of strength, patience, and resilience."



TEACHING INSTRUCTIONS

Period : **One (01)**

Type : **Practice**

Year : **2nd Year SD/SW**

Conducting Officer : **Associate NCC Officer**

Training Aids : **Script/Lesson Plan: OHP, Screen, Presentation**

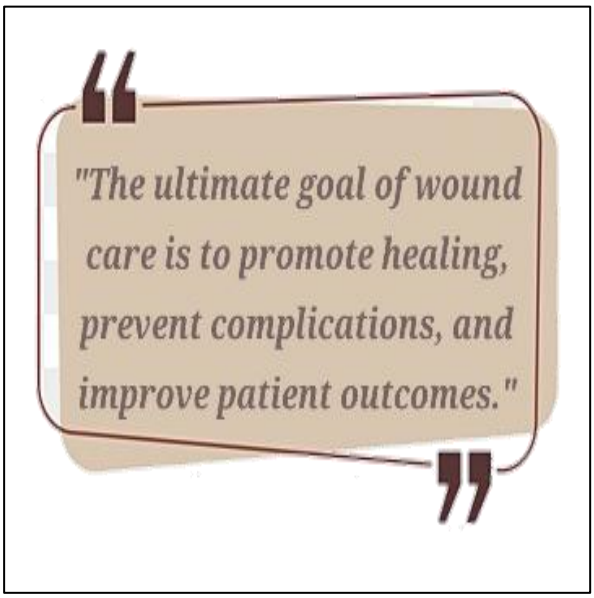
Time Plan

- Introduction : 02 Mins
- Part I : 04 Mins
- Part ll : 10 Mins
- Part III : 09 Mins
- Part IV : 12 Mins
- Conclusion : 03 Mins



INTRODUCTION

1. Wounds, fractures, and minor injuries are common in everyday life, especially during outdoor activities and training. Knowing how to recognise different types of wounds and provide appropriate First Aid helps prevent complications, infections, and promotes faster recovery.



LEARNING OBJECTIVES

- Understand the types and classification of wounds.
- Learn First Aid techniques for cuts, bruises, burns, and fractures.
- Apply basic bandaging and dressing methods.
- Know how to handle minor animal and insect bites.
- Recognise items in a First Aid kit and their uses

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I : Types of Wounds

(b) Part II : Bleeding & Wound Care

(c) Part IV : Bandaging Techniques

(d) Part V : First Aid Kit Essentials

INTERESTING FACTS

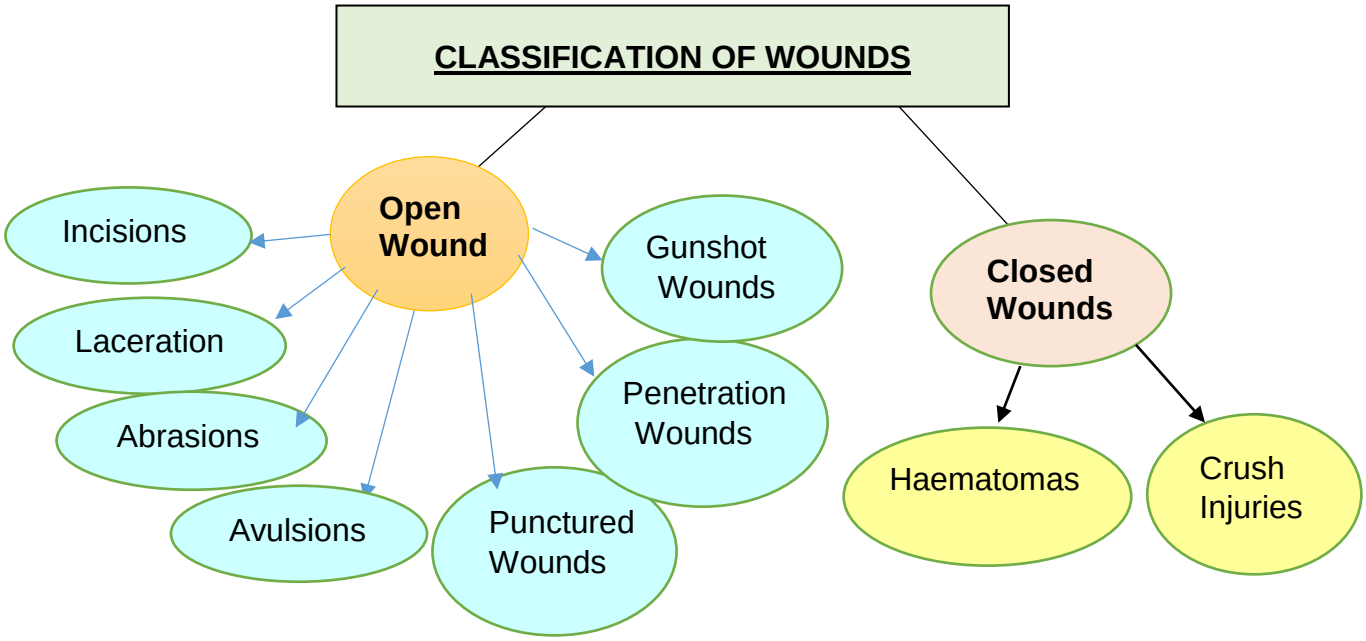
- Plasters were made from a number of materials, including clay or mud, herbs, and plants.
- Oil was another common ingredient used in plasters and provided some protection against infection, which also prevented the bandage from sticking to the wound.

PART I : INFECTIOUS DISEASES

2. **Wounds.** Wounds are breaks in the skin or muscle membrane caused by violence. Wounds can be classified in several ways based on their cause, depth, severity, and degree of contamination.

Primary Classification of Wounds

3. Primary classification of wounds, as open and closed wounds, is **based on their cause(and the resulting injury)** :-



4. **Open Wounds.** Open wounds can be categorised based on the object that caused the wound:-

(a) **Incisions.** Caused by a clean, sharp-edged object such as a knife, razor, or glass splinters.

(b) **Lacerations**. A wound characterised by torn or uneven edges, typically resulting from the impact of a blunt instrument. Lacerations and incisions can be linear (regular) or stellate (irregular). The term laceration is often incorrectly used to refer to incisions.

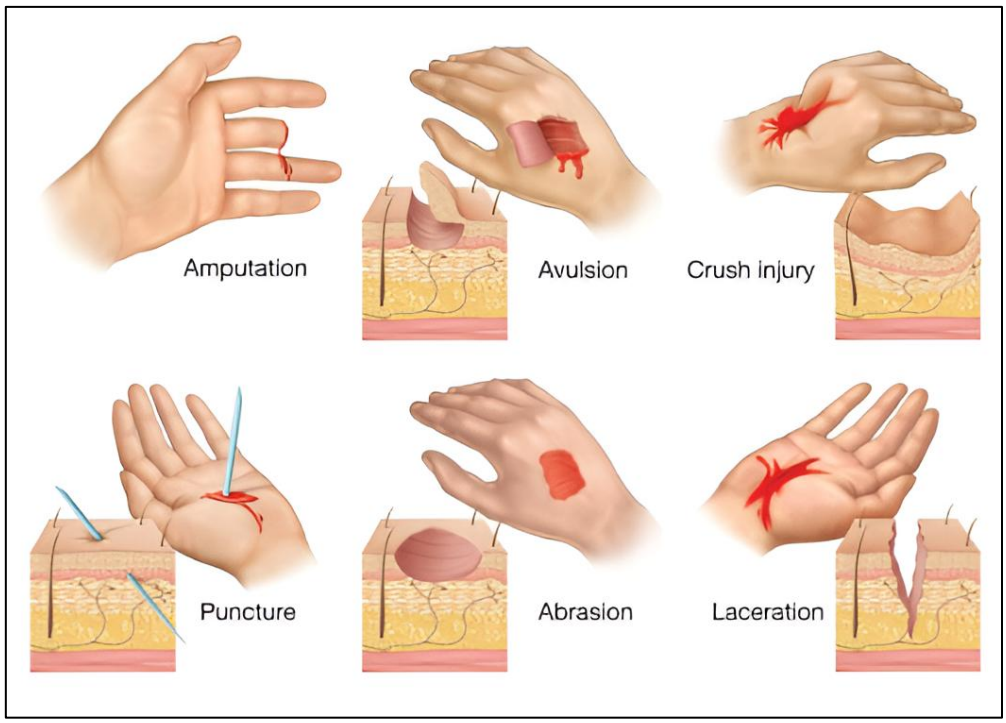
(c) **Abrasions (Grazes).** These are superficial injuries where the outermost layer of the skin (the epidermis) is removed. Abrasions typically occur due to sliding falls onto rough surfaces.

(d) **Avulsions.** Injuries where a body part is forcibly removed from its usual attachment, where an extremity is pulled off rather than cut off.

(e) **Puncture Wounds.** Caused by an object puncturing the skin, such as a splinter, nail, or needle.

(f) **Penetration Wounds.** Caused by an object, such as a knife piercing through the skin.

(g) **Gunshot Wounds.** These are caused by a bullet or similar projectile penetrating or passing through the body. The sites of entry and exit may result in two wounds, known as a "through-and-through."



5. **<u>Closed Wounds</u>.** A Closed Wound or Contusion is a bruise where tissues are torn or ruptured without breaking the skin, typically caused by a blow or by a blunt instrument. Closed wounds, though fewer in categories than open wounds, pose significant risks. The basic types of close wounds include:-

(a) Haematomas (also known as blood tumours).

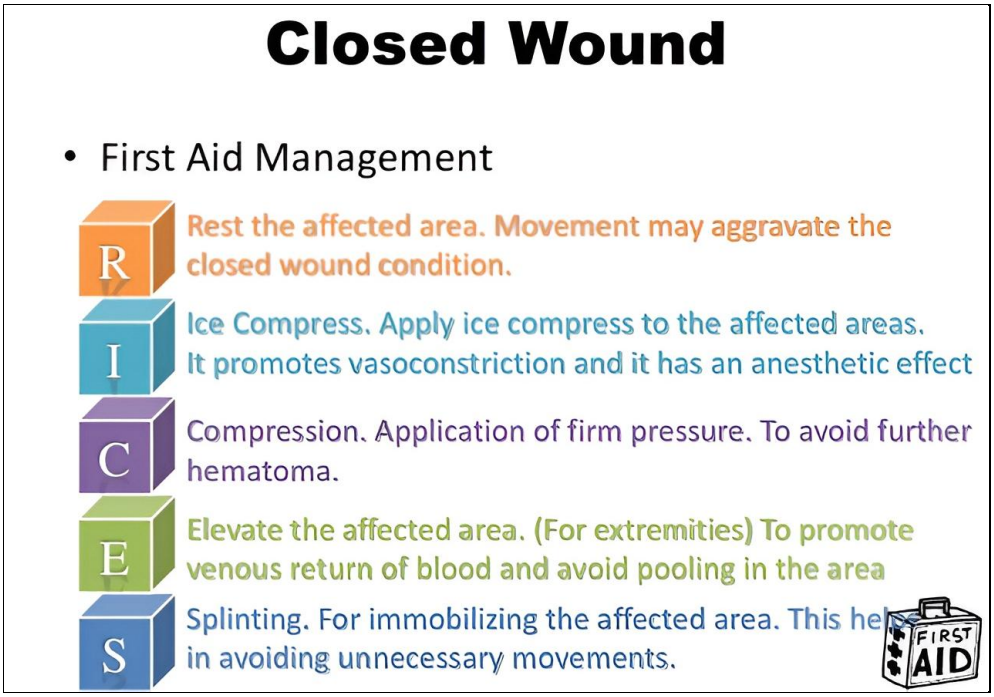
(b) Crush Injuries.

Haematomas (or Blood Tumours)

- **Caused by damage to a blood vessel, which in turn causes blood to collect under the skin.**
  - **Haematomas that originate from internal blood vessel pathology are petechiae, purpura, and ecchymosis. The different classifications are based on size.**
  - **Haematomas that originate from an external source of trauma are contusions, also commonly called bruises.**

Crush Injuries

- **Caused by a great or extreme amount of force applied over a long period of time.**



Secondary Classification of Wounds

6. Open and Closed Wounds may further be classified by degree of depth, severity, contamination, acuteness and time taken for healings.

7. **Classification by Depth.**

(a) **Superficial Wound (First-degree).** Only affects the outer layer of the skin (epidermis), such as in minor abrasions or sunburns.

(b) **Partial-thickness Wound (Second-degree).** Affects both the epidermis and part of the dermis, seen in burns or blisters.

(c) **Full-thickness Wound (Third-degree).** Extends through the dermis and affects deeper tissues, possibly reaching muscles or bones, such as deep lacerations or severe burns.

8. **Classification by Severity.**

(a) **Minor Wound.** Small and superficial wounds that usually heal on their own without major complications.

(b) **Moderate Wound.** Deeper or more extensive damage that may require medical attention but doesn't pose an immediate threat to life or major function.

(c) **Severe Wound.** Large, deep, or life-threatening injuries that may require emergency medical intervention, surgery, or other intensive care.

9. **Classification by Contamination.**

(a) **Clean Wound.** A wound with minimal contamination, often caused by surgical incisions or clean-cut injuries.

(b) **Clean-Contaminated Wound.** Wounds that occur in areas where bacteria are present (e.g., gastrointestinal tract, respiratory tract) but have been treated to minimize infection.

(c) **Contaminated Wound.** Wounds with gross contamination or exposure to bacteria, such as from dirt, debris, or animal bites.

(d) **Infected Wound.** A wound that has already been contaminated and shows signs of infection, like redness, warmth, swelling, and pus.

10. **Other Classifications.**

(a) **Chronic Wounds.** Wounds that do not heal properly or take a long time to heal, often due to underlying conditions like diabetes or poor circulation.

(b) **Acute Wounds.** Wounds that heal in a predictable manner over time without complications, typically from trauma or surgery.

PART II: BLEEDING AND WOUND CARE

11. **Treatment of Wounds.** The treatment of wounds involves medical and nursing practices to promote healing, prevent infection, and restore tissue function. It includes evidence-based procedures to manage wounds, reduce complications, and improve patient outcomes.

12. **Key Components of Treatment of Wounds/Injuries.**

(a) **Assessment.** Evaluate the wound's size, depth, location, and severity.

(b) **Cleaning and Debridement**. Removing debris, bacteria, and dead tissue to promote healing.

(c) **Dressing and Bandaging**. Protecting wounds, promoting healing, and preventing infection.

(d) **Wound Care.** Providing ongoing care includes monitoring for signs of infection, managing pain, and adjusting treatment plans as necessary.

(e) **Rehabilitation**. Restoring function and mobility to the affected areas.

13. **Management of Wounds/Injuries.** The following procedures should be followed to provide effective first aid for managing wounds and injuries:-

(a) Position the patient comfortably.

(b) Control any bleeding.

(c) Remove visible foreign bodies if they can be easily extracted.

(d) Apply a sterile dressing to prevent germs. For minor cuts and bruises, clean with antiseptic lotion and bandage.

(e) Rest the part by using a sling.

(f) Immobilise the affected area if the wound is extensive or complicated by a fracture.

POINTS TO REMEMBER

- Severe bleeding must be stopped immediately.
- Broken bones must be stabilised if the casualty is to be moved. Do not move a casualty with a suspected spine injury unless urgently required.
- Reassure the casualty and treat any other injuries as required.

Injuries to Internal Organs

14. **Meaning and Symptoms**. Internal injuries are not visible and must be suspected when bleeding occurs within the chest or abdomen, potentially rupturing vital organs like the heart, lungs, liver, or spleen. Symptoms of internal injuries include:-

(a) Cold, clammy skin.

(b) Weak or rapid pulse.

(c) Shallow breathing.

(d) Pale, pinched face.

(e) Sunken eyes with dark rings.

(f) Restlessness, anxiety, possible loss of consciousness.

15. **Recognition**.

(a) Trouble breathing

(b) Shallow breathing

(c) Tenderness at injury site

(d) Chest deformity and bruising

(e) Discomfort during movement, deep breathing, or coughing

(f) Lips or nail beds appearing dusky or blue

(g) May cough blood

(h) Crackling sensation on the victim’s skin

16. **Treatment**.

(a) Call an ambulance

(b) Help the victim get comfortable

(c) Conduct a secondary survey

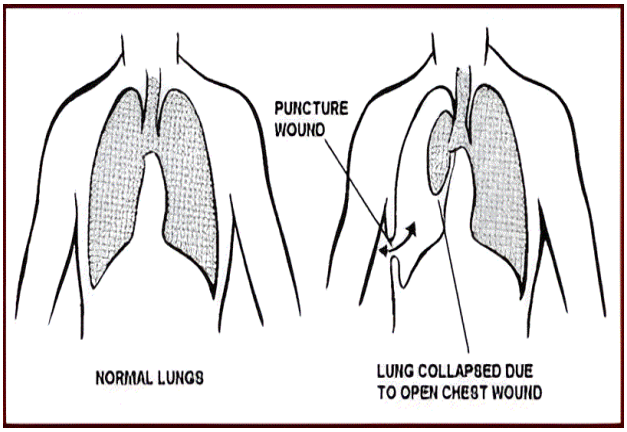
(d) Monitor the patient's condition

(e) Stay vigilant and alert for any changes.

(f) If a flail segment is suspected, secure a bulky dressing tightly to stabilise the injury.

Open Chest Wounds

17. **Meaning and Symptoms**. An open pneumothorax or sucking chest wound occurs when the chest wall is penetrated by an object such as a knife, bullet, or sharp item.



18. **Recognition**.

(a) An open chest wound – air escaping.

(b) Entrance and potential exit wound (exit wounds are typically more severe).

(c) Difficulty breathing.

(d) A sound is produced as air moves through an opening in the chest wall.

(e) Blood or blood-stained bubbles may be released with each exhalation.

(f) Coughing up blood.

19. **Treatment**.

(a) Assess CABs and intervene if needed(As explained in previous chapter).

(b) Do not remove embedded objects.

(c) Contact emergency medical services immediately.

(d) Position the victim laterally with the injured side facing downward.

(e) Treat for shock.

(f) Conduct a secondary survey.

(g) Monitor vitals.

Burns and Scalds

20. **Symptoms**. Burns can be caused by dry heat (fire or flame), wet heat (hot water, fluids, steam), or chemicals (strong acids or alkalis). Scalding refers to burns from moist heat.

21. **First Aid Treatment for Burns**.

(a) For chemical burns, ensure that water from a pipe or hose flows continuously over the affected area to thoroughly wash away the chemical.

(b) Cover the burn with a clean towel or cloth.

(c) Ensure the patient gets plenty of rest and fluids.

(d) Provide reassurance to the patient and arrange for the patient's transportation to the nearest hospital or medical dispensary.

(e) Do not apply ice, butter, or toothpaste.



Fractures

22. **Fractures**. Fractures or broken bones, are common injuries that can occur due to various reasons, such as falls, accidents, or direct blows to the body. They can range from simple breaks that cause minimal damage to the surrounding tissues, to more severe types that involve multiple fragments or pierce through the skin. Understanding the different types of fractures and their appropriate first aid measures is crucial in providing effective care and preventing further complications.

23. **Types of Fractures.** To effectively address fractures, one must first understand the various types that can occur. There are four types of fractures:

(a) **Simple (Closed) Fracture**. The bone is broken but the skin is intact.

(b) **Compound (Open) Fracture**. The broken bone pierces through the skin.

(c) **Greenstick Fracture**. Partial fracture, common in children.

(d) **Comminuted Fracture**. A comminuted fracture is a broken bone that's broken into more than two pieces. It's also known as a multi-fragmentary fracture.

24. **First Aid for Different Type of Fractures.**

(a) **Do not move** the injured area unnecessarily.

(b) **Immobilise** using splints (e.g., stick, plywood, cardboard) or any firm support.

(c) **For open fractures**, cover the wound with a sterile dressing.

(d) **Control bleeding** with gentle pressure if necessary.

(e) Use padding to prevent further injury during transport.

(f) Keep the patient calm and seek medical attention immediately.

PART III : ANIMAL AND INSECT BITES

Snake Bite

25. **Snake Bite**. Snakes inject poison through hollow, grooved fangs, often biting exposed limbs like hands, feet, and lower legs. **Scorpion Bites** are also Treated like snake bites.

26. **First Aid Treatment**. Treat all snake bites as if they are poisonous. First aid should include the following steps:-

(a) Keep the victim still

(b) Clean the bite area gently with water.

(c) Offer reassurance, don't allow panic to set in.

(d) Monitor vital signs for shock, difficulty breathing, dizziness, nausea, or rapid pulse

(e) Remove tight clothing or jewellery near bite site allowing for swelling and prevent constriction

(f) If breathing stops, begin artificial respiration.

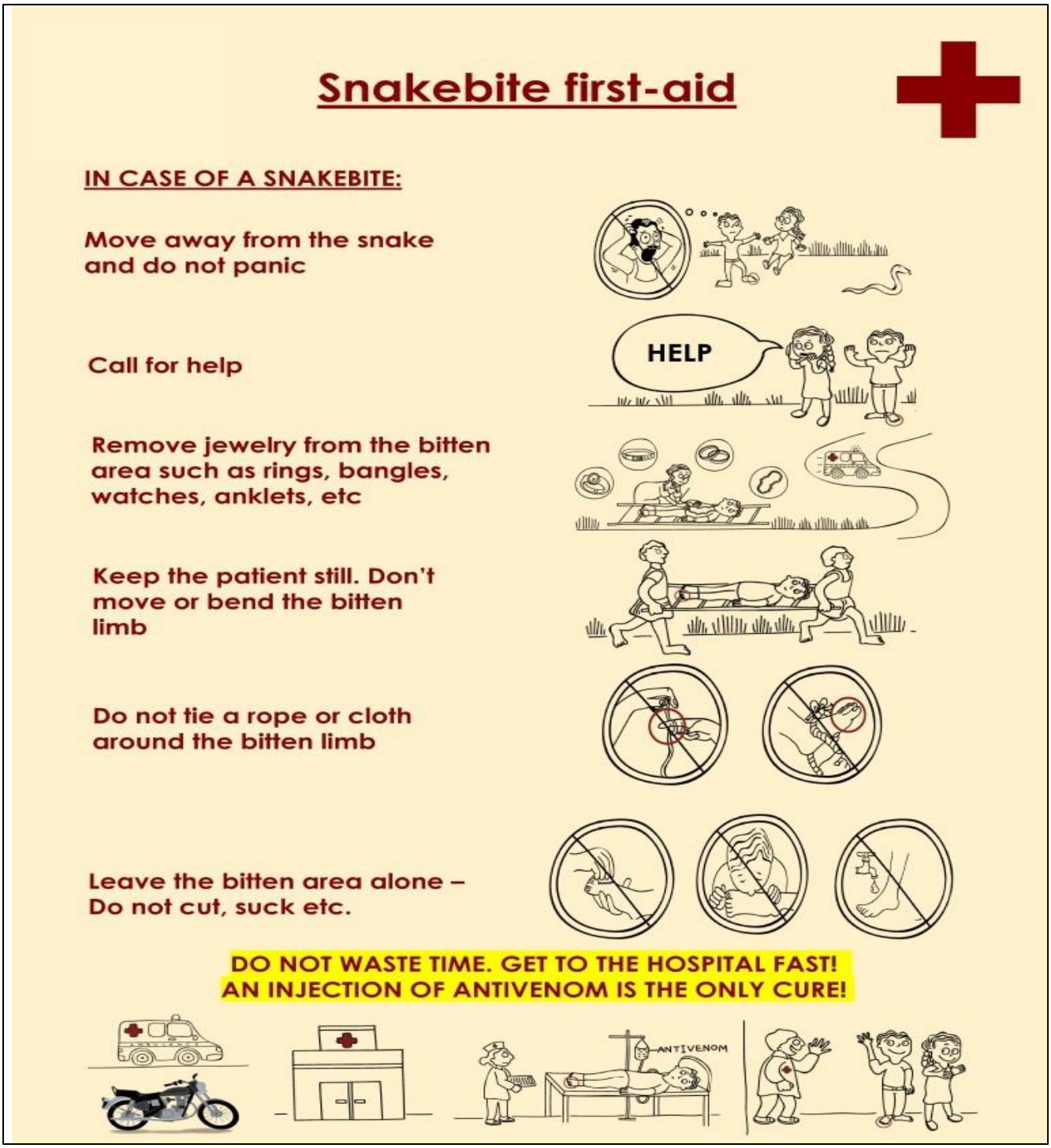
(g) Quickly transport the patient to the nearest hospital or dispensary.

(h) **Applying a tourniquet is no longer recommended in the case of a snake bite. This approach has been shown to cause more harm than good because of venom accumulation, increased tissue damage and limited effectiveness.**

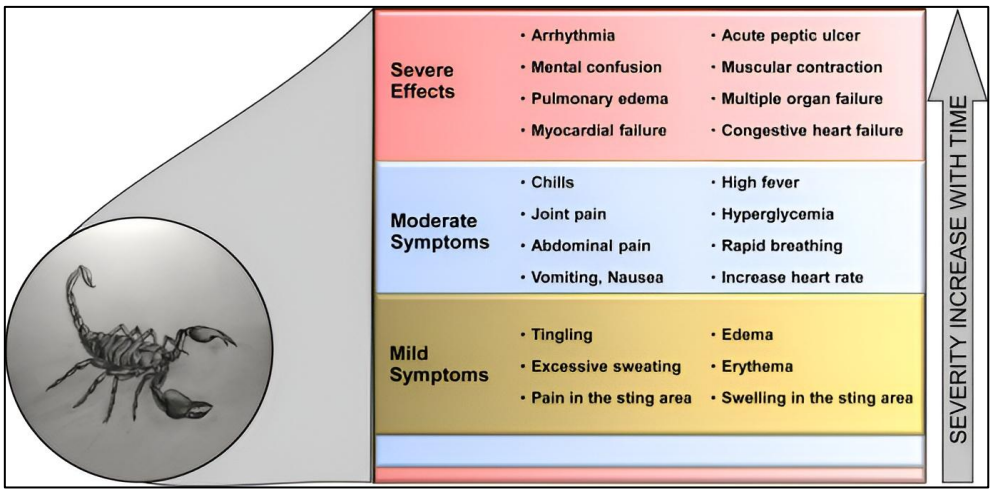
27. **Snake Bites: Dos and Don'ts**. Are as given:-

(a) **Dos**. Keep the victim calm and still. Provide strong reassurance to alleviate fear of death. Seek **emergency medical help** immediately. Call an ambulance. Remove any items that are constricting the area, such as a ring or watch. Apply a **loose bandage** above the bite to allow slow venom spread. Raise the affected area above heart level. If breathing stops, initiate artificial respiration.

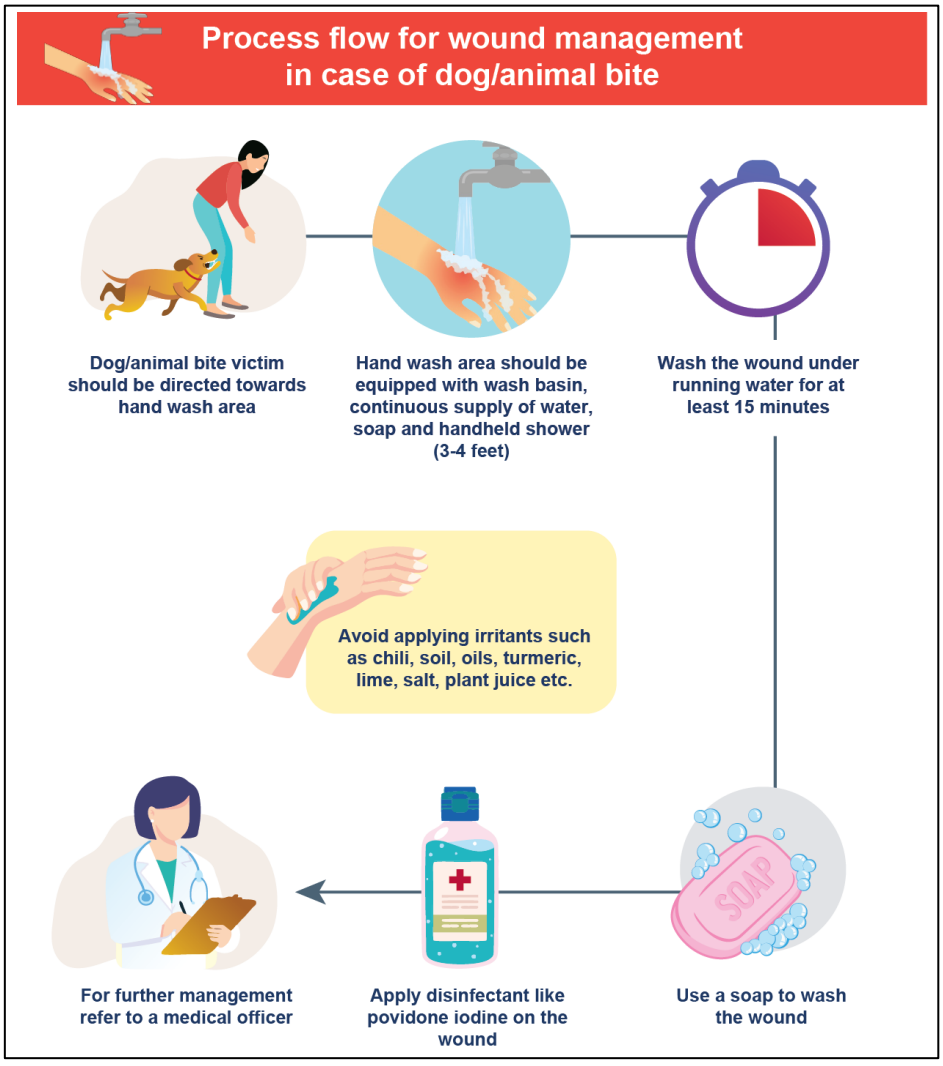
(b) **Don'ts**. Do not apply a tourniquet. **Do not cut or suck** the wound. Do not apply ice, as it may damage the tissue. Refrain from using a commercially available extraction device. Avoid using electrical therapy. Avoid using any lotion or ointments.



28. **First-Aid.** Immobilise the patient until medical help arrives In case of bleeding, clean the wound thoroughly with water antiseptic lotion. Do not apply pressure on the area of the sting or cut the wound open.



29. **Rabid Dog Bite**. Rabies, transmitted by rabid dogs, is preventable with an anti-rabies vaccine.



30. **First Aid Treatment**.

(a) Thoroughly cleanse the bite area with a substantial amount of water and soap immediately.

(b) Clean the wound using an available antiseptic.

(c) The patient is recommended to receive an anti-rabies injection course at the hospital.

(d) Observe both the dog and the patient for at least 10 days.

PART IV: BANDAGING TECHNIQUES

Bandaging

31. **General Principles and Rules.** Bandages made from flannel, calico, elastic net, or special paper, can also be improvised from stockings or ties. They serve various purposes:-

(a) Applying direct pressure on a dressing can help control bleeding.

(b) Keep dressing and sling in place.

(c) To minimise swelling.

(d) Offer support for a limb or joint.

(e) Restrict the movement.

32. **Application of Bandages**. Bandages should be applied with sufficient firmness to secure the dressing and splints in place, while ensuring they are not so tight as to cause harm or obstruct blood circulation. Indicators that bandages may be excessively tight include a bluish tinge in fingers or nails, and loss of sensation.

33. **Types of Bandages**. According to shape, there are two types of bandages:-

(a) Triangular Bandages.

(b) Roller Bandages.

34. **General Rules of Bandaging**.

(a) Position yourself in front of the patient, except when bandaging the head or eyes.

(b) Support the part being bandaged.

(c) Use the correct width of bandage.

(d) Place the outer side of the bandage on the area to be bandaged.

(e) Start with an oblique turn, then secure the bandage with the next turn.

(f) Start inside and work outwards and upwards.

(g) Unroll a few inches of bandage at a time.

(h) Cover two-thirds of previous turns evenly.

(j) Finish the bandage using a reef knot, safety pin, or special fastening.

(k) Avoid placing the knot on an injury or a bone.

35. **Common Mistakes While Using a Bandage**. Every effort should be made to avoid the following mistakes while applying a bandage:-

(a) A wet bandage will shrink when it dries.

(b) Uneven bandage tension can cause issues. A tight bandage restricts blood flow, causing tissue death. A loose bandage may shift out of place.

(c) Avoid reversing turns over a prominence or wound, as this can cause discomfort and pain for the patient.

(d) Excessive bandage use causes discomfort and increases cost.

(e) Improperly securing bandage ends can cause harm.

36. **Types of Turns of Bandages**.

(a) **Circular**. A circular turn involves wrapping the bandage around the body part in a circular motion, typically in one place or over a small area. It is often used to cover small injuries or to anchor the beginning and end of the bandage. Its purpose is to Secure the bandage in place and prevent it from unravelling.

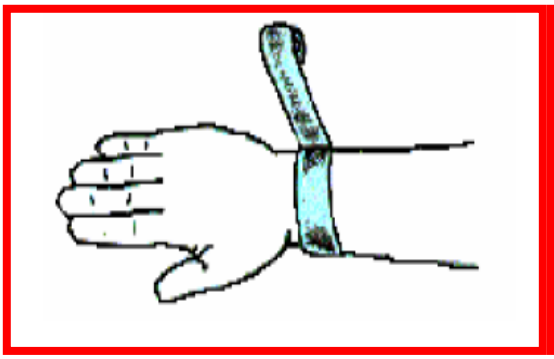
(b) **Spiral**. The bandage follows a helical or spiral path around the limb and is typically used on cylindrical body parts like the arms, legs, or fingers.

(c) **Reverse Spiral**. The reverse spiral is similar to the spiral turn but direction of the bandage is alternated at each pass, creating a zig-zag pattern. It is typically used on conical-shaped body parts, such as the lower leg or forearm. Its purpose is to help provide more secure coverage and a more fitted bandage around the limb's tapering shape.

(d) **Figure-of-Eight.** This type of bandaging involves wrapping the bandage around the body part in a figure-eight pattern, crossing over the joint or the area that needs to be secured. Its commonly used for joints like the knee, elbow, or ankle. The figure-of-eight turn provides better immobilization for joints.

(e) **Recurrent.** A recurrent turn is a type of bandage turn used to cover a stump or a rounded area, such as the head, residual limb, or a circular body part like the wrist or ankle. It involves wrapping the bandage back and forth over the area in a series of overlapping turns, creating a pattern that repeats on itself. It is used for covering areas that do not have a clear cylindrical or conical shape, like the ends of limbs (stumps) or the head.

Circular



Spiral and Reverse-Spiral

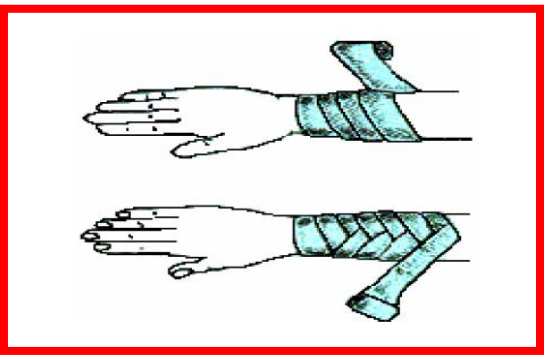
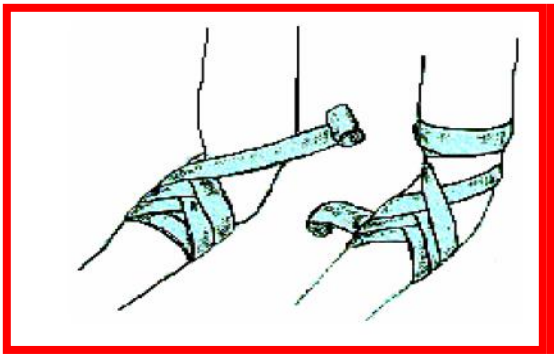
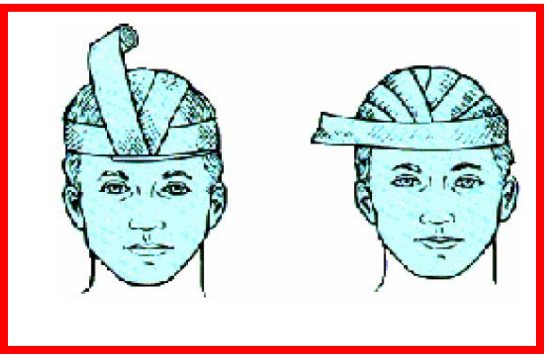


Figure-of-Eight



Recurrent



PART V : FIRST AID KIT ESSENTIALS

37. **Practice and Raise Awareness**. Cadets should practice First Aid techniques for various wounds/injuries daily and promote health awareness in society. The following paragraphs summarise the basic injuries/wounds and their First Aid techniques.

38. **Essential First Aid Techniques**.

S. No.	Wound/Injury	First Aid technique
(a)	**Cuts and Lacerations.** Injuries caused by sharp objects, such as knives, glass, or metal.	(i) **Bleeding Control.** • **Direct Pressure**. Apply direct pressure to the wound using a clean cloth or gauze. • **Elevation**. Raise the injured limb above heart level to decrease blood flow. • **Tourniquet**. Use a tourniquet if direct pressure and elevation don't control bleeding. (ii) **Wound Care.** • **Clean the Wound.** Rinse with clean water and mild soap. • **Apply Antibiotic Ointment.** Use a thin

		layer to prevent infection. • **Cover the Wound**. Apply a sterile bandage or dressing.
(b)	**Bruises and Contusions**. Injuries from falls or blows.	(i) Keep the person still. Have them lie down with their head and shoulders slightly elevated. Only move if absolutely necessary. (ii) Control bleeding by applying firm pressure to the wound using sterile gauze or a clean cloth. (iii) Monitor changes in breathing and alertness. If the individual shows no signs of circulation, such as no breathing, coughing, or movement, begin CPR.
(c)	**Sprains and Strains**. The overstretching or tearing of muscles and ligaments.	The primary treatment for sprains and strains is **RICE**: **Rest, Ice, Compression** and **Elevation.**
(d)	**Fractures and Dislocations** • **Fractures**. Bone breaks in arms, legs, ribs, etc. • **Dislocations**. Misaligned joints like shoulders, elbows, etc.	(i) **Immobilise the Injured Area**. Use a splint or sling to stabilise. (ii) **Apply Cold Packs**. Use cold packs to reduce pain and swelling.
(e)	**Head Injuries**. Cuts, bruises, or fractures to the head.	(i) Apply firm pressure to the wound with sterile gauze or a clean cloth. Don't apply direct pressure to the wound if you suspect a skull fracture. (ii) Use sterile gauze or a clean cloth to apply firm pressure to the wound. Avoid applying direct pressure if a skull fracture is suspected.
(f)	**Burns and Scalds** (i) **Thermal Burns**. Burns from heat, flames, or hot surfaces. (ii) **Scalds**. Burns from hot liquids or steam.	(i) **Cool the Burn**. Run cool water over the burn for 10-15 minutes. (ii) **Apply Topical Ointment**. Use antibiotic ointment to prevent infection. (iii) **Cover the Burn**. Protect with a non-stick, sterile dressing.
(g)	**Ankle and Knee Injuries**. Twists, sprains, or strains.	(i) Apply an ice pack (or frozen vegetables in a tea towel) to the injury for 20 minutes every 2-3 hours. (ii) Wrap a bandage around the injury for

		support and keep it elevated on a pillow.
(h)	**Back Injuries.** Strains, sprains, or fractures.	(i) If unsure, treat any injury as a fracture. (ii) Elevate the injured area. (iii) Get medical help.
(j)	**Heat-Related Illnesses.** Heat exhaustion or heat stroke.	Relocate the person to a cool, shaded area and give them water.
(k)	**Cardiac Emergencies.**	**(i) CPR (Cardiopulmonary Resuscitation):** Perform if unresponsive and not breathing. **(ii) Use an AED (Automated External Defibrillator):** Use if available.
(l)	**Seizures.**	Ensure the person's safety and comfort during the seizure.
(m)	**Allergic Reactions.**	Use an 'EpiPen' or auto-injector to administer measured dose of epinephrine.
(n)	**Eye Injuries.** Cuts, scratches, or foreign objects in the eyes.	(i) Avoid rubbing your eyes. Lift the upper eyelid over the lower and roll your eyes. (ii) Rinse your eyes with water, keeping them open. (iii) Repeat until the object is removed. (iv) See a doctor to ensure no debris remains and eyes are undamaged. (v) If an object is embedded, do not remove it. Cover the eye and get medical help.
(o)	**Dental Injuries.** Tooth damage or loss.	(i) Rinse mouth with warm water. (ii) Hold the tooth by the crown and wash it in water. (iii) Place the tooth back in the socket using gauze. (iv) If that fails, store the tooth in milk, saliva, or lukewarm salt water. (v) Control bleeding with a sterile cloth. (vi) Seek emergency medical or dental help.

First Aid in Cadets' Day-to-Day Life

39. **During NCC Training and Camps**. The significance of learning First Aid and its applications by cadets during training activities includes:

(a) **Injury Response**. First Aid training equips NCC cadets to handle injuries

like sprains, strains, and cuts during training.

(b) **Emergency Situations**. NCC cadets are trained to manage emergency situations such as cardiac arrests, severe bleeding, and choking.

(c) **Camping and Outdoor Activities**. First Aid training is crucial for NCC cadets engaged in camping and outdoor activities, where medical assistance may be delayed.

40. **In Daily Life**. First Aid techniques benefit in their daily activities:-

(a) **Home and Family**. NCC students are able to apply First Aid skills to assist family members or friends in emergency situations.

(b) **School and Community**. First Aid training equips NCC students with the skills to handle medical emergencies in school or community settings.

(c) **Personal Safety**. NCC students learn basic First Aid for minor cuts and burns.

41. **During Social and Community Service**. Cadets may be required to apply First Aid techniques during various other outdoor activities:-

(a) **Disaster Response**. First Aid training equips NCC students to handle emergencies like natural disasters and accidents.

(b) **Community Service**. NCC students can use First Aid skills in community service projects like blood donation camps or health fairs.

(c) **Social Responsibility**. First Aid training educates NCC cadets on the importance of assisting others in need.



42. **First Aid Kit.** Everyone should have a well–stocked first aid kit handy. The contents of your kit will vary depending upon the number of people it is designed to protect, as well as special circumstances where it will be used.

43. **Essential Contents for a First Aid Kit**.

(a) Adhesive bandages (plasters)

(b) Sterile gauze pads and rolls

(c) Antiseptic solution (e.g., Dettol or Betadine)

(d) Scissors and tweezers

(e) Safety pins and adhesive tape

(f) Gloves, cotton, and a digital thermometer

(g) First Aid manual or instructions

44. **Recommended Additional Contents.**

(a) Hydrogen Peroxide
(b) Injection morphine
(c) Pain medication
(d) Diarrhoea medication (anti-diarrhoea tablets, ORS)
(e) Denatured spirit
(f) Antacid
(g) Antibiotic ointment
(h) Calamine lotion
(j) Adhesive strip bandages of assorted sizes
(k) Adhesive tape
(l) Sterile gauze pads of various sizes
(m) Dressings
(n) Triangular bandages
(o) Cotton balls
(p) Cotton swabs
(q) Disposable gloves
(r) Insect repellent
(s) Matches
(t) Salt (Used against leeches)
(u) Soap
(v) Splint (Cramer wire)
(w) Syringes-disposable

45. **Practice in Cadets' Daily Life and Spread Awareness**. After learning the various types of wounds/injuries and their First Aid techniques, every cadet should practice in their daily life and spread awareness towards achieving better quality of health in the society. The basic types of injuries/wounds and their First Aid techniques are illustrated in the succeeding paragraphs.

HIGHER ORDER THINKING SKILLS (HOTS)

- **Imagine you are in a field training exercise and a cadet suffers both, a laceration and a suspected fracture. Which injury would you address first, and why?**
- **You are a first responder at a scene where a person has both a burn injury and a broken arm. What steps would you take to treat both injuries at the same time?**
- **Imagine a scenario where an unconscious person is lying on the road. What key observations and actions would you take before administering First Aid?**

CONCLUSION

46. Mastering wound care, including accurate identification, proper cleaning, and effective bandaging, is crucial not only for individual recovery but also for overall group safety in NCC training and daily life.

47. This chapter has provided practical guidance on handling various types of wounds, treating burns and fractures, and managing minor injuries efficiently. By understanding and applying these techniques, cadets can significantly reduce complications, prevent infections, and ensure better healing outcomes. This knowledge also reinforces the value of preparedness, leadership, and community service in emergency situations.

SUMMARY

- **Wound Classification.** Distinguishing between open and closed wounds is essential for choosing the correct treatment.
- **Bleeding & Wound Care.** Immediate actions such as applying direct pressure and maintaining cleanliness are critical to prevent infections.
- **Burns & Scalds.** Cooling the wound with water and covering it with a sterile dressing helps mitigate tissue damage.
- **Fractures & Dislocations.** Proper immobilization and stabilization are vital for reducing further injury and ensuring proper healing.
- **Animal & Insect Bites.** Appropriate First Aid for bites involves cleaning, applying cold compresses, and seeking further medical intervention when needed.
- **Bandaging Techniques.** Effective bandaging provides support, controls bleeding, and minimizes swelling, while adherence to best practices prevents additional harm.
- **First Aid Kit Essentials.** A well-stocked kit is fundamental to successfully manage minor injuries in the field.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. Which of the following best describes an abrasion?

(a) A deep puncture wound

(b) A superficial wound caused by scraping

(c) A clean, straight cut

(d) A tissue tear due to blunt trauma

Q2. What is a key characteristic of a laceration?

(a) Clean, straight edges (b) Irregular or jagged edges

(c) A superficial scrape (d) A wound with minimal bleeding

Q3. Which type of fracture is characterised by the bone breaking into several fragments?

(a) Simple fracture (b) Compound fracture

(c) Greenstick fracture (d) Comminuted fracture

Q4. For an open fracture, what is the first aid recommendation?

(a) Immediately realign the bone

(b) Cover the wound with a sterile dressing

(c) Apply direct pressure to the bone

(d) Massage the affected area

Q5. What is the main purpose of applying a splint to a fractured limb?

(a) To reduce pain by massaging the area

(b) To immobilise the fracture and prevent further injury

(c) To allow immediate movement of the injured limb

(d) To cool the affected area

Q6. Which of the following is a proper step in treating burns?

(a) Apply ice directly to the burn

(b) Cool the burn under running water

(c) Cover the burn with butter

(d) Wrap the burn tightly with a bandage

Q7. In the management of animal bites, what is an essential first step?

(a) Immediately apply a tourniquet

(b) Wash the area thoroughly with soap and water

(c) Squeeze the bite to remove venom

(d) Apply a hot compress

Q8. What is the purpose of using a figure-of-eight bandage?

(a) To secure a dressing on a circular wound

(b) To support a joint like the ankle or elbow

(c) To immobilize a fractured finger

(d) To cover a large area uniformly

Q9. Which component is NOT typically found in a First Aid kit for wound care?

(a) Adhesive bandages (b) Sterile gauze pads

(c) Antiseptic solution (d) Cooking oil

Q10. What is a haematoma?

(a) A type of open wound

(b) A collection of blood outside blood vessels

(c) A superficial scrape

(d) An infection caused by bacteria

Q11. How should you control bleeding from a wound?

(a) Apply direct pressure with a clean cloth

(b) Elevate the limb only

(c) Immediately apply a tourniquet in all cases

(d) Use water to rinse the wound first

Q12. What is the primary goal of dressing a wound?

(a) To cover and protect the wound

(b) To stop the pain

(c) To increase blood flow

(d) To remove all bacteria from the wound

Q13. Which statement about abrasions is true?

(a) They always require surgical treatment

(b) They are deep wounds that affect muscles

(c) They usually involve only the epidermis

(d) They require immobilization with a splint

Q14. What does the term “bandaging” refer to?

(a) Cleaning the wound only

(b) The application of a dressing secured by a bandage

(c) Administering pain medication

(d) Performing CPR

Q15. Why is it important to regularly check the circulation beyond a bandaged area?

(a) To ensure the bandage is very tight

(b) To make sure the dressing does not shift

(c) To avoid constriction that can lead to tissue damage

(d) To promote increased blood flow immediately

Short Answer Questions

Q1. Define the difference between open and closed wounds with examples.

Q2. List and explain the steps you would take to manage a superficial abrasion.

Q3. Describe the key differences between a simple fracture and a compound fracture.

Q4. Explain why proper bandaging is crucial in wound management.

Q5. What are the essential items in a First Aid kit for treating minor injuries, and why is each important?

Long Answer Questions

Q1. Explain in detail the classification of wounds (open vs. closed) and discuss the primary first aid treatments for each type, including the importance of preventing infection and promoting healing.

Q2. Describe different bandaging techniques (circular, spiral, figure-of-eight, recurrent) and discuss the rules for proper bandaging. Explain how improper bandaging can affect wound healing.

Q3. Discuss the types of fractures (simple, compound, greenstick, comminuted) and elaborate on the first aid procedures for managing fractures. Include how you would immobilise the injury and manage any associated bleeding.

Q4. Compare and contrast the first aid treatments for snake bites, dog bites, and scorpion/insect bites. Explain the rationale behind each treatment step and the potential complications if not managed properly.

Q5. Evaluate the importance of having a well-stocked First Aid kit. Describe the key items that should be included and justify how each item contributes to effective emergency care.

PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW VI: MENTAL WELLNESS

"Action dispels fear"



TEACHING INSTRUCTIONS

Period	**:**	**Three (03)**
Type	**:**	**Lecture**
Year	**:**	**One in 1st Year and Two in 3rd Year SD/SW**
Conducting Officer	**:**	**Associate NCC Officer**
Training Aids	**:**	**Script/Lesson Plan; OHP, Screen, Pointer Staff, Presentation/Chart.**

Time Plan

- Introduction : 05 Mins (1st Year)
- Part I : 35 Mins
- Part ll : 20 Mins (2nd Year)
- Part III : 20 Mins
- Part IV : 20 Mins (3rd Year)
- Part V : 15 Mins
- Conclusion : 05 Mins



INTRODUCTION

1. The Latin quote, **'mens sana in corpore sano**!'(A healthy mind in a healthy body) emphasizes the interconnectedness of physical and mental well-being, suggesting that a healthy body contributes to a healthy mind and holistic well-being. Mental wellness is the foundational element of overall wellness as it shapes how individuals interpret experiences, manage emotions, and navigate challenges.

2. Friendships are vital to emotional and mental wellness, with real-life connections offering deeper, more meaningful support than virtual ones. Excessive screen time, especially among young people, can disrupt sleep and create a harmful cycle that affects mental health. Taking moments of pause and seeking healing through nature, alongside traditional approaches, can help restore balance in today's fast-paced world.



LEARNING OBJECTIVES

- Understand the role of Emotional Wellness.
- Learn the importance of Real Friends vis-à-vis Virtual Friends.
- Understand the positive effects of balanced screen time.
- Make lifestyle adjustments and time management for better sleep hygiene.
- Connect with nature for a healthy state of mental wellness.

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I : Emotional Wellness

(b) Part II : Real Vs Virtual Friends

(c) Part III : Reducing Screen Time

(d) Part IV : Sleep Hygiene

(e) Part V : Connect with Nature

INTERESTING FACTS

- People with strong social connections live longer, healthier lives.
- Humans can survive longer without food than they can without sleep.
- Just 20 mins in a park lower stress levels and improve one's mood.
- A healthy diet supports mental wellness while a poor diet can exacerbate symptoms like anxiety and depression.

PART I : MENTAL WELLNESS

3. Mental Wellness is a holistic concept made up of interconnected elements like spiritual practices, healthy eating, exercise, safety, substance abuse prevention, environmental care, disease prevention, and stress management, all required for overall wellness. Mental wellness refers to a state of psychological and intellectual balance that enables an individual to think clearly, make informed decisions, and maintain concentration.



4. Wellness falls into seven key components: physical, emotional, mental, social, environmental, occupational, and spiritual. Mental wellness is the foundational element of overall wellness as it shapes how individuals interpret experiences, manage emotions, and navigate challenges.

5. **Factors Affecting Mental Wellness.** Mental wellness is influenced by various factors that shape how we manage stress, build relationships, and cope with life's challenges. Understanding these factors can help us nurture a healthier and more balanced mental state. The factors include:-

(a) **Biological Factors.** Mental wellness can be influenced by genetic predispositions, chemical imbalances in the brain, and chronic or neurological health conditions.

(b) **Emotional Regulation.** The ability to manage stress and emotions, and to bounce back from challenges, plays a key role in maintaining mental health.

(c) **Lifestyle and Habits.** Good mental health is supported by sleep hygiene, proper nutrition, regular physical activity, and avoiding harmful substances like alcohol and drugs.

(d) **Social Environment.** Positive relationships with real friends and a sense of belonging promote wellness, while isolation, discrimination, or toxic interactions can negatively impact mental health.

(e) **Psychological Factors.** Healthy self-esteem, constructive thinking patterns, and the ability to process past trauma contribute to a stronger mental state.

(f) **Environmental Conditions.** Living in a safe, clean, and stable environment with access to natural spaces enhances mental well-being, while constant exposure to stressors can be harmful. So, connecting with nature is very important.

(g) **Spiritual and Philosophical Outlook.** A sense of purpose, alignment with personal values, and inner peace cultivates resilience and emotional balance.

Dealing with Mental Distress

6. **Signs of Mental Distress.** State of mental distress can be figured out when an individual does the following:-



(a) Avoids eye contact and interaction.

(b) Have an aversion to a physical affection.

(c) Does not smile in situations calling for such a response.

(d) Does not show any remorse despite showing unwarranted behavior.

(e) Carries no emotional reaction when left alone.

(f) Shows anger in the form of tantrums or passive aggressive behavior.

(g) Lacks interests in recreational activities.

7. **Techniques to Manage One's Mental State.** The following are the skills/techniques to identify one's present state of mental wellness, and practices to be followed by any person to manage it:-



(a) **Self-Introspection**. Self-introspection enables individuals to understand the reasons behind their responses to situations or interactions. This capacity can assist in resolving conflicts by differentiating the emotion from the specific circumstances.

(b) **Acceptance**. It is the capacity to acknowledge one's feelings without excessive contemplation. This can facilitate more rational responses to situations that elicit particular mental behaviour.

(d) **Perspective**. Adopting a more expansive viewpoint can assist in managing your mental responses by considering them within a larger frame of reference.

(e) **Empathy**. Empathy, the ability to understand and share the feelings of others based on one's own experiences, is vital for building fulfilling relationships and avoiding conflicts in any form of settings.

8. **Benefits of Mental Wellness.**

(a) **Better Stress Management**. Helps in managing and reducing stress more effectively. By being mentally aware, one can identify the early signs of stress and apply techniques such as deep breathing, mindfulness, or problem- solving to

mitigate its impact, preventing stress from becoming overwhelming.

(b) **Stronger Relationships**. Establishes deeper and more meaningful connections with others. Mental wellness enhances ones emotional expressions that lead to better empathy and understanding, allowing people to engage more deeply in relationships.



(c) **Boosts Self-Esteem.** Boosts self-awareness and confidence, leading to a more positive self-image. Mental wellness boosts confidence, helping people accept themselves and pursue their goals with a positive outlook and strong belief in their abilities.

(d) **Increased Resilience.** Encourages the ability to recover from setbacks and challenges with a positive attitude. Mental wellness strengthens resilience, making it easier to face and overcome life's challenges.

(e) **Sharper Mental Focus.** Enhances clarity of thought and improves decision-making skills. With mental wellness, individuals experience better clarity in their thoughts and decision-making processes. When emotions are well-managed, there is less mental clutter, it improves concentration and productivity.



(f) **Emotional Stability**. Mental wellness ensure mental balance and reduces feelings of anxiety or depression. With emotional stability, individuals are better equipped to handle everyday stress without feeling overwhelmed.

(g) **Healthier Body**. Lowers the likelihood of stress-related physical ailments by maintaining good emotional health. Mental wellness is closely linked to physical health. When an individual’s mental state remains balanced, the body experiences less tension and stress, lowering the likelihood of developing stress-related ailments such as headaches, high blood pressure, or digestive problems.

(h) **Improved Communication**. Mentally balanced individuals are better at expressing their emotions clearly and listening with empathy. This leads to healthier communication, which fosters stronger relationships.

(j) **Enhanced Coping Mechanisms**. Mental wellness equips individuals to cope with life’s inevitable challenges through problem-solving skills, emotional regulation techniques, or seeking social support.

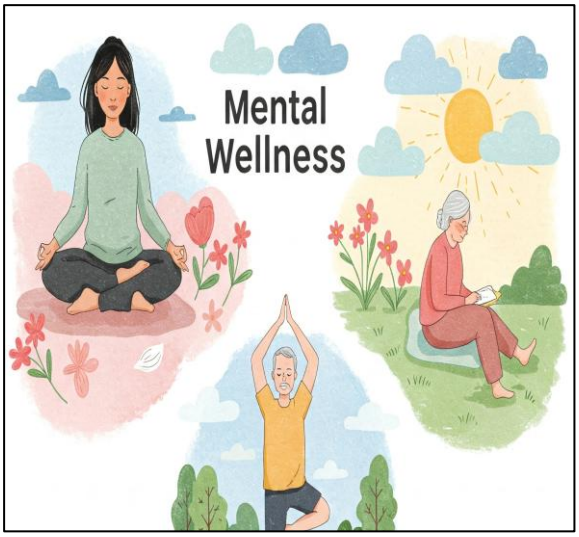
(k) **Sound Sleep**. Mental balance ensure restful, regular sleep that leaves one feeling refreshed. A good state of mental wellness promotes relaxation, reduces stress and anxiety, and helps regulate mood-related chemicals like serotonin and melatonin which are essential for quality sleep.

(l) **General Feeling of Hope, Optimism and Motivation**. When the mind is well- balanced, individuals demonstrate greater resilience, a clear sense of purpose, and increased confidence in overcoming life's challenges.

Case Study : 'Smarty' Shines Again

9. **Background**. Cdt 'Smarty' was a motivated NCC cadet, studying in third year of graduation. Holding aspirations of becoming an Army Officer, Smarty was a regular in NCC classes and participated actively in miscellaneous NCC events. Being good in sports apart from academics, he was a fairly known figure in his college, as also on social media, where he had built a following of over 80,000 fans through daily workout tips, motivation clips, and transformation reels. Known for his energy and charisma, Smarty was not just a content creator—he became a teen influencer in his city. Over the past two months however, his participation in NCC activities had diminished and his absence from parades had started increasing. This had his ANO, Mr 'Keen-Eye,' really concerned, because Smarty was not being seen around much otherwise also, not even with his best friend, Cdt 'Sincere'.

10. **The 'Crisis'.** ANO Keen-Eye decided to meet Smarty to know what was going on. After several attempts, he could finally locate him one evening sitting alone in the football field. Smarty seemed reluctant to talk but ANO Keen-Eye was persistent. Eventually he opened up and revealed that till about three-four months back everything was going fine but then a viral video he made was misunderstood and labelled "toxic motivation". Backlash spread quickly, and within weeks, his followers dropped by nearly 50,000. The algorithm also stopped favouring his content and his popularity plummeted. He went through a period of intense self-doubt and identity crisis. **"Am I not good enough anymore?"** Since he had started tying his **self-worth to the numbers of 'likes'**, a dip in that triggered intense self-questioning. It felt like he was losing his purpose. While he could handle the trolls who mocked him with jabs like' You have lost it!, it was when those that he had considered as his most **loyal fans** de-subscribed/drifted away, that Smarty started feeling totally '**abandoned'**. **The increased pressure to stay 'relevant' led to bouts of anxiety**, **depression**, **and eventually to social withdrawal**. He stopped posting entirely, started skipping classes & workouts, became withdrawn and avoided friends including Cdt Sincere.



11. **The Turning Point.** ANO Keen-Eye had a good understanding of inter-personal relations. He listened to Smarty patiently and then spoke - reminding him about his purpose as to why he started; that name and fame can't be chased and were often a by-product of following one's passion and still not certain; that while having followers on social media was ok, it was a bad idea to derive one's sense of self-worth from them; that between his family and a staunch friend like Cdt Sincere, Smarty still had his 'support system' intact and that, when the spotlight dims, it often reveals what we are made of underneath.

12. **<u>The 'Bounce-Back'</u>.** The conversation sparked something in Smarty. He 'reconnected' with his purpose and decided to start afresh and stay true to his core values. He started slow, doing it for himself, focussing on fitness not for amassing numbers but as a therapy, as its own reward. He stayed off social media, choosing to spend more time with Sincere. ANO Keen-eye mentored him personally and with his guidance, over the course of next few weeks, Smarty was able to build a core community of like-minded enthusiasts. With regained confidence, Smarty returned to content—but this time with a shift. He launched a new series: **"Fitness After Failure"**, focusing on mental health and physical recovery. The content was more authentic**,** honest, and slow-paced. His audience, though smaller now, was far more engaged. He ended up gaining back 10,000 followers, but this time numbers mattered little to him. He had gained himself back!

13. **<u>Key Lessons</u>.**

(a) The idea of 'acceptance' by others is misplaced. Sense of self-identity and self-esteem should stem from our core values and what we are from inside.

(b) Seeking validation from 'Numbers' can crush a young creator's self-worth, but healing/recovery is possible.

(c) Emotional resilience isn't built in the spotlight—it's built in quiet recovery. Life's hardest battles are often fought against oneself.

(d) Integrity of purpose is vital. Redefining success around purpose and effort, is a powerful comeback tool. 'Name/Fame' are incidental and variable.

(e) Support of friends and well-wishers is a great source of strength.

<u>PART II : REAL VERSUS VIRTUAL FRIENDS</u>

14. Friendships are key to emotional wellness and personal growth. As people grow, their relationships help shape their character. Impersonal virtual connections may have their value in terms of getting to know people and networking, however it is the real-life friendships that provide deeper & meaningful connections and tend to act as a safety valve against life's ups and downs.



15**.** **<u>Real Friend</u>**. A real 'friend' is someone you like meeting & spending time with, can confide in, and have a personal conversation sharing your joys and sorrows with equal ease. They tend to act as your support system during tough times. They could be anyone including your family members.

16. **<u>Virtual Contacts</u>**. **This term is used for virtual or online 'contacts', which social media generously terms as 'friends'**. Social and professional networking sites facilitate

the building of ‘online acquaintances’. While social media may term these as a ‘friend’, they hardly qualify to be so in absence of any personal connect.



17. Here a distinction between the two is warranted. **It is not to say that one can’t meet or connect with a real friend online or in a virtual mode, or that connecting with our best friend online, turns him/her into a ‘virtual contact’**. Distance or medium of connection hardly matters in real friendships and they remain strong despite such barriers. **The distinction lies in the depth and understanding we have with our real friends as compared to our virtual contacts.** One of the reason why interactions with virtual contacts may remain mostly ‘superficial’ is the **medium of exchange of information, which is mostly text/message**. Even when it is a call, its either in formal settings for anyone to bother or is so infrequent that the **listener can’t differentiate between the caller’s usual self or him/her sounding different on a particular day, a cue which a real friend is unlikely to miss and probe.**

18. You would have noticed that despite having many people in your physical social circle, real friendships are restricted to very few, probably just one or two. This holds even more true for ‘online/virtual contacts’ as lack of ‘in-person’ interaction prevents fostering of deep and meaningful friendships

19. Another reason why interactions with virtual contacts may remain mostly ‘superficial’ is the **medium of exchange of information, which is mostly texts/messages**. Even when it is a call, its either in formal settings for anyone to bother or it is so infrequent that the **listener can’t differentiate between the caller’s usual self or him/her sounding different, a cue which a real friend is unlikely to miss.**

20. Mental/emotional heaviness occurs when we overthink, magnify our problems multi-fold and stay bottled up. Sharing our concerns or issues with a willing listener is like already addressing half the problem. This is where ‘real’ friends often come to our rescue who can listen empathically, can relate to what we are going through and provide a safe space for us to vent out our feelings and lighten our mental burden. While impersonal virtual connections may give a temporary high with ‘**likes**’ or ‘**comments**’, they firstly can never give a lasting joy and secondly tend to put us into an addictive ‘**validation loop’**, eventually placing our self-esteem in other’s hands.

21. For making personal connections stronger, remember that ‘**a call is better than a text message, and meeting personally is even better than a call’.**

22. **Effects of Excessive Virtual Interactions.** Can be as under:-

(a) **Loss of Physical Health.** Spending long hours interacting with virtual contacts negatively affects physical health, leading to **sedentary lifestyles and general weariness.**

(b) **Erosion of Real-World Social Skills.** Over-reliance on virtual interactions can lead to social awkwardness or **discomfort in face-to-face communication**. **The real loss is of the urge to work on real-world relationships to make them work, and finding online connections more 'convenient'.**

(c) **Impact on Mental Well-Being.** Excessive online interaction may turn individuals into **social-recluses, leading to real-time social isolation**.

(d) **Dopamine Overload.** Likes, comments, and messages provide a quick dopamine hit—the brain's reward chemical. This can create addiction-like behaviour, leading to **irritability or low mood when offline.**

(e) **Increased Anxiety and Stress.** Constant notifications, messages, and the pressure to stay "online" can lead to **mental fatigue**. Fear of missing out **(FOMO)** and the need to stay connected can **heighten anxiety levels**.

(f) **Lowered Self-Esteem.** Continuous exposure to 'curated', filtered lives of others can trigger **comparison traps**. Feeling "less successful" or "not good enough" can lead to **self-doubt and insecurity**.

(g) **Increased Loneliness.** Ironically, being constantly connected online can make people feel **more isolated** in real life. The lack of authentic and empathetic interaction can leave one with a sense of **emotional emptiness.**

Comparison Between a Real-Life Friend and a Virtual Contact

23. In today's digital age, making online contacts is easy. Although they can also provide companionship and enjoyment, they differ from friends in terms of interaction, communication, and personal connection and one must not confuse a Virtual Contact (**called Contact hereafter**) for a Real Life Friend (**called Friend hereafter**).

24. **Differences.** The difference between *Friends* and *Contacts* can be compared through the following characteristics:-

(a) **Social Interaction and Engagement.** In-person interactions with *Friends* allow for more complex, layered conversations where both individuals can engage in more meaningful exchanges about personal experiences, feelings, and future plans. But the nature of connection with *Contacts* limit personal conversations, seldom going into life's challenges or deep emotions.

(b) **Emotional Support.**

(i) *Friends* offers immediate, tangible support, such as comforting one during stressful times, offering physical presence in difficult moments (e.g., during family issues or personal crises), and engaging in face-to-face activities to cheer (e.g., going to the movies, having lunch together). This type of emotional support has a strong physical and psychological impact.



(ii) Relations with *Contacts* seldom reach the depth for them to provide emotional support. Even when they do offer sympathy, it is mostly formal or semi-formal, primarily through messages or voice calls. **In times of distress, one hug from a close friend is more reassuring than a million messages on social media!**



(c) **Communication Style.**

(i) Conversations with *Friends* are often face-to-face, allowing for a richer, more immediate exchange of emotions, body language, and physical cues (hugs, gestures, etc.). Misunderstandings are less frequent as tone, facial expressions, and physical proximity help clarify emotions. In-person meetings provide a deeper sense of connection, even if it's not daily.

(ii) Communication with *Contacts* occurs through text, voice messages, or video calls, relying on written words or audio to convey emotions. The absence of physical presence can sometimes lead to misinterpretation of tone or intent.

(d) **Trust and Reliability.**

(i) Trust with *Friends* generally develops over years of shared joys and sorrows. Physical presence and real-world interactions give a deeper sense of comfort, security and reliability.

(ii) Although with *Contacts,* one may have a sense of trust built over time, the virtual nature of the engagement means there are fewer opportunities for *Contacts* to understand the other person fully, making the relation stay on the surface.

(e) **<u>Impact on Mental Health and Well-being</u>.**

(i) In-person interactions with *Friends* have a positive impact on mental health, providing moments of joy, relaxation, and comfort. The physical presence and non-verbal communication offer a more holistic form of emotional well-being.

(ii) Though *Contacts* can provide a sense of community, especially during times of stressfulness, the lack of an emotional connect may provide limited sense of companionship.

25. Contacts do have their value in terms of enhancing our social or professional network. Their numbers also may be more than *Friends* but connection and communication with *them* stays restricted and limited. ***Friends*, even when far lesser in numbers, can provide deep emotional support and social interaction that enhances our well-being in tangible ways.**

PART III : REDUCING SCREEN TIME





26. **<u>Screen Time</u>**. The amount of time spent using a device with a screen, such as a smartphone, computer, television, video game console, or a tablet, is referred to as screen time.



<u>Harmful Effects of Excessive Screen Time</u>

27. There is a correlation between screen time and mental and physical harm in an individual's development. The positive or negative health effects of screen time are influenced by the level of exposure. These can have an impact on both physical and mental well-being.

28. **Physical Health Effects.** Excessive screen time can have the following physical health effects:-

(a) **Eye strain (Computer Vision Syndrome)**. Dry, irritated eyes, blurred vision, and headaches.

(b) **Neck & Back Pain**. From looking down at screens ("tech neck").

(c) **Poor Posture**: Slouching becomes the default.

(d) **Headaches**. Often triggered by eye-strain or constant exposure to screens.

(e) **Disrupted Body Clock**. Again, that blue light at night can throw off your circadian rhythm.

(f) **Overeating or Mindless Snacking**. Especially when watching shows or gaming.

(g) **Sluggishness**. Reduced physical activity leading to muscle loss and an increased risk of weight gain

29. **Mental Health Effects**. Excessive screen time can have the following mental health effects:-

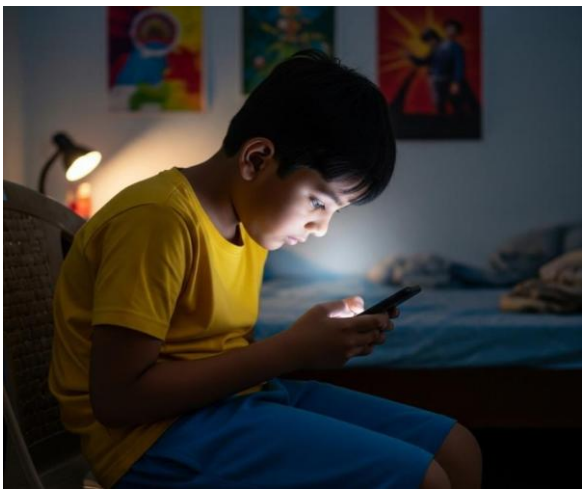
(a) **Delayed Development (in Toddlers)**. Too much passive screen time can affect speech and social skills.

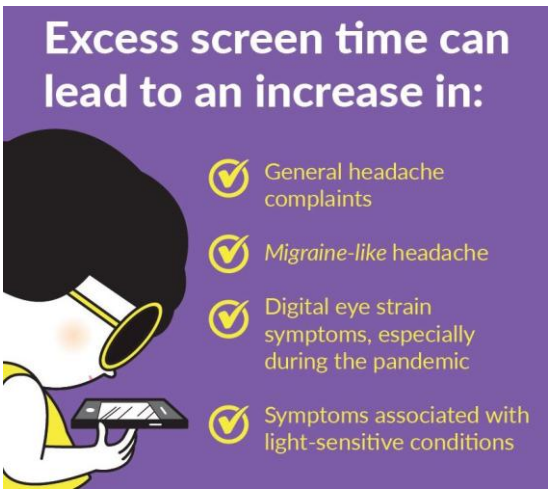
(b) **Lower Academic Performance**. Happens when screen use replaces sleep, study, or physical activity.

(c) **Addictive Behaviour.** Games and social media are designed to keep you hooked.

(d) **Reduced Attention Span**. Constant notifications and fast-paced content can rewire your brain for shorter bursts of focus.

(e) **Digital Burnout**. Feeling mentally exhausted or irritable after long sessions.





Balancing Screen Time

30. Maintaining an optimal level of screen time is crucial for overall wellness. Balanced screen usage can provide the following benefits:-

(a) **Physical Health Advantages**. Increased physical activity through balanced screen time can lower the risk of obesity. Similarly, by moderating screen usage, one can improve sleep quality. Furthermore, limiting prolonged screen exposure mitigates eye strain and vision problems.

(b) **Mental and Emotional Health Enhancements**. Modulating screen time can yield positive cognitive and emotional outcomes. Reduced screen exposure has been associated with lower levels of stress and anxiety, potentially by facilitating relaxation and calmness. Balancing screen time enhances concentration and mental focus.

(c) **Improvements in Academic Performance**. Balancing screen time enables students to maintain focus on educational activities. It aids in students' development for effective time management strategies. Hence, it also facilitates students' ability to complete tasks efficiently.

(d) **NCC-Specific Advantages**. Regulating screen time enables NCC cadets to dedicate more attention to physical training and other activities, thereby enhancing their endurance and agility. Decreased screen usage promotes in-personal interactions, by means of taking up tasks involving teamwork and camaraderie among NCC cadets. It allows NCC cadets to allocate more time to developing critical leadership skills, such as communication, problem-solving, and decision-making.

31. **Tips to Balance Screen Time**.

(a) **Screen-Free Zones**. Designate screen-free zones, such as the dining room or bedrooms, to limit exposure to digital devices.

(b) **Screen-Free Time**. Establish specific screen-free times, such as during meals, before bed, or during family activities, to promote healthier habits.

(c) **Physical Activities**. Engage in physical activities, like sports or outdoor activities, to reduce overall screen time. Utilize monitoring applications to track and manage screen time effectively.

(d) **Hobbies**. Cultivate alternative hobbies, including reading, drawing, or playing musical instruments, as a means to decrease dependence on digital screens.

(e) **Lead by Example**. Be an active role mode by monitoring your own screen time. Make for time to be active with friends and family every day.





PART IV : SLEEP HYGIENE

32. Exposure to digital screens suppresses melatonin, which is responsible for regulation of sleep wake cycles. Excessive screen time among young people impacts sleep and well-being, creating a vicious cycle. These may in turn drive one to seek distraction or escapism by relying more on to the screen. Therefore, it becomes mandatory to break this cycle and improve both screen habits and sleep hygiene for better mental wellness.

33. In our fast-paced society, sleep is frequently compromised. Nonetheless, upholding proper sleep hygiene, encompassing consistent sleep schedules and a conducive sleeping environment, is paramount. Sleep deprivation can result in a myriad of health issues, including obesity and mental health issues.

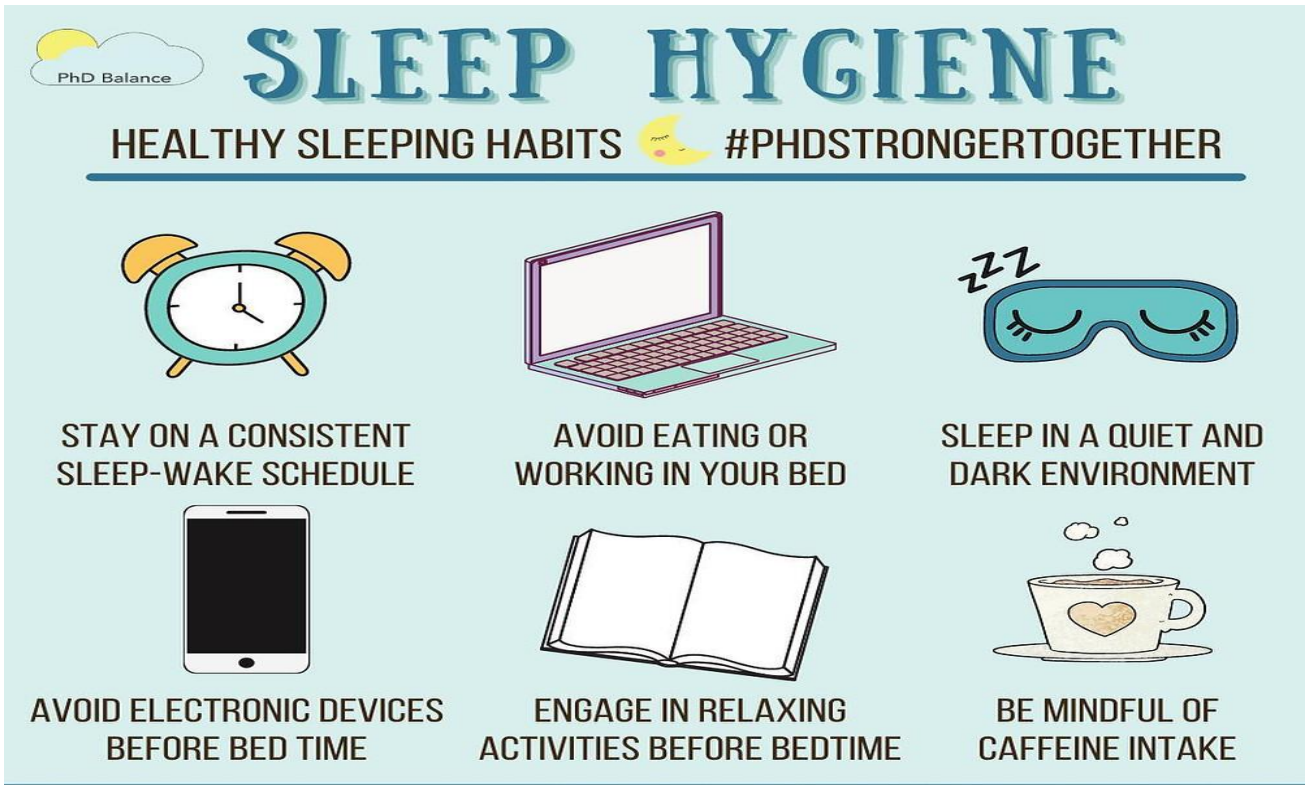
34. **Key Components of Sleep Hygiene.** Important components of sleep hygiene are as follows:-

(a) **Environment.** Maintaining an appropriate sleep environment is crucial for quality rest. Specifically, one should ensure the bedroom remains dark, quiet, and cool/comfortable. Removing electronic devices from the sleeping space is recommended, as they can disrupt sleep patterns.

(b) **Routine**. Maintaining an optimal sleep environment is crucial for achieving quality rest. Establishing a regular sleep schedule, developing a calming pre-sleep routine, and maintaining a consistent morning routine are strategies to regulate the sleep-wake cycle effectively. schedule, developing a calming pre-sleep routine, and maintaining a consistent morning routine are strategies to regulate the sleep-wake cycle effectively. to regulate the sleep-wake cycle effectively.

(c) **Lifestyle**. Engage in regular physical exercise that promotes better sleep, but refrain from intense workouts within 2-3 hours of the designated sleep period. Abstain from consuming caffeine and nicotine in close proximity to bedtime. Complete eating at least 2-3 hours before the intended sleep time. Avoid using screen at least one hour before sleep.





35. **<u>Factors affecting Sleep Quality</u>**. The elements influencing sleep quality can be categorized as:-

(a) **Environmental Factors**. Residing in a noisy neighborhood can disrupt sleep due to the presence of noise and disturbances. Change of location or weather conditions also affects sleep.

(b) **Behavioral Factors**. Lack of exercise can cause sleep problems. Cadets may struggle to sleep due to pressure and anxiety. Using excessive coffee or tea to stay awake harms sleep quality. Overuse of screens before bed reduces sleep quality.

(c) **Psychological Determinants**. Staying in unfamiliar living spaces like hostels or PG accommodations can induce homesickness among cadets, leading to emotional distress and impaired sleep quality. In certain life-situations, cadets may experience social stressors and peer pressure, which can adversely impact their sleep patterns. Furthermore, academic pressures can contribute to increased anxiety and stress, further disrupting their sleep quality.

36. **Benefits of Quality Sleep**. The benefits of quality sleep are as follows:-



(a) **Physical Health Benefits**. Adequate sleep enhances physical performance. Quality sleep bolsters the immune system, reducing the risk of illnesses and infections.

(b) **Cognitive Function Benefits**. A well-rested individual can concentrate more effectively during daily tasks and can display improved alertness and focus. Sleep facilitates enhanced memory and learning, enabling individuals to retain information more readily. Quality sleep enhances critical reasoning and problem-solving skills.

(c) **Emotional Health Benefits**. Good sleep plays a crucial role in emotional wellness by stabilizing mood, reducing stress, and enhancing emotional resilience.

(d) **Academic Performance Benefits**. Adequate sleep enhances concentration, memory, and learning, ultimately resulting in enhanced academic performance.

37. **Strategies for Improving Sleep Quality**. Following are the guidelines for improving the sleep quality:-



(a) **Pre-Sleep Routine**. Incorporating relaxation techniques, such as deep breathing, progressive muscle relaxation, or meditation, can help calm the mind and body before sleep. Engaging in soothing activities like reading or listening to calming music can also aid in unwinding. Additionally, journaling to process thoughts, feelings, and experiences can help clear the mind prior to sleep.

(b) **Sleep-Conducive Environment**. Maintain a dark, tranquil, and comfortable sleeping environment. Investing in a supportive mattress and pillows can enhance physical comfort and facilitate better sleep.

(c) **Lifestyle Adjustments**. Engaging in regular physical activity is recommended, but strenuous exercise should be avoided within 2-3 hours of bedtime. Fueling the body with a balanced diet while limiting heavy meals close to bedtime is advised. Maintaining adequate hydration throughout the day is important, but excessive fluid intake immediately before sleep should be avoided. Limit or avoid the use of tea/coffee in daily routine.

(d) **Time Management**. Prioritize obtaining adequate sleep each night. Allocate 7-9 hours for sleep and maintain a consistent sleep schedule. Set boundaries by learning to decline nonessential activities that may disrupt your sleep. Utilize free time during the day productively by engaging in relaxing, reading, or other calming activities.

(e) **Avoid Screens Before Bed**. Avoid the use of electronic devices for at least one hour prior to the desired bedtime.

(g) **Seek Support**. Communicate your sleep-related issues with trusted peers, relatives, or an adult mentor to obtain guidance and assistance.

38. Sleep and mental wellness are deeply intertwined. Good sleep promotes emotional regulation, enhances mood stability and improves overall mental resilience. It helps the brain process emotions effectively, contributing to a more balanced and positive emotional state.

PART V : CONNECT WITH NATURE

39. **Nature and Mental Wellness**. In today's fast-paced, chaotic world, it's essential to take moments of pause and find peace. Instead of relying solely on medical treatments for issues like depression, insomnia, and anxiety, we should also look to nature too for healing and relaxation. Engaging with natural environments reduces stress, enhances mood, and boosts cognitive function, promoting emotional resilience. This text explores the benefits and suggests practical ways to incorporate nature into their daily routines for better mental health.

40. **How to Connect with Nature.** Connecting with nature serves as a powerful means for individuals to restore inner calm. It can be accomplished through a tranquil walk beneath leafy trees, listening to the gentle songs of morning birds, or noticing the breeze as it brushes past. These quiet enriching encounters help ground a person in the present. Nature’s rhythm encourages a slower pace, offering clarity and emotional balance along the way.

Ways to Connect with Nature

41. **Combine Nature with Creative Expression**. Expressing creativity in natural environments—through photography, painting, or writing—serves both therapeutic and

artistic purposes. Creative immersion in nature can relieve stress, improve mood, and inspire fresh perspectives.

42. **Establish a 'Sit Spot'**. Designating a regular outdoor place for quiet observation encourages consistency in nature connection. It sharpens awareness of natural cycles, weather changes, and the behavior of local flora and fauna.



43. **Observe Animal Behavior & Practice Birdwatching.** Watching wildlife, even the smallest creatures, can provide insights into survival, adaptation, and coexistence. Such observations promote empathy, curiosity, and a stronger ecological bond. Birdwatching helps one cultivate focus, patience, and appreciation for biodiversity.

44. **'Interact' with Plants.** Spending time with plants, whether gardening, visiting botanical gardens, or simply observing—strengthens one's relationship with the environment. It encourages attentiveness to life processes and cultivates calmness.



45. **Walk, Run, or Hike in Nature**. Physical activities such as walking or hiking offer dual benefits, enhanced physical health and deeper nature appreciation. Exploring varied landscapes or inviting companions adds social value to the experience.





46. **Sit Under the Night Sky**. Stargazing invites reflection on the universe and one's place within it. It is a powerful way to build emotional connection with the broader natural world and nurture existential insight.

47. **Enjoy a Nature-Based Picnic**. Picnicking in green spaces combines leisure with environmental appreciation. It offers a casual setting for bonding with others while enjoying scenic beauty and outdoor recreation.

48. **Watch the Sunrise or Sunset**. These daily events offer profound stillness and visual splendor. Regularly observing them encourages contemplation, emotional balance, and alignment with natural rhythms.



49. **Go Camping.** Camping provides full immersion in the natural world, free from technological intrusions. It enhances resilience, strengthens social ties, and promotes a grounded connection with nature.

50. **Participate in Environmental Protection**. Conservation efforts—like tree planting, habitat restoration, or community cleanups—instill a sense of purpose. They affirm the individual's role as a steward of the environment.

Benefits of Connecting with Nature

51**.** **Reduces Stress and Anxiety.** A strong connection to nature leads to improved mood and lower anxiety, with benefits even from simple interactions like spending time in parks or watching nature documentaries. Spending 20-30 minutes in nature lowers cortisol, reducing stress and promoting relaxation. Nature provides a mental break, helping people disengage from daily pressures and motivating mindfulness through activities like walking or listening to natural sounds.



52**.** **Improving Mood and Happiness.** Exposure to nature increases serotonin levels, improving mood and fostering joy and creativity. Studies show that people who regularly connect with nature feel happier and more fulfilled, making nature an essential component of emotional wellness.

53**.** **Boosting Attention and Memory.** Nature helps restore mental clarity by reducing cognitive fatigue. Exposure to nature improves focus and memory, benefiting individuals with any sort of stress. Outdoor activities also enhance cognitive function and performance.

54**.** **Cultivating Awe and Connection**. Connecting with nature inspires a feeling of wonder and a sense of being interconnected such that it encourages empathy and a deeper connection to both people and the environment. This experience promotes prosocial behaviors and a desire to protect nature.

55**.** **Physical Activity in Nature**. Exercising outdoors amplifies nature's mental health benefits. Physical activity in natural settings reduces stress and improves mood and sleep. For those struggling with anxiety and depression, outdoor exercise in nature provides additional therapeutic effects.

56. **Psychological and Physiological Benefits**. Nature helps reduce stress and improve mood by activating the parasympathetic nervous system and boosting serotonin. It also enhances mindfulness, reduces rumination, and strengthens the immune system, promoting overall well-being.

57. Nature's positive impact on mental wellness is undeniable. It reduces stress, improves mood, and enhances cognitive function. We can easily incorporate nature into our routines through parks, outdoor activities, and nature-based experiences, improving mental health and a flourishing mind.

HIGHER ORDER THINKING SKILLS (HOTS)

- **Analysis Exercise. Ask your friends to reflect on their daily screen time and discuss the various ways to balance screen time with their benefits.**

CONCLUSION

58. Emotional and mental wellness are crucial for leading a balanced and fulfilling life. Various factors, including biological, psychological, and social elements, shape our emotional well-being. In today's digital age, managing screen time, maintaining real-life connections, and practicing self-awareness are essential for good mental health. By adopting healthy habits such as proper sleep, stress management, and emotional regulation, individuals can improve their resilience and overall quality of life. Prioritizing emotional wellness not only enhances personal well-being but also strengthens relationships and social interactions.

SUMMARY

- Emotional and mental wellness are essential for a balanced life.
- Factors like biology, psychology, and social environment influence emotional health.
- Practicing self-awareness helps in regulating emotions effectively.
- Excessive screen time can negatively affect mental well-being.
- Real-life friendships contribute more to emotional wellness than virtual ones.
- Good sleep hygiene and stress management improve mental health.
- Building resilience through healthy habits leads to overall well-being.

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ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. A cadet's ability to effectively tackle a problem is most directly influenced by their:-

(a) Physical strength and stamina

(b) Level of social media influence

(c) Mental wellness, including reasoning ability

(d) Access to financial resources

Q2. Which statement best reflects the relationship between stress and performance?

(a) Any level of stress invariably impairs performance

(b) Optimal performance occurs in the absence of any stress

(c) A small amount of stress can be beneficial for performance

(d) Stress only affects physical, not mental performance

Q3. In the context of managing mental wellness, self-introspection is crucial for:-

(a) Predicting external events with certainty

(b) Understanding the underlying causes of one's emotional responses

(c) Suppressing emotional expression to maintain composure

(d) Avoiding social interactions to minimize emotional triggers

Q4. Which of the following lifestyle factors has the most complex interaction with mental wellness?

(a) Regular physical activity

(b) Sleep hygiene

(c) Social environment

(d) Nutrition

Q5. The concept of "Eustress" is best exemplified by:-

(a) Experiencing chronic financial strain

(b) Feeling motivated by an upcoming challenge

(c) Enduring prolonged relationship issues

(d) Reacting to a sudden, traumatic event

Q6. In Smarty's case, the decline in his mental wellness was primarily triggered by:-

(a) Physical exhaustion from over-exercising

(b) A perceived loss of social media validation

(c) Increased pressure from his family to succeed

(d) Academic difficulties in his third year

Q7. Which of the following is a core component of developing emotional resilience?

(a) Emotional detachment

(b) Self-awareness

(c) Suppression of negative emotions

(d) Social isolation

Q8. Real-life friendships are emphasized for their ability to provide:-

(a) A larger network of contacts

(b) More superficial but frequent interactions

(c) Deeper emotional support and a "safety valve" effect

(d) Opportunities for professional advancement only

Q9. The 5-4-3-2-1 grounding technique is primarily used to:-

(a) Enhance long-term memory

(b) Stimulate the nervous system for increased alertness

(c) Reduce immediate stress or anxiety by focusing on the present

(d) Identify deep-seated emotional traumas

Q10. Which of the following statements about stress is most accurate?

(a) Stress is always detrimental to an individual’s well-being

(b) The impact of stress is solely determined by its duration

(c) An individual’s perception of a stressor can influence its effect

(d) Physiological responses to stress are entirely avoidable

Q11. In the context of the documents, a balanced mood and reduced anxiety are most indicative of:-

(a) High social media engagement

(b) Emotional stability, a benefit of mental wellness

(c) Complete absence of stressors

(d) Suppression of emotional expression

Q12. What did ANO Keen-Eye's intervention primarily address in Smarty's situation?

(a) Smarty's financial instability

(b) Smarty's lack of physical fitness

(c) Smarty's misplaced sense of self-worth

(d) Smarty's academic performance

Q13. Which of the following is NOT a recommended strategy for managing stress emotions?

(a) Deep breathing exercises

(b) Engaging in physical activity

(c) Positive self-talk

(d) Emotional suppression

Q14. According to the texts, what is a key differentiator between real and virtual connections:-

(a) Virtual connections are more effective for emotional support

(b) Real connections offer a depth of personal interaction often lacking in virtual ones

(c) Virtual connections are essential for developing empathy

(d) Real connections hinder networking opportunities

Q15. What is the long-term implication of unmanaged chronic stress?

(a) Enhanced emotional resilience

(b) Improved cognitive functions

(c) Potential harm to physical health

(d) Increased social connectivity

Short Answer Questions

Q1. What does the Latin quote ‘Mens sana in corpore sano’ mean?

Q2. Name any four domains of wellness.

Q3. What triggered Smarty’s mental distress?

Q4. What is the role of self-awareness in managing mental state?

Q5. Why are real friendships better than virtual contacts for mental health?

Long Answer Questions

Q1. Explain how mental wellness is interconnected with other domains of wellness.

Q2. Discuss the various factors affecting mental wellness.

Q3. Describe the techniques used to manage one’s mental state.

Q4. Analyze the key lessons from Smarty’s case study.

Q5. Compare and contrast real friendships with virtual contacts. How do they impact mental well-being?

PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW VII : STRESS MANAGEMENT

"Feelings are much like waves. We can't stop them from coming, but we can choose which ones to surf."



TEACHING INSTRUCTIONS

Period : **Three (03)**

Type : **Lecture and Practice**

Year : **One in 2nd Year and Two in 3rd Year SD/SW**

Conducting Officer : **Associate NCC Officer**

Training Aids : **Script/Lesson Plan: OHP, Screen, Pointer Staff, Presentation, Charts/Posters.**

Time Plan

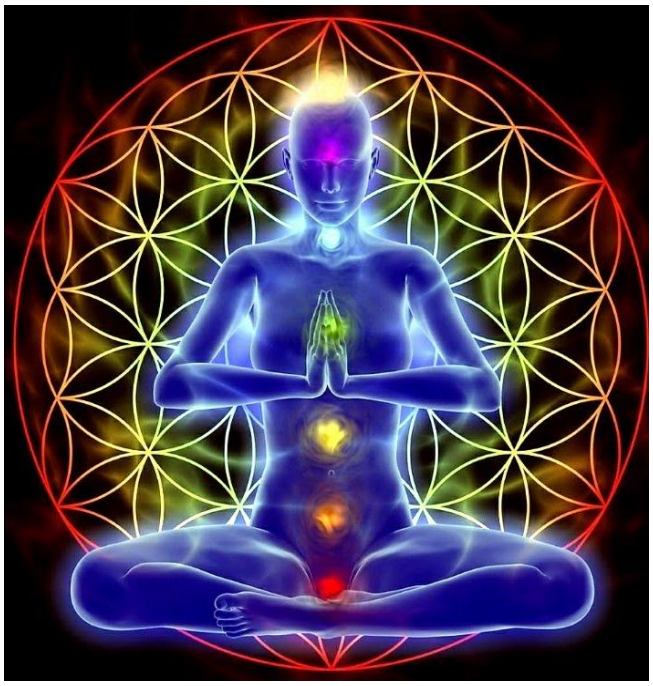
- Introduction : 03 Mins
- Part I : 15 Mins
- Part ll : 22 Mins
- Part III : 40 Mins
- Part IV : 17 Mins
- Part V : 20 Mins
- Conclusion : 03 Mins



INTRODUCTION

1. Stress is a natural response to life's pressures, but how we manage it can determine its impact on our mental and physical health. In this lesson, we will explore stress management techniques, the importance of emotional awareness, and the benefits of meditation for stress relief.

2. Stress can be managed by number of techniques including mindfulness to cultivate the ability to recognise our thoughts and senses without bias or attachment. For cadets, it is the equilibrium between academic tasks and personal life pursuits.



LEARNING OBJECTIVES

- Understand the causes, effects, and types of stress
- Explore various stress management techniques
- Recognize emotions and feelings to manage stress effectively.
- Understand the connection between spirituality, social wellness, and stress management
- Practice meditation, mindfulness, and work-life balance techniques to reduce stress.

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I : Understanding Stress

(b) Part II : Stress Management

(c) Part III : Meditation & Mindfulness

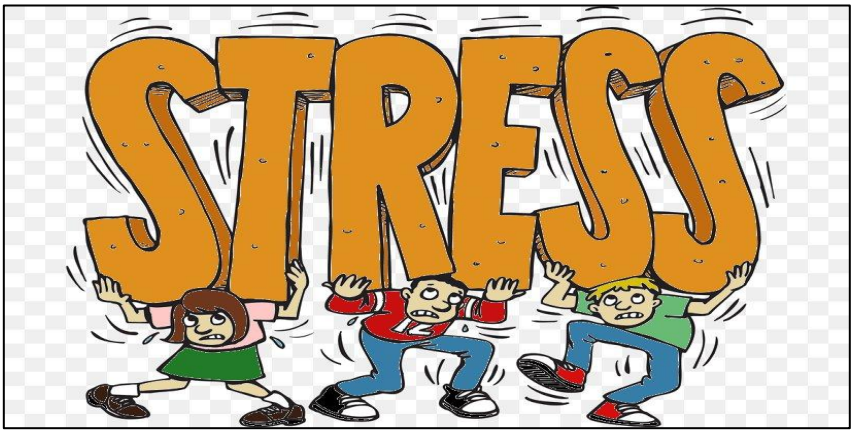
(d) Part IV : Positive Thinking

(e) Part V : Affirmations -The Power Within

INTERESTING FACTS

- Chronic Stress can contribute to premature aging and wrinkles.
- Taking breaks boosts productivity by recharging brain and thereby improving focus.
- Laughter reduces Stress by lowering cortisol levels and boosting one's mood.

PART I: UNDERSTANDING STRESS



3. **Stress**. Stress is a normal emotional and physiological response to a situation that makes you feel overwhelmed, anxious, or pressured. It's like a warning signal that alerts your body to prepare for action. It reflects a normal state of mind. Stress is not always bad as a little amount of it is required for a better performance.

4. If one imagines body as a rubber band and stretches it a little, it will retract back. This is analogous to small challenges but if one stretches it too much, it might break or stay stretched out. That's what happens with too much stress.

5. **Physiology of Stress.** When a situation is perceived with some sort of stress, our bodies release cortisol and adrenaline hormones that increase our heart rate, and breathing, preparing one for a reaction. This is a normal hormonal process but if this is extended for long periods, it becomes harmful.

6. **Eustress.** It refers to positive stress that proves to be beneficial as it motivates an individual, helps one improve performance, and creates a sense of excitement. It's all about how we perceive a stressor! If we perceive it as a challenge than a threat, distress becomes eustress.

Types of Stress

7. Depending upon the duration of stress, there can be different types of stress which are as follows:-

(a) **Acute Stress.** It is a **short term stress**, like before a presentation or before speaking on stage.

(b) **Episodic Acute Stress.** It is when one has frequent acute stress from some **constant worry or pressure**.

(c) **Chronic Stress.** It is a **long-term stress, due to accumulated factors** such as problems related to financial difficulties or relationship issues. Chronic Stress is most damaging to health.

External Factors of Stress

8. Understanding the root causes of stress is essential for managing and mitigating its effects. The causes of stress can be broadly categorized into external and internal factors. External stressors come from outside influences, including life events and environmental conditions that challenge an individual's ability to cope. **Types of External factors** are given in succeeding paragraphs.

9. **Academics Related Stress.**

(a) High workload and tight submission deadlines

(b) Fear of failure/not doing well in competitions/exams

(c) Interpersonal conflicts at school, and poor management

(d) Mental burnout from excessive responsibilities

(e) Physically demanding training/routine resulting in fatigue

10. **Financial Pressure.**

(a) Struggling to meet basic financial needs

(b) Debt and economic instability

(c) Unexpected expenses and emergencies

11. **Relationship Challenges.**

(a) Conflicts with family, friends, or partners

(b) Lack of emotional support

(c) Separation from, or loss of, a loved one

12. **Major Life Events.**

(a) Separation of parents

(b) Anxiety of relocating to a new place/surroundings

(c) Pressure of leaving studies or getting Married

(d) Struggle in recovery from an Injury/disease

13. **Environmental and Social Factors.**

(a) Noise, pollution, and crowded spaces

(b) Discrimination, social injustice, and bullying

(c) Global crises such as pandemics or wars

Internal Factors for Stress

14. Internal stressors arise from personal thoughts, emotions, and behaviours that contribute to stress. **Types of Internal factors** are given in succeeding paragraphs.

15. **Negative Thought Patterns.**

(a) Overthinking and excessive worrying

(b) Low self-esteem and self-doubt

(c) Fear of failure and perfectionism

(d) Battling misplaced feelings of ‘inadequacies/worthlessness’

16. **Unrealistic Expectations.**

(a) Pressuring oneself to meet high standards

(b) Comparing oneself to others

(c) Feeling responsible for everything

17. **Health Concerns and Emotional/Psychological Triggers.**

(a) Chronic illness or injury

(b) Lack of sleep and poor nutrition

(c) Substance abuse and unhealthy coping mechanisms

(d) Suppressing emotions rather than addressing them

(e) Anxiety and depression

(f) Difficulty adapting to change

Signs of Stress

18. Recognising the signs of stress can help you take timely action to manage it. Various symptoms of stress are as follows:-

(a) Feeling anxious, worried, or overwhelmed

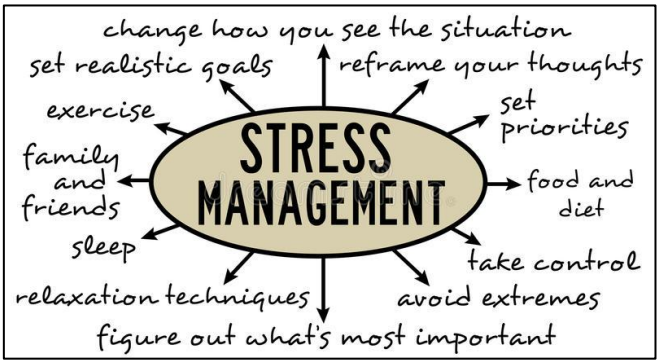
(b) Difficulty sleeping or concentrating

(c) Physical symptoms like headaches, stomach aches, or muscle tension

(d) Irritability, mood swings, or emotional outbursts

(e) Avoiding activities or situations that trigger stress

PART II: STRESS MANAGEMENT



Stress Management Techniques

19. As an NCC student, it's essential to develop healthy coping mechanisms to manage stress. The stress management techniques are as follows:-



Thoughts, Emotions and Feelings

20. **Thoughts.** Thoughts are the mental processes that involve reasoning, analyzing, remembering, or considering something. They are often conscious and can be either positive or negative.

(a) **Origin.** They originate from the brain's cognitive processes and can be influenced by external events, past experiences, or current situations.

(b) **Example.** "I think I should start exercising more."

21. **Emotions.** Emotions are intense, automatic responses to stimuli, often involving a physiological reaction (like a racing heart or sweating) and an immediate mental reaction. They tend to be short-lived and are generally more instinctual.

(a) **Origin**. Emotions arise from deep-seated, primal brain centers (like the limbic system), triggered by thoughts, external events, or physiological states.

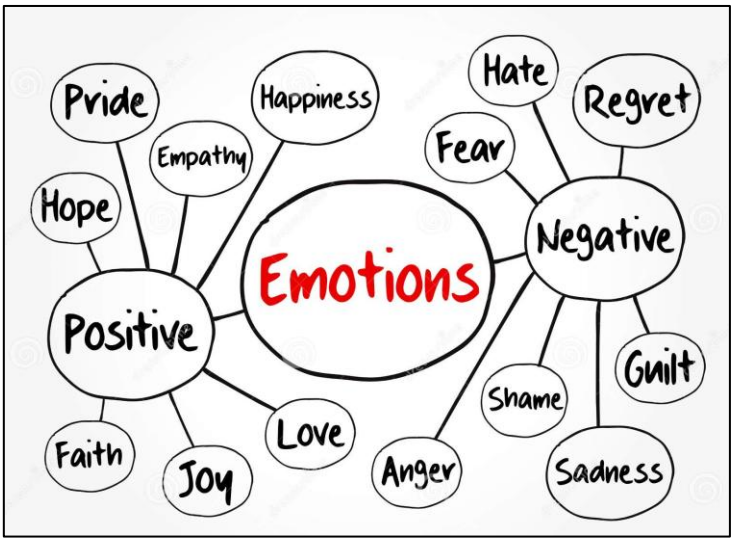
(b) **Example**. "I was angry when my car was hit in the morning."

22. **Feelings**. **Feelings are the union of thoughts and emotions.** Feelings are more prolonged and are the subjective experience of emotions. They are how we interpret and make sense of our emotions. Both emotional experiences like comfort, fear, or happiness as well as bodily sensations like pain, warmth, or cold can be referred to as feelings.

(a) **Origin.** Feelings are based on our emotional responses but are processed through our thoughts and personal understanding of the emotion.

(b) **Example.** "I am still angry over the ill-mannered and disrespectful behaviour of the driver."

23. **Difference between Emotions and Feelings.** Emotions, e.g. anger, are an internal response triggered by an event (may be a perceived threat, injustice etc). How we interpret and perceive that event over a period of time, becomes a feeling. While **emotions are often more instinctive, feelings are how we reflect on those emotions over time, infusing them with thought and giving them energy**. Since **stress is caused by feelings**, recognizing and separating them from emotions (the immediate reaction), can help us see how we are carrying their load internally & tackle them better, thereby reducing stress.



24. **Understanding Stress Emotions**. The stress emotions are as follows:-

(a) **Anxiety**. Feeling worried, nervous, or fearful about the future.

(b) **Irritability**. Feeling short-tempered, snappish, or impatient.

(c) **Overwhelm**. Feeling overwhelmed, hopeless, or helpless.

(d) **Frustration**. Feeling stuck, blocked, or unable to achieve goals.

25. **Managing Stress Emotions**. Dealing with stress emotions involves the following:-

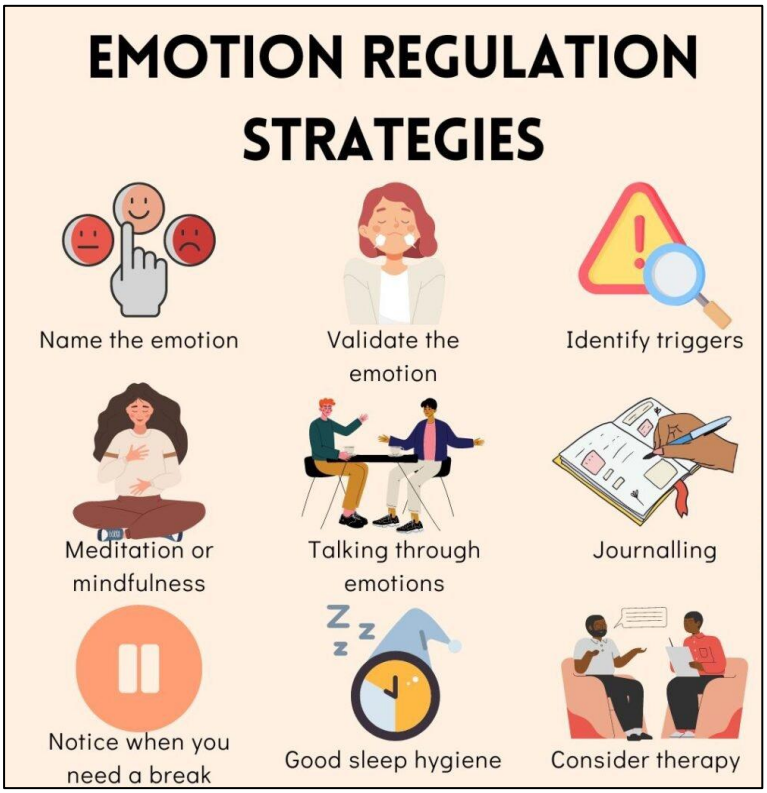
(a) **Deep Breathing**. For mental and physical tranquillity, breathe slowly and deeply.

(b) **Physical Activity**. Engage in physical activities like exercise, sports, or yoga to release tension.

(c) **Mindfulness**. Focus on the present moment, without judgement, to reduce stress and anxiety.

(d) **Positive Self-Talk**. Inspire yourself with positive affirmations to feel better.

(e) **Seek Support**. Talk to trusted friends, family, or mentors about your stress emotions.



26. **Developing Emotional Resilience**. In order to cope with stress and emotion, one must practice the following ways to develop emotional resilience:-

(a) **Self-Awareness**. Develop a better understanding of your emotions, strengths, and weaknesses.

(b) **Self-Regulation**. Learn to manage your emotions, thoughts, and behaviours.

(c) **Motivation**. Find purpose and meaning in your life to stay motivated and focused.

(d) **Empathy**. Practice understanding and compassion towards others to build stronger relationships.

Dealing with 'Stressful Feelings' (Lingering Negative Emotions)

27. Tackling stressful feelings can be challenging, but it's entirely possible with the right approach. The goal is not to repress the emotions but to **experience** and **release** them in a way that doesn't hold you back or negatively impact your well-being. Here are some effective strategies to move past stressful feelings and free yourself from their grip:-

28. **Acknowledge and Accept the Emotion.** The first step in managing negative emotions is acknowledging them. Denying or suppressing emotions only makes them stronger over time. Acceptance doesn't mean you approve of the emotion, but rather that you give yourself permission to feel it without judgment.

29. **Mindfulness and Self-Awareness**. Mindfulness allows you to be present with the emotion and observe it without getting caught up in it. This practice helps you distance yourself from the emotion, reducing its power over you. Practice mindful breathing, paying attention to your breath as you inhale and exhale.

30. **Journaling.** Writing can be a therapeutic way to process your emotions. When you write down your feelings, you externalize them, which helps you gain perspective and release the emotional weight.

31. **Cognitive Reframing.** It is the practice of changing the way you perceive a situation in order to alter your emotional response to it. By viewing things from a different angle, you can reduce the intensity of negative emotions. Challenge the negative thought patterns that accompany your emotions. Ask yourself questions like, "Is this thought accurate?" or "Can I look at this from another perspective?"

32. **Engage in Physical Activity.** Exercise helps release **endorphins**, the brain's natural mood boosters, and can alleviate the physical tension that often accompanies negative emotions. Moving your body also helps you process lasting emotions and release pent-up energy.

33. **Practice Self-Compassion.** Negative emotions often stem from over self-criticism or harsh judgment. Being kind to yourself, especially when you're experiencing difficult emotions, helps reduce the intensity of those emotions and promotes healing. When you're struggling with negative emotions, treat yourself as you would a close friend. Offer words of encouragement, understanding, and reassurance.

34. **Letting Go Through Release Techniques.** Sometimes, the best way to release lingering negative emotions is through physical or emotional release techniques. This could be crying to release the pent-up feelings, using art to express how you feel (painting, dancing, writing a letter you never send) or even yelling into a pillow.

35. **Seek Support.** Talking about lingering emotions with someone you trust can provide clarity, relief, and support. Sharing your feelings with others can help you process them more effectively. Talk to someone you are close to or to your institution's ANO.

36. **Forgiveness.** Holding on to resentment or anger toward others or yourself can keep negative emotions lingering. **Forgiveness isn't about excusing bad behaviour but about freeing yourself from the grip of those emotions**. If someone hurt you, rather than ruminating on the injustice, remind yourself that forgiveness is a tool for **your healing**, not theirs.

DID YOU KNOW?

The 5-4-3-2-1 Grounding Technique to Reduce Stress

The 5-4-3-2-1 grounding technique is a simple yet powerful tool to help calm your mind, reduce stress and anxiety, and bring yourself back to the present moment. Here's how it works:-

The Technique

- **5**: Notice 5 things you can see around you, such as a chair, a book, or a picture.
- **4**: Acknowledge 4 things you can touch or feel, like your feet on the ground, the chair beneath you, or the air on your skin.
- **3**: Identify 3 things you can hear, such as a clock ticking, a bird chirping, or the hum of a machine.
- **2**: Recognise 2 things you can smell, like a scented candle, fresh air, or a fragrant flower.
- **1**: Focus on taking 1 deep breath and feeling the air flow in and out of your body.

Advantages

- Distracts from stressful thoughts: By focusing on your surroundings, you shift your attention away from worrisome thoughts.
- The approach decreases heart rate, blood pressure, and relaxes the nervous system.
- Increases mindfulness: You become more aware of your surroundings, encouraging you to live in the present moment.
- Reduces anxiety and panic: By grounding yourself in reality, you can reduce feelings of anxiety and panic.

Social Support

37. **Social Support**. Social support refers to the interconnectedness and positive relationships we have with others in our community and the world around us. It encompasses our ability to build and maintain healthy relationships, communicate effectively, show empathy and compassion, and contribute positively to society. Social Support is important for our physical and mental wellness since humans are social creatures who need connection and belonging. This dimension is important because the development of strong relationships is an essential part of achieving fulfilment in life.





38 **Key Aspects of Social Support**. Key aspects of social support are as follows:-



(a) **Quality of Relationships**. One key aspect of social support is the quality of our relationships with family, friends, colleagues, and acquaintances. Strong, supportive relationships offer belonging, emotional support, and a support network in times of need, that boosts our happiness and well-being.



(b) **Communication**. Clear and open communication helps development of understanding, trust, and mutual respect in relationships. It allows us to express our thoughts and feelings honestly, listen actively to others, and resolve conflicts in a constructive manner.

(c) **Empathy and Sympathy**. Empathy and sympathy are fundamental qualities that underpin social support. Empathy involves the ability to understand and share the feelings of others, while sympathy involves showing kindness, compassion and concern for others' well-being. These qualities enable us to connect with others on a deeper level, offer support and encouragement, and promotes a sense of unity and solidarity within our communities. By practicing empathy and sympathy in our interactions, we can create a caring society where individuals feel valued, understood, and supported.

39. **Positive Implications of Social Support.** Social support promotes positive changes in leading a good, healthy life and leads towards achieving fulfilment in life. Some of the positive implications of social support are as follows:-



a) **Strong Relationships**. Social support helps in maintaining strong family and friendship ties, rewarding interactions, and acceptance of diversity in society.

(b) **Good Communication**. Social support encourages communication, which is a component of conflict management and human interaction.

(c) **Social Interaction**. Qualities that demonstrate wellness in this dimension include improved sociability, strong relationships, participation in social activities, strong friendships, and excellent communication skills.

(d) **Improved Social Confidence**. People with optimal social support express themselves assertively, engage in community activities, create strong social networks, like fun, and are respectful.

(e) **Positive Influence on Physical and Mental Wellbeing**. Social support has a great impact on physical health. Social connectedness has direct implications on the functioning of the immune system, motor skill retention, and overall wellness. Social interactions facilitate the expulsion of stress and emotions.

40. **Ways to Improve Social Support**. As under:-

(a) **Make Social Connections**. Scientists are finding that social connections can affect our health positively. Social relationships with family, friends, and neighbours affect our mental well-being.

(b) **Get Active Together**. Exercise has various benefits. It boosts energy, happiness, and health. Friends, family, and other social relationships keep the motivation on to get active for improvement of health.

(c) **Build Healthy Relationships**. You may develop healthy connections with your parents/siblings/kids by being compassionate, responsive, consistent, and available. One must learn social skills to form and sustain connections. Relationships can be improved at any age. Knowing what makes a healthy relationship and how to maintain one is crucial.

(d) **Healthy Habits**. Cadets must develop healthy habits of reading, gaining knowledge through interactions, eating well, and consistent physical activities that last throughout their lives.

PART III: MEDITATION & MINDFULNESS

Meditation & Practice

41. **Meditation**. Meditation is a powerful concentration technique that helps calm your mind, relax your body, and focus your attention.

42. **Focused Breath-work and Meditation**. Breathing techniques like pranayama help regulate the body's energy by calming the nervous system and supporting healing for reducing stress.

43. **Benefits of meditation**. Benefits of meditation accrued are as under:-

(a) Physical Benefits.

(b) Emotional Benefits.

(c) Cognitive Benefits.

44. **Physical Benefits**. The physical benefits of meditation are as follows:

(a) **Reduces Stress and Anxiety**. Meditation helps calm the mind and body, reducing stress hormones like cortisol.

(b) **Improves Sleep**. Regular meditation can help students get better sleep, leading to improved physical and mental health.

(c) **Boosts Immune System**. Meditation has been shown to strengthen the immune system by reducing inflammation and increasing antibody production.

(d) **Reduces Chronic Pain**. Meditation can help reduce chronic pain by increasing the brain's pain tolerance and decreasing emotional reactivity.

45. **Emotional Benefits**. The emotional benefits of meditation are as follows:-

(a) **Increases Emotional Regulation**.Meditation improves students' emotional intelligence and well-being.

(b) **Enhances Self-Awareness**. Students gain self-awareness and understand their actions through regular meditation.

(c) **Develops Resilience**. Meditation helps students develop resilience, enabling them to better cope with challenges and setbacks.

(d) **Improves Relationships**. By increasing empathy and understanding, meditation can help students build stronger, more positive relationships.

46. **Cognitive Benefits**. The cognitive benefits of meditation are as follows:-

(a) **Improves Focus and Concentration**. Meditation improves attention and focus, enabling students to perform better academically and in co-curricular activities.

(b) **Enhances Problem-Solving Skills**. Regular meditation practice can improve problem-solving skills, creativity, and critical thinking.

(c) **Boosts Memory and Learning**. Meditation enhances memory, learning, and academic achievement.

(d) **Increases Grey Matter**. Meditation boosts grey matter in attention, emotion regulation, and memory areas.

47. **Practical Applications of Meditation**.

(a) **Before Major Events or Competitions**.

(i) **Pre-Training Meditation**. Practice meditation before training events/competitions to improve focus, concentration, and mental clarity.

(ii) **Event Preparation**. Meditate before events like drills, parades, or camps to reduce anxiety and boost confidence.

(b) **During Academic Studies**.

(i) **Study Breaks**. Take short meditation breaks during study sessions to refresh your mind and improve retention.

(ii) **Exam Preparation**. Practice meditation before exams to reduce stress, improve focus, and enhance performance.

(c) **For Personal Development**.

(i) **Daily Routine**. Practice meditation everyday to develop self-awareness, mental discipline, and emotional management.

(ii) **Goal Setting**. Meditate on your goals and aspirations to clarify your thoughts, boost motivation, and develop a positive mindset.

(d) **For Team Building and Leadership**.

(i) **Team Meditation**. Practice meditation with your team to improve communication, build trust, and enhance teamwork.

(ii) **Leadership Development**. Meditate on leadership qualities like discipline, responsibility, and decision-making to develop your leadership skills.

48. **Simple Meditation Techniques**. Some of the simple meditation techniques every student must practice and spread awareness of are as follows:

(a) **Deep Breathing**. Focus on slow, deep breathing to calm your mind and body. **You may like to concentrate on your breath, as it is an actual action always happening in the 'present'.** This can be done either with closed eyes or by keeping them partially open and concentrating on the tip of the nose.

(b) **Mindful Walking**. Pay attention to your walking, noticing the sensation of your feet touching the ground, the movement of your legs, and the rhythm of your breath.

ACTIVITY

- ANO to make cadets sit/stand and practise a 'mind-calming' drill. Suggested routine is as under:-
  - **Stand or sit with** spine straight, shoulders relaxed.
  - **Close your eyes** gently and bring your focus to your breath.
  - **Inhale slowly through the nose** (count 1...2...3).
  - **Hold** the breath gently (count 1...2).
  - **Exhale slowly through the mouth** (count 1...2...3...4).
  - With each breath, mentally say:
  - Inhale: *"I am focused."*
  - Exhale: *"I am calm."*
- Repeat this cycle for **5 deep breaths**, staying in the 'present' moment.

Mindfulness & Work Life Balance

49. **Mindfulness**. Mindfulness involves becoming aware of one's thoughts rather than focussing on nothingness. Seeing thoughts arise, dwell, and fade helps one accept and move on. Unlike putting thoughts away, we may see and comprehend others better when we are mindful.

50. Mindfulness involves being 'present' and not letting ideas distract us. The mind is wired to ponder, analyse, and figure things out, therefore, if left to its own devices, it will always seek new stimuli, thoughts, and ways to escape reality. Mindfulness gently restrains the mind to stay in the present moment.

51. **Beginning to Practice Mindfulness**. To move towards mindfulness, it is suggested to begin gradually as follows:-

(a) As a beginner, one should set some time aside when they feel relatively calm.

(b) One can do this at home or when out for a walk. When one is ready, they should begin by observing the present moment as it is, without judgement or desire to change what is.

(c) One should notice how their thoughts come and go and pull themselves back into the present if the mind starts to wander.

(d) One should not cast judgement on any thought that arises. Remember, one is not trying to "silence" the mind. Instead, when a thought arises, one should recognise it and let it go before returning to the present.

52. **Developing Mindfulness as a Habit**. The best part of mindfulness is its portability. Whether one misses a bus or is in the locker room before a major game, mindfulness can help. Finding a tranquil place to start helps one stay present. Once one is comfortable with mindful-thinking basics, one can slowly apply them wherever they are.



53. **Benefits of Mindfulness**. Mindfulness and its effects on mind and body are still being studied. However, some of the benefits of mindfulness are as follows:-

(a) **Mental Wellness**. Mindfulness has a positive impact on mental health. Stressful thoughts fluctuate in frequency. Practicing mindfulness may help to keep one in the present. Mindfulness helps one to recognise and observe thoughts and feelings without getting caught up in them and judging oneself for how one feels.

(b) **Improving Alertness**. Being attentive is noticing what's around, which makes one more alert and present in the moment. Once one stops worrying about the past and future, one can fully enjoy the present. It can help one in one's day-to-day life, work, relationships, and overall well-being

(c) **Concentration and Memory**. By helping one focus on the present, mindfulness can help to concentrate on tasks and remember the things going on around.

(d) **Sleep**. Mindfulness can help manage distracting thoughts that may stop one from sleeping and can assist one to physically relax. Mindfulness can help if one struggles to sleep or if one has anxiety about falling asleep.

(e) **Healthy Eating**. Mindful eating can help one focus on their eating experiences. This can help improve one's relationship with food.

(f) **Relationships**. Being mindful can also help one in relationships. In busy live, its easy to not notice the different perspectives of those around. Being more mindful may help one understand these issues and connect differently with people.

(g) **Reduce Stress and Anxiety**. Mindfulness helps one to separate one's thoughts and how one reacts to them. This puts one in the best place to manage emotions calmly and effectively. This can help reduce stress and anxiety.

(h) **Sense of Happiness**. Mindfulness can also help one to notice and engage with things around them that they may take for granted. This can help one to feel happy.

(j) **Managing Mental Health Conditions**. Practicing mindfulness can also help one notice warning signs of mental health challenges. This way, one can take care of oneself and seek help earlier if needed. Mindfulness can also help manage mental health conditions like depression, anxiety disorders, addiction, eating disorders, etc. It may also help with anger management.

Work-Life Balance

54. **Work-Life Balance**. Balance between work and life is crucial. It reduces work stress and maintains mental and physical health. Social, personal, and professional lives should be valued equally. Work-life balance means balancing work and other aspects of life in a positive way.

55. **Importance of Maintaining Work-life Balance**. Maintaining work-life balance is crucial for stress-free living. First, it boosts work productivity. A healthy work-life balance helps you focus and get more done faster. Second, it prevents burnout which is generally a consequence of chronic stress.

56. **Ways to Maintain Work-Life Balance**. Work-life balance can be achieved in different ways. Here are some tips:-

(a) **Make Time for Yourself**. Make sure to set aside some time each day for oneself. This is time that you can use to do things that you enjoy and that make you happy. This can be something as simple as reading a book, taking a walk, or taking a yoga class.

(b) **Set Boundaries**. It is important to set boundaries between work and life. This means making sure that one is not working all the time and that they are taking time for themselves and their loved ones.

(c) **Delegate**. If one is feeling overwhelmed by work, try to delegate some of one's tasks to other people. This will help one to focus on the most important tasks and free up some time for oneself.

(d) **Take breaks.** Breaks are essential at work. These help prevent burnout and rejuvenate, preparing one to tackle the next assignment.

(e) **Learn to Say No**. If you are overworked, politely decline further jobs or projects. This helps preserve work-life balance and prevent burnout.

(f) **Finding Support.** Create a network of friends, family, and mentors who understand your aspirations and obstacles.

57. **Advantages of a Healthy Work-life Balance**. Students and cadets often struggle to balance academic life, miscellaneous commitments, and personal life. They need a healthy work-life balance for success and well-being. The advantages of maintaining good work-life balance are as follows: -

(a) **Motivation**. Cadets are more motivated to perform their duties since they do not get overworked. This is advantageous to both institution/family and cadets.

(b) **Reduced Stress**. Understanding what matters most to one to allocate time to studies, work, and activities that align with one's goals and values. Learning to prioritise effectively can significantly reduce stress and improve productivity.

(c) **Effective Time Management**. Utilise planners, apps, or calendars to schedule one's days and weeks. Break down tasks into manageable chunks, set realistic deadlines, and stick to them. Time management is key to avoiding last-minute rushes and maintaining quality in one's work.

(d) **Optimum Performance**. Taking care of your mental and physical health boosts your energy levels and concentration, which lets you perform optimally.

(g) **Finding Support**. Sharing experiences and recommendations can provide innovative work-life balance solutions.

(h) **Reflection and Adjustment**. Being self aware and flexible lets you adjust to changing priorities and duties without sacrificing your health.

58. Finding a student work-life balance is difficult but possible with the right tools. You can succeed both academically and personally by defining priorities, managing time, caring for yourself, setting boundaries, leveraging resources, finding support, and being adaptable.

PART IV : POSITIVE THINKING

59. **The Power of Positive Thinking.** Positive thinking is more than just maintaining a cheerful attitude; it is a mindset that influences mental, emotional, and even physical well-being. By fostering an optimistic outlook, individuals can improve resilience, reduce stress, and enhance overall life satisfaction.

Importance of Positive Thinking

60. A positive mindset can transform the way we approach challenges and setbacks. **When we approach stressful situations with a positive mindset, we are less likely to become overwhelmed by them**. Positive thinking encourages us to see challenges as opportunities for growth or learning, as opposed to a heavy burden, causing stress. Gains from cultivating positive thinking are given in succeeding paragraphs.

61. **Enhances Mental Health.**

(a) Reduces stress and anxiety

(b) Lowers the risk of depression

(c) Improves emotional regulation and resilience

62. **Boosts Physical Well-being.**

(a) Strengthens the immune system

(b) Lowers blood pressure and reduces the risk of heart disease

(c) Promotes better sleep and overall energy levels

63. **Improves Relationships and Social Life.**

(a) Encourages healthier communication and emotional intelligence

(b) Fosters stronger personal and professional relationships

(c) Helps in conflict resolution and emotional support

64. **Increases Productivity and Success.**

(a) Enhances problem-solving skills

(b) Boosts motivation and perseverance

(c) Helps in achieving personal and professional goals

How to Cultivate Positive Thinking

65. Developing a positive mindset takes practice and intentional effort. Here are some effective strategies.

66. **Practice Gratitude.**

(a) Keep a gratitude journal to note daily positive experiences

(b) Express appreciation to others regularly

(c) Focus on what you have rather than what you lack

67. **Reframe Negative Thoughts.**

(a) Challenge negative self-talk with constructive alternatives

(b) Look for lessons in setbacks rather than dwelling on failure

(c) Shift focus from problems to potential solutions

68. **Surround Yourself with Positivity.**

(a) Engage with optimistic and supportive people

(b) Consume uplifting books, music, and media

(c) Minimize exposure to negativity and pessimism

69. **Practice Mindfulness and Disciplined Self-Care.**

(a) Engage in meditation or deep breathing exercises

(b) Prioritize sleep, healthy eating, and physical activity

(c) Take breaks and manage stress through relaxation techniques

70. **Set Realistic Goals and Celebrate Progress.**

(a) Break large goals into smaller, achievable steps

(b) Recognize and celebrate personal achievements

(c) Stay patient and acknowledge growth over time

71. Positive thinking **cultivates** an optimistic mindset and lays the foundation for deeper mental practices. Building on this base, affirmations serve as an equally important and more focused tool, offering a conscious way to reshape thoughts and effectively manage stress.

PART V : AFFIRMATIONS-THE POWER WITHIN

Affirmations



72. Affirmations are uplifting and empowering phrases that individuals repeat to themselves to foster self-confidence, reinforce positive thinking, and develop a constructive mindset. **Affirmations are centred on the idea that one's thoughts and beliefs create one's reality**. The concept of Affirmations is also frequently referred as the **'Law of Attraction'**.

73. **Gist of the Law of Attraction.** Our experiences, and the resulting feelings from them, lead to the building of new neural pathways by the brain, a process known as **neuroplasticity.** What Affirmations try to achieve is reverse the cycle, by first forcing the brain to create the neural pathways by 'believing and feeling' that a desired event has already happened, in order to 'attract' that event to us.

Scientific Explanation of Affirmations/Law of Attraction

74. It is believed that we are based in a Quantum Field, an 'invisible field of intelligent energy/infinite possibilities', its around us, its inside us, its everywhere. We are always connected with it, though it cannot be measured or quantified in conventional terms. **The language of human emotions, thoughts, beliefs and feelings is what allows us to communicate with the 'field'.**

75. **The Quantum Field and Consciousness**. In quantum theory, everything is made up of **energy and vibration**—including thoughts and feelings. Though it is not a 'settled science' yet, it is believed that in the quantum field, all potential realities exist. So for each of our 'failed' endeavour, a version of reality exists where it's already fulfilled. For every negative outcome, a positive one also exists at the same time. Whenever we set an 'intention' firmly, we are aligning with and 'attracting' those positive possibilities, among many potential outcomes.

76. Our heart is more than just an organ—it's a powerful energy center, said to be far stronger (100 times electrically and 5000 times magnetically) than the brain. Though emotions are processed in the brain, we often *feel* them in the heart due to its strong link to physical sensations like a racing pulse or chest tightness. The heart's **electromagnetic signal, more powerful than the brain's,** can influence the body and potentially interact with the environment in subtle ways. Emotions like gratitude, love, or joy **create coherent heart rhythms**, which studies show, puts us "in sync" with desired outcomes.

77. **Feeling, as the union of thought and emotion, carries the strongest vibration**. While the brain imagines and visualizes, it's the heart that brings these visions to life through feelings. True shift in our reality comes from '*feeling from the outcome*' as if has already happened, and being genuinely grateful for it. This is what sends powerful waves into the universal field and 'draws/attracts' the outcome to us.

How to Wish/Affirm

78. **Feeling is the Secret.** The emotional charge behind an affirmation is what gives it true power. Without genuine feeling, thinking or repetition alone has little effect. Feeling is the fuel that activates transformation. **The universe responds not to what we say we want, but to what we're energetically aligned with**.

79. **Law of Assumption**. According to the **Law of Assumption**, embodying the feeling of your desire already fulfilled attracts its manifestation. It means that you should mentally and emotionally live as though your desired outcome has already come true. **Be enveloped in that 'assumed reality'**, with every little detail thrown in.

80. **Focus Area**. Try to feel and see 'from' the desired outcome, not of the obstacles in achieving it. Like a martial artist aiming to break a set of concrete blocks is not thinking about his hand hitting the bricks, because he knows it's going to hurt. Instead, he is focused on what happens after his hand has passed through the block. **He focusses on a place at the bottom of the block, as if it has already happened and his hand is already in that place.**

Why do Affirmations Generally Fail to Manifest

81**.** Affirmations can be powerful, but they often fall short for a few key reasons. Few reasons why they usually fail to manifest are given in succeeding paragraphs.

82. **Lack of Belief**. If affirmations don't match your true feelings, your subconscious resists. Inner doubt overrides empty words.

83. **Too Vague or Unrealistic**. Affirmations like "I attract abundance" are ineffective without clarity. Use specific, measurable, and realistic goals (think SMART).

84. **No Action Taken**. Affirmations set the mindset, but results need action. Saying "I am fit" means little without healthy habits.

85. **No Emotion Involved**. Words without genuine feeling don't impact the brain. Emotion makes affirmations effective.

86. **Coming from Lack**. Affirming, "I am rich", while feeling poor, reinforces scarcity. Start from a mindset of abundance.

87. **Mismatch with Identity**. If your self-image contradicts your affirmation, change won't stick. Align affirmations with a shifted identity, not just words.

Origin of Potent Affirmations and Examples

88. **The Need to 'Let Go'**. The world will mirror back what we are feeling deep in our hearts. So if its anger, hate, jealous, rage etc., we're going to see them in our body, in our health, in our relationships. The world mirrors our inner emotional state. Similarly, feelings of peace and healing attract positive outcomes. Gratitude and forgiveness are essential to move forward—they shift our internal chemistry and help us transcend past hurt. Ultimately, our outer reality reflects who we've become from inside. The more de-cluttered and 'filtered' we feel from within, the stronger our connection with the 'field of creation' becomes.

89. **Gratitude for Fulfillment**. Seeking a true wish is like it's giving thanks as if the desire has already been fulfilled. Begin with gratitude for small things already there in your life, then shift to feeling grateful for one's desired goal having manifested. The individual must be honest, ego-free, and fully immersed in the reality of the wish fulfilled. This heartfelt alignment is the true language of effective 'ask'.

90. **Example.** So for a student wanting to excel in studies, affirming that:-

(a) "*I am confidently studying 3 hours every day and am confident of completing all my syllabus in time. I retain concepts easily and am fully prepared to score above 85% in my final exams.*" is fine.

(b) However, better would be to 'feel the desired reality' from the end-state. Such an affirmation could be, ***"Having mastered my syllabus with ease and clarity, I consistently studied 3 focused hours a day, and it has paid off — I have scored 90% in my exams! My family, my teachers and my friends are feeling proud of me. I am feeling extremely thankful, motivated, confident, and completely aligned with my highest potential."***

How do Affirmations Help in Stress Management

91. **Creating a Sense of Control.** In being able to 'attract' a desired outcome, positive Affirmations place individuals in a position of initiative, and promote a shift in mindset, from being 'a victim of circumstances', to someone with a sense of fair degree of control over one's life, fostering a sense of empowerment.

92. **Shifting from Negative to Positive Thinking.** Many individuals experience negative self-talk. Affirmations guide the mind to shift focus from problems to possibilities, encouraging a solution-oriented mindset.

93. **Building a Growth Mindset.** Affirmations that emphasize growth help individuals view challenges as opportunities to learn and evolve. This mindset reduces the sense of being overwhelmed and promotes adaptability.

94. **Reducing Negative Self-Talk.** Stress often intensifies inner criticism. Affirmations counter this by fostering self-compassion, thereby easing emotional distress and building the confidence needed to face adversity.

95. **Enhancing Relaxation.** Affirmations centered on peace, calm, and present-moment awareness can activate the body's relaxation response. This helps alleviate physical symptoms of stress such as muscle tension and shallow breathing.

96. **Boosting Self-Esteem.** By reinforcing a sense of self-worth, affirmations empower individuals to believe in their ability to handle challenges. This internal strength provides a buffer against stress.

97. **Mindfulness and Focus.** The repetitive nature of affirmations helps anchor attention in the present. This mindfulness practice prevents the mind from drifting into worry, reducing overall stress levels.

98. **Positive Emotion Regulation.** Affirmations enhance emotional resilience, and promoting a sense of calm and inner control. It encourages positive thinking, which stimulates the brain's release of dopamine and serotonin. These "feel-good" chemicals help uplift mood and diminish feelings of anxiety or stress.

99. **Stress Reduction.** Through consistent practice, affirmations can help lower cortisol—the stress hormone—by activating the parasympathetic nervous system, which promotes relaxation and counters the fight-or-flight response.

100. **Self-Perception and Self-Esteem.** Affirming personal strengths and desired outcomes reshapes one's self-image. This enhanced perception leads to more confident behavior and a proactive approach to life's challenges.

CONCLUSION

101. **Mindfulness** helps us feel better by changing how we think, feel, and see the world. It supports a healthy balance between work and personal life, letting us enjoy time with family, friends, and hobbies. In today's busy world, practicing **meditation daily** can help us stay calm and emotionally balanced.

102. **Managing stress** is important for staying mentally healthy. Some helpful ways include **relaxation**, **time management**, **meditation**, and **positive thinking**.

103. Using **positive affirmations**, repeating encouraging thoughts, can boost confidence and help us stay focused and calm during tough times. They build emotional strength and help us handle stress more clearly and positively.

SUMMARY

- Stress is a natural response; management is key. Stress is a normal response to feeling overwhelmed. Eustress is positive stress.
- Stress Management Techniques include reframing, positive self-talk, exercise, deep breathing, muscle relaxation, prioritizing tasks, and social connection.
- Thoughts are mental assessments, emotions are raw, immediate reactions to stimuli, and feelings are the conscious experience and interpretation of those emotions.
- Negative feelings or, lingering negative emotions in other words, can often be released by acknowledging them and processing them in healthy ways.
- Spiritual practices can help manage stress. Spirituality can provide purpose, connection, growth, and peace.
- Social support involves positive relationships and community connection.
- Meditation calms the mind and relaxes the body. Its benefits include reduced stress, emotional wellness and improved cognitive processes.
- Mindfulness is awareness of thoughts without judgment. It involves being present and not distracted by thoughts.
- Work-Life Balance is important for reducing stress and maintaining health. Making time for oneself, setting boundaries, delegating, taking breaks, saying no, and finding support can help attain it.
- Managing stress, pursuing wellness, practicing mindfulness, and maintaining work-life balance are essential for well-being.
- Affirmations are an effective counter-stress tool. Regular practice of affirmations can reprogram mental patterns, enhance self-worth, and maintain focus on personal objectives.
- Affirmations contribute to the building of new neural pathways, a process known as neuroplasticity, which is the brain's ability to reorganize itself by forming new neural connections in response to learning, experience, or repeated behaviours.
- Affirmations play a crucial role in building a positive mindset, supporting transformation, and promoting overall well-being.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the term for positive stress that can be beneficial?

(a) Distress (b) Acute stress

(c) Eustress (d) Chronic stress

Q2. Which type of stress is long-term and results from accumulated factors like financial difficulties or relationship issues?

(a) Acute stress (b) Episodic acute stress

(c) Chronic stress (d) Physical stress

Q3. Which statement best reflects the emotional mechanism behind successful affirmations?

(a) The more complex the affirmation, the better the result

(b) Only affirmations spoken aloud work effectively

(c) The emotional charge behind the affirmation determines its power

(d) Logical reasoning must accompany every affirmation

Q4. Which of the following is NOT a stress emotion?

(a) Anxiety (b) Irritability

(c) Happiness (d) Frustration

Q5. Which of the following is considered a spiritual practice?

(a) Physical exercise (b) Mindfulness

(c) Eating healthy (d) Playing video games

Q6. Which type of spirituality is associated with organized religions and their doctrines?

(a) Mystical spirituality (b) Religious spirituality

(c) Nature-based spirituality (d) Humanistic spirituality

Q7. What does social wellness refer to?

(a) Physical health

(b) Positive relationships and community

(c) Financial stability

(d) Career success

Q8. Which of the following is a key aspect of social wellness?

(a) Competitive behavior (b) Clear communication

(c) Emotional detachment (d) Independence

Q9. How do affirmations help in stress management beyond cognitive restructuring?

(a) They function mainly as a placebo to distract from stressors

(b) They lower blood sugar levels and reduce digestive issues

(c) They trigger the parasympathetic nervous system, reducing cortisol, inducing calm

(d) They mimic the effects of physical exercise on the brain

Q10. Which of the following is a simple meditation technique?

(a) Running (b) Deep breathing

(c) Weightlifting (d) Playing video games

Q11. What does mindfulness involve?

(a) Focusing on the past

(b) Ignoring one's thoughts

(c) Being aware of one's thoughts without judgment

(d) Suppressing emotions

Q12. Which of the following is a benefit of taking breaks?

(a) Decreased focus (b) Increased stress

(c) Boosted productivity (d) Reduced brain activity

Q13. What is a key aspect of mindfulness?

(a) Focusing on the past (b) Controlling thoughts

(c) Being present (d) Avoiding emotions

Q14. Which of the following best captures the principle behind the “Law of Assumption” in the context of affirmations?

(a) You must repeat your desire 108 times daily to attract it

(b) Manifestation occurs when you detach completely from desire

(c) Feeling and visualizing a goal as already achieved aligns you with its reality

(d) Speaking affirmations out loud is more powerful than writing them

Q15. What does practicing empathy involve?

(a) Showing compassion

(b) Understanding and sharing the feelings of others

(c) Showing indifference

(d) Social discussions and arguments

Short Answer Questions

Q1. List the main types of stress that students may experience?

Q2. Briefly explain three techniques to identify your feelings.

Q3. Name and briefly explain three physical benefits of meditation.

Q4. Differentiate spirituality from religion.

Q5. Give any three ways to improve social wellness.

Long Answer Questions

Q1. Describe the physiology of stress, explain the difference between eustress and distress, and discuss the various types of stress.

Q2. Explain the importance of managing stress emotions, discuss techniques for doing so, and outline how to develop emotional resilience.

Q3. Discuss the concept of spirituality, differentiate between spirituality and religion, and explain the importance of spirituality, detailing its various categories.

Q4. Explain the concept of mindfulness, discuss its benefits for mental wellness, and describe techniques for practicing mindfulness.

Q5. Discuss the importance of work-life balance, especially for NCC cadets, and describe effective strategies for achieving and maintaining work-life balance, including its advantages.

PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW VIII : DIET AND NUTRITION

"***Your body is a reflection of what you consume.***"



TEACHING INSTRUCTIONS

Period : **One (01)**

Type : **Lecture and Practice**

Year : **2nd Year SD/SW**

Conducting Officer : **Associate NCC Officer**

Training Aids : **Script/ Lesson Plan; Presentation on Computer Charts/Posters on Diet and Nutrition.**

Time Plan

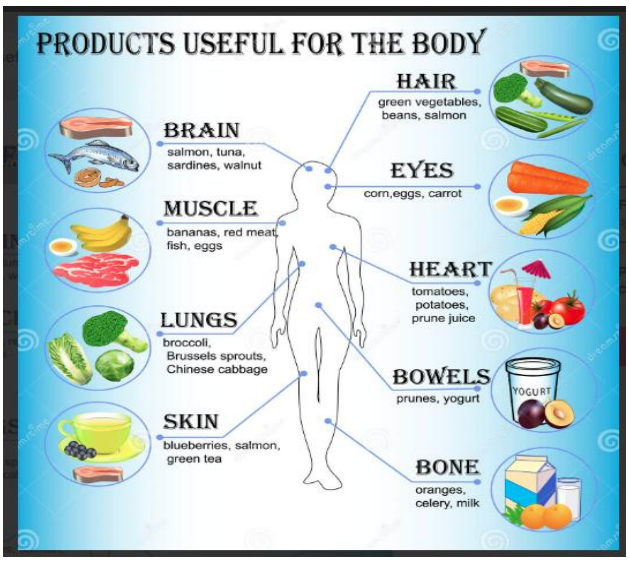
- Introduction : 02 Mins
- Part I : 17 Mins
- Part ll : 18 Mins
- Conclusion : 03 Mins



INTRODUCTION

1. Diet and nutrition play an essential role in our overall health and well-being. Nutrition refers to the process by which the body absorbs and utilizes the nutrients found in food. Diet, on the other hand, is the type and quantity of food we consume daily. Together, these elements work to fuel our body, support growth, repair cells, and maintain various bodily functions.

2. A balanced diet supplies your body with the essential nutrients it needs to operate efficiently. Without proper nutrition, the body becomes more vulnerable to illness, fatigue, infection, and decreased performance. In this chapter, we will delve into the significance of a balanced diet, its key components, and the ways it benefits overall health.



LEARNING OBJECTIVES

- Understand the importance of nutrition and balanced diet.
- Understanding functions of macronutrients and micronutrients
- Develop healthy eating habits.
- Learn to incorporate meal planning and the adoption of healthy cooking methods

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I : Diet and Nutrition

(b) Part II : Overview Of Dietary Guidelines

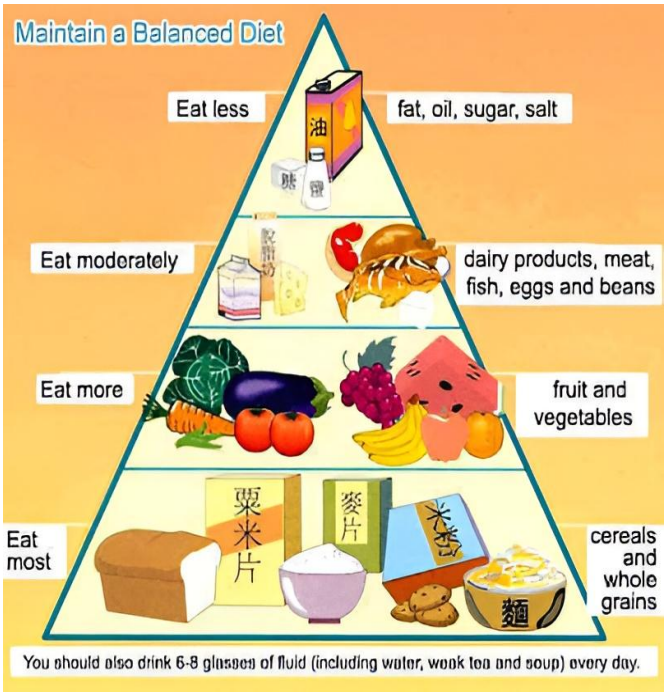
INTERESTING FACTS

- Eating a variety of colorful fruits and vegetables provides essential vitamins, minerals, and antioxidants. Aim for 5 servings a day to support overall health and well-being.
- Drinking enough water can boost your metabolism and increase fat-burning by up to 30%.

PART I : DIET AND NUTRITION

3. **Diet**. A diet consists of the food and drinks we consume daily. It can also refer to a specific eating plan or regimen designed for health, weight management, or medical reasons. It includes a variety of foods that provide essential nutrients—such as carbohydrates, proteins, fats, vitamins, and minerals—to support overall health and well-being.

4. **Composition of Food**. Food consists of both energy-providing and non-energy components. The energy is delivered in the form of calories, which are essential for the proper functioning of our body cells. The energy components of food include:-



(a) **Macronutrients** include:-

(i) **Carbohydrates**. The main source of energy for the body. They are broken down into glucose, which fuels our cells, tissues, and organs.

(ii) **Protein**. The 'building blocks'**,** crucial for growth, repair, and maintenance of body tissues, as well as for immune function and enzyme production.

(iii) **Fats.** Fats provide energy, support cell structure, aid in the absorption of fat-soluble vitamins (A, D, E, and K), protect organs, and help produce essential hormones.

(b) **Micronutrients** include:-

(i) **Vitamins.** These are essential for various biochemical processes such as immunity, energy production, and cell function.

(ii) **Minerals.** Minerals support key body functions such as bone health, nerve transmission, and muscle function.

(iii) **Dietary Fiber.** Aids in digestion by promoting regular bowel movements and maintaining intestinal health.

(c) **Water**. Water is neither a macro nor a micro nutrient in the classical sense. It is vital for hydration, digestion, nutrient transport, temperature regulation, and waste elimination. Water doesn't provide energy (calories) like the others, but it's **essential for life and needed in large quantities(3-4 Litres), which is why it is usually grouped with macro nutrients.**

5. **Nutrition.** It is the process by which the body takes in, absorbs, and utilizes the nutrients from food to maintain health, growth, and energy. It involves the intake of essential substances like carbohydrates, proteins, fats, vitamins, minerals, and water, which support various bodily functions, promote cell repair, and help prevent diseases. Proper nutrition is crucial for overall well-being, physical and mental performance, and disease prevention.

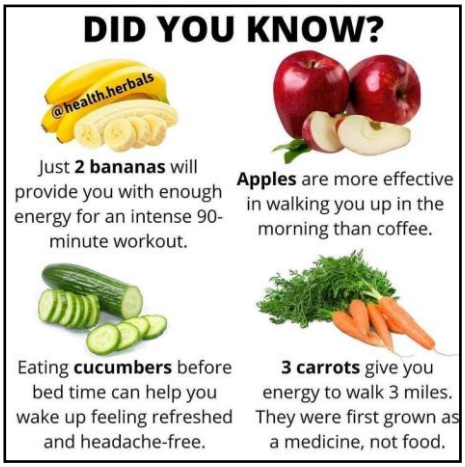
6. **Balanced Diet.** A balanced diet is one that includes a variety of foods from all the food groups in the right proportions to provide the necessary nutrients including carbohydrates, proteins, fats, vitamins, minerals, and water which are needed for the body to function effectively.

7. **Significance of Maintaining A Well-Balanced Diet**.

(a) **Provides Essential Nutrients**. Ensures the body receives the right balance of carbohydrates, proteins, fats, vitamins, minerals, and water to function properly.

(b) **Supports Immune Health.** Strengthens the immune system, helping the body fight off infections and illnesses.

(c) **Prevents Chronic Diseases**. Reduces the risk of developing conditions like heart disease, diabetes, obesity, and high blood pressure.



(d) **Promotes Growth and Repair.** Supports the growth, repair, and maintenance of cells, tissues, and organs in the body.

(e) **Boosts Energy Levels**. Helps maintain stable energy throughout the day, preventing fatigue and enhancing physical and mental performance.

(f) **Improves Mental Health**. Contributes to better brain function, mood regulation, and cognitive abilities.

(g) **Aids in Weight Management**. Helps in maintaining a healthy weight by regulating calorie intake and supporting metabolic processes.

(h) **Promotes Longevity**. A well-balanced diet can increase life expectancy and improve overall quality of life.

8. **Components of a Balanced Diet**. The various types, functions, and sources of components of a balanced diet are as given: -

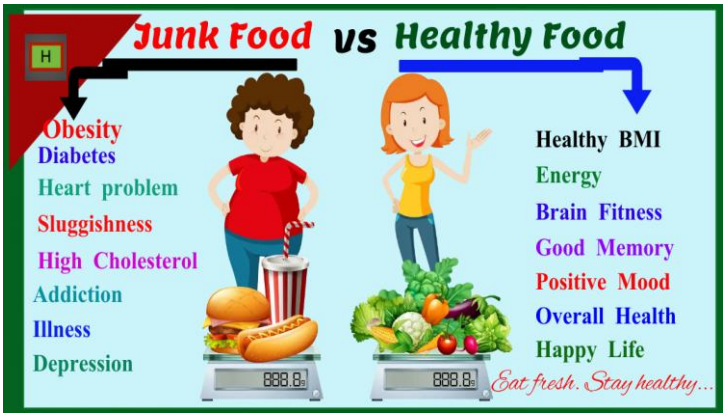
Component	Function	Types of Component	Sources
Carbohydrates	Provide energy for the body. Spare proteins from being used for energy, allowing them to be used for growth and repair. Aid in the functioning of the brain and nervous system.	**Simple Carbohydrates** (sugars that provide quick energy)	Fruits (e.g., bananas, apples), dairy (e.g., milk, yogurt), honey, and table sugar.
		Complex Carbohydrates (starches that provide sustained energy)	Whole grains (e.g., brown rice, oats, quinoa), vegetables (e.g., potatoes, sweet potatoes), legumes

			(e.g., beans, lentils), and whole-wheat bread.
Proteins	Build and repair tissues. Produce enzymes and hormones. Support immune function and muscle building. Provide a secondary source of energy.	**Complete Proteins.** These contain all nine essential amino acids (usually animal-based sources).	Meat, fish, poultry, eggs, dairy products.
		Incomplete Proteins. These lack one or more essential amino acids (usually plant-based sources).	Legumes (e.g., beans, lentils), nuts, seeds, tofu, quinoa, and whole grains.
Fats	• Provide long-term energy storage. • Protect organs and support cell structure. • Help absorb fat-soluble vitamins (A, D, E, and K). • Regulate hormones and help with brain function.	**Unsaturated Fats.** (Healthy fats that improve heart health and support cell function)	Olive oil, avocados, nuts, Fatty fish, flaxseeds, walnuts.
		Saturated Fats. (Found in animal products and some plant oils. They can raise cholesterol levels if consumed in excess.)	Butter, fatty cuts of meat, coconut oil.
		Trans Fats. These are artificially produced fats and are considered unhealthy.	Processed foods, margarine, baked goods.
Vitamins	Vitamins are essential for various metabolic processes. They support immune function, skin health, vision, and overall growth and development.	Vitamin A	Carrots, sweet potatoes, spinach, kale.
		Vitamin C	Citrus fruits (oranges, lemons), strawberries, bell peppers, tomatoes.
		Vitamin D	Sunlight, fortified dairy, fatty fish.
		Vitamin E	Nuts, seeds, spinach, broccoli.
		Vitamin K	Leafy greens (e.g., kale, spinach), broccoli, Brussels sprouts.
Minerals	Minerals are important for maintaining bone health, nerve function, fluid balance, muscle function, and enzyme activity.	Calcium	Dairy products (e.g., milk, cheese, yogurt), leafy greens (e.g., kale, bok choy), fortified plant-based milks.

		Iron	Red meat, poultry, seafood, legumes, spinach, fortified cereals.
		Potassium	Bananas, potatoes, spinach, tomatoes, oranges.
		Magnesium	Nuts, seeds, leafy greens, whole grains.
Fiber	Fiber is important for digestive health, regulating bowel movements and lowering cholesterol levels	-	Whole grains, fruits, vegetables, legumes and seeds.
Water	Water is essential for hydration, temperature regulation, nutrient transport, and waste removal.	-	Drinking water, fruits (e.g., watermelon, oranges), vegetables (e.g., cucumbers, lettuce), and other fluids like herbal teas.

PART II : OVERVIEW OF DIETARY GUIDELINES AND MEAL PLANNING

9. **Overview of Dietary Guidelines and Strategies for Balanced Meal Planning.** Dietary guidelines are designed to promote overall health, prevent chronic diseases, and maintain a healthy weight. They provide advice on the types and amounts of foods to eat for maintaining a balanced, nutrient-rich diet. Applying these guidelines through meal planning can help ensure that you're meeting your nutritional needs. Below is an overview of the key dietary guidelines and strategies to incorporate them for balanced meal planning. To achieve balanced meal for the body, following guidelines must be adhered to:-



(a) **Eat a Variety of Foods.** Include a variety of Foods from all food groups to ensure you get all the necessary nutrients.

(b) **Hydrate Adequately**. Drink plenty of water throughout the day to stay hydrated. Include electrolyte-rich foods like bananas (potassium), dates (potassium), and coconut water (electrolytes) to help maintain electrolyte balance.

(c) **Limit Processed and Junk Foods**. Try to limit your intake of processed and junk foods that are high in sugar, salt, and unhealthy fats.

10. **Meal Frequency and Timings**.

(a) **Space Out Meals**. Space out your meals to maintain energy levels and support muscle growth and repair.

(b) **Include a Balanced Breakfast**. Start your day with a balanced breakfast that includes complex carbohydrates, lean protein, and healthy fats.

(c) Post-workout meal**. Consume a balanced meal with complex carbohydrates, lean protein, and healthy fats within 30-60 minutes after physical training.**

11. **Practical Tips for Maintaining a Balanced Diet**.

(a) **Portion Control.** Pay attention to portion sizes to avoid overeating.

(b) **Whole Foods.** Focus on whole, unprocessed foods as they are more filling and nutritious.

(c) **Regular Meals.** Eat at regular intervals to maintain steady blood sugar levels and avoid binge eating.

(d) **Hydration.** Drink plenty of water throughout the day to help control hunger and stay hydrated.

(e) **Plan Meals.** One should take time to plan meals for the week to including a variety of nutrients in the diet.

(f) **Read Labels.** When shopping, food labels should be read to make informed choices about what one is consuming.

(g) **Cook at Home.** Meals should be prepared at home, where one can control the ingredients and portion sizes.

(h) **Limit Processed Foods.** Minimize the intake of processed foods, which often contain high levels of salt, sugar and unhealthy fats.

12. **Foods to Focus on**. In a nutritious diet the foods to focus on are:-

(a) **Whole grains**. Brown rice, whole wheat bread and oats.

(b) **Lean proteins**. Chicken, fish, beans and lentils.

(c) **Fresh fruits and vegetables**. Aim for 5 servings per day.

DID YOU KNOW?

- Our body can store carbohydrates as glycogen for later use, providing a quick energy source during physical activity

(d) **Low-fat dairy**. Milk, yogurt and cheese.

(e) **Healthy fats**. Nuts, seeds, avocados and olive oil.

13. **Foods to Limit**. Must limit consumption of following food items:-

(a) **Sugary Drinks**. Soda, sports drinks and energy drinks.

(b) **Fast Food and Processed Snacks**. Limit or avoid foods high in salt, sugar and unhealthy fats.

(c) **Saturated and trans fats**. Limit or avoid foods high in saturated and trans fats such as butter, lard and partially hydrogenated oils.

14. **Food Label**. Food label offers critical information about the nutritional composition of the food and is designed to help consumers make better choices.

15. **Reading and Interpreting Food Labels**. The skill of reading food labels is crucial for making informed dietary decisions. By understanding the serving sizes, calorie content, nutrient breakdown, and ingredients list, you can more effectively assess the healthiness of a product. With time and practice, interpreting food labels becomes simpler, allowing you to take charge of your nutrition and make choices that support your health goals. Packaged foods include a standardized “Nutrition Facts" label to help guide these decisions.



16. On a food label, the undermentioned components are found and each of these components provides insight into the healthiness of the food one is consuming:-

(a) Serving Size

(b) Calories

(c) Nutrients (Fats, Carbohydrates, Proteins, etc.)

(d) Percent Daily Values (%DV)

(e) Ingredients List

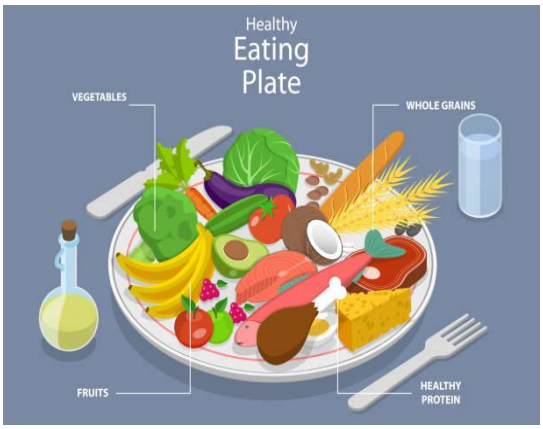
17. The information details found on a food label are as tabulated.

Ser No	Component	What It Means	Why It Matters	Tip
(a)	**Serving Size. The Foundation of Understanding Nutrition**	The serving size is the amount of food that the nutritional information refers	If you eat more than the listed serving size, you’ll consume more calories and nutrients than indicated on the label. E.g.	Always check the serving size first to gauge whether your portion aligns

		to. The listed serving size may not be the same as what you consider a typical portion.	if a bag of chips lists a serving size of 1 ounce but you eat 3 ounces, you'll need to multiply all the values on the label by three.	with the listed amount.
(b)	**Calories. Energy and Weight Control**	Calories represent the energy you get from the food. Understanding your calorie intake is key to managing your weight and energy levels.	Excessive calorie consumption can lead to weight gain, while inadequate calories can result in fatigue and malnutrition. Caloric needs vary based on factors such as age, gender, and physical activity level. A typical adult requires about 2,000 to 2,500 calories per day, but individual needs may differ.	Pay attention to "calories per serving," not just the total calorie count, to ensure you're not consuming more than intended.
(c)	**Nutrients.** Balancing Fats, Carbohydrates and Protein			
	Fats	Total fat includes all the fat in one serving, broken down into Saturated Fat and Trans Fat.		
	Saturated Fat	Fat found in animal products and some processed foods.	High levels of saturated fat can increase cholesterol levels, raising the risk of heart disease.	Check the ingredients list for "partially hydrogenated oils" to avoid hidden trans fats.
	Trans Fat	Often listed as "partially hydrogenated oils,"	These are artificial fats that can lead to an increased risk of heart disease. Even small amounts are harmful, so it's best to avoid them completely.	
	Carbohydrates	Total carbohydrates include sugars, fiber and starches.	Focus on foods high in dietary fiber and low in added sugars.	Look for "added sugars" on the label and aim to limit foods with more than 10% of your daily sugar intake in one serving.
	Protein	Watch for processed foods that add isolated	Protein is essential for building and repairing	The Nutrition Facts label will show the

		proteins like soy or pea protein, which are not as healthy as whole food sources.	tissues.	amount of protein in grams (g) per serving.
(d)	**Percent Daily Values (%DV). A Quick Reference for Nutrient Needs**	The % Daily Value (DV) indicates how much of a particular nutrient one serving of the food contributes to the total daily intake, based on a 2,000-calorie diet.	The %DV helps you determine whether a food is high or low in a particular nutrient. For example, 5% or less of a nutrient is considered low, while 20% or more is considered high. This can guide you toward or away from foods depending on your dietary goals.	Use the %DV to ensure you're not exceeding the recommended intake for nutrients like sodium, saturated fat, and added sugars.
(e)	**Ingredients List. Decoding What's Inside**	The ingredients list provides a detailed breakdown of what's in the food, listed in order of quantity by weight. The first few ingredients are the most abundant in the product, so they are the most important to pay attention to.	Many processed foods contain additives, preservatives, and artificial ingredients that may not be immediately obvious from the front label. A long ingredients list with many unrecognizable names can indicate a heavily processed food with fewer whole, natural ingredients.	Focus on foods with shorter ingredients lists that contain whole foods and avoid products with a lot of artificial additives.

18. **Meal Planning**. Meal planning is the process of preparing a week's worth of meals in advance, allowing individuals to make thoughtful, health-conscious decisions about their diet. It involves creating a menu, shopping for ingredients, and organizing meals to meet specific nutritional needs or lifestyle goals, such as weight management or fitness targets. By planning ahead, meal prepping reduces the stress of last-minute cooking, minimizes food waste, and helps maintain a balanced diet. It also offers financial benefits, as buying ingredients in bulk and sticking to a plan can cut down on unnecessary purchases.



19. **Meal Planning During NCC Camps**. NCC Camps are typically periods of intense physical activities including PT, games, shooting, obstacle training etc. and place a

nutritional demand on the body. Accordingly the meals during camps, have to be rich in proteins, apart from carbohydrates, and low in oil/fat content. For **moderate to high activity**, aim for:-

(a) **2,800–3,500 calories/day**

(b) **Carbs:** 50–60% (main fuel)

(c) **Protein:** 15–20% (for muscle recovery)

(d) **Fats:** 20–30% (long-lasting energy)

(e) **Hydration**: Minimum 3–4 litres/day(more with sweat or heat).



Know Your Spices

20. **Adoption of Healthy Cooking Methods**. Adopting healthy cooking methods is essential for preserving the nutritional value of food while reducing the intake of unhealthy fats, sugars, and salts. Here are some tips for healthier cooking:-

(a) **Use Healthy Oils**. Opt for oils that are rich in healthy fats, such as olive oil or avocado oil, instead of butter or margarine.

(b) **Grilling, Steaming, and Baking**. These methods help retain the nutrients in food without adding extra fat. Grilling and baking can also enhance the flavor of food without excessive oil.

(c) **Sautéing with Minimal Oil**. When sautéing vegetables or proteins, use only a small amount of oil or choose non-stick cookware to reduce the amount of oil needed.

(d) **Slow Cooking and Stewing**. These methods allow for tenderizing tougher cuts of meat and infusing flavor without excessive use of fats.

(e) **Avoid Deep Frying**. Instead of deep frying, use alternative methods such as baking, grilling, or air frying, which can achieve a similar crispy texture without excess oil.

(f) **Boiling and Poaching**. These are excellent techniques for cooking vegetables, eggs, and fish, allowing the natural flavors to come out without added fats or calories.

(g) **Incorporate Fresh Ingredients**. Prioritize fresh vegetables, lean proteins, and whole grains to ensure your meals are nutrient-dense.

(h) **Control Seasoning**. Use herbs, spices, and citrus to enhance flavor rather than relying on high-sodium condiments like soy sauce or pre-made sauces.

21. **Nutritious Eating Habits**. Nutritious eating habits include the following:-

(a) Eat a variety of foods, including fruits, vegetables, whole grains, protein foods, and fat-free or low-fat dairy.

(b) Enjoy fruits and vegetables for key nutrients and dietary fiber.

(c) Be mindful of nutrient needs.

(d) Go easy on salt.

(e) Shift toward healthier sweet options.

(f) Strive for a healthy weight.

HIGHER ORDER THINKING SKILLS (HOTS)

- **How would you compare the benefits of eating fruits and vegetables with the risks of eating too much junk food?**
- **Why is it important to include both proteins and carbohydrates in your daily diet? How do they work together to keep you healthy and energized?**
- **Imagine you are planning a healthy meal for your family. How would you plan include all the important nutrients like vitamins, minerals, and protein?**

CONCLUSION

22. In conclusion, diet and nutrition are vital components of maintaining overall health and well-being. A balanced diet, rich in essential macro- and micronutrients such as carbohydrates, proteins, fats, vitamins, minerals, and water, supports the body's functions, boosts energy, promotes growth and repair, and reduces the risk of chronic diseases.

23. By following dietary guidelines, practicing portion control, and focusing on whole, nutritious foods, individuals can improve their physical and mental health, manage their weight, and enhance their quality of life. It is crucial to adopt healthy eating habits, stay hydrated, and make informed choices based on food labels, ensuring that the body receives the necessary nutrients for optimal performance and longevity.

SUMMARY

- Diet is the type and quantity of food we consume, while nutrition refers to how the body absorbs and utilizes nutrients from food.
- A diet includes the food and drinks consumed daily and can refer to a specific eating plan for health or medical reasons.
- Proper diet and nutrition support growth, repair cells, and maintain bodily functions.
- A balanced diet is crucial for efficient body function, preventing illness, fatigue, infection, and decreased performance.
- It includes essential nutrients like carbohydrates, proteins, fats, vitamins, and minerals.
- Nutrition is the process by which the body absorbs and uses nutrients to maintain health and prevent diseases.
- Balanced Diet Benefits Provides essential nutrients for proper body function. Supports immune health, reduces disease risk, promotes growth, boosts energy, improves mental health, aids weight management, and enhances longevity.
- Dietary guidelines are designed to promote overall health, prevent chronic diseases, and maintain a healthy weight.
- Food Labels provide nutritional information to help make informed choices.
- Meal Planning helps in saving time, meeting nutritional goals, and reducing grocery costs.
- Adopting healthy cooking methods is essential for preserving the nutritional value of food while reducing the intake of unhealthy fats, sugars, and salts.

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ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the primary source of energy for the body?

(a) Carbohydrates (b) Proteins

(c) Fats (d) Fiber

Q2. Which nutrient is essential for building and repairing muscles, bones, and tissues?

(a) Carbohydrates (b) Proteins

(c) Fats (d) Vitamins

Q3. What is the recommended daily intake of water?

(a) 3-4 Litres (b) 5-6 Litres

(c) 1-2 Litres (d) 4-5 Litres glasses

Q4. What is the recommended percentage of protein for a balanced diet?

(a) 55—65% (b) 15-20%

(c) 135% (d) None of the above

Q5. What is the function of fiber in the body?

(a) Helps with digestion.

(b) Regulates blood sugar levels.

(c) Supports heart health.

(d) All of the above.

Q6. What does fat include?

(a) Saturated Fat (b) Trans Fat

(c) Sugar (d) (a) and (b) above

Q7. Benefits of a balanced diet include:-

(a) Improved immune function

(b) Lethargy

(c) Acidity

(d) All of the above.

Q8. Micronutrients include:-

(a) Carbohydrates (b) Vitamins and Minerals

(c) Proteins (d) Fats

Q9. **Vitamin important for vision is:-**

(a) Vitamin D (b) Vitamin B

(c) Vitamin A (d) All of the above.

Q10. **Which food items should be limited in consumption?**

(a) Sugary drinks

(b) Fast food and processed snacks

(c) Saturated and trans fats

(d) All of the above.

Q11. **The food component which is the building block for our body is:-**

(a) Vitamins (b) Fats

(c) Proteins (d) All of the above.

Q12. **Micronutrients are nutrients that are required to:-**

(a) Provide energy, growth and maintenance of the body.

(b) Regulate various bodily functions, such as immune function, nerve function and bone health.

(c) None of the above.

(d) All of the above.

Q13. **Food label includes the following:-**

(a) Ingredients List

(b) Percent Daily Values (%DV)

(c) Nutrients (Fats, Carbohydrates, Proteins, etc.)

(d) All of the above.

Q14. **Function of minerals is to:-**

(a) Regulate body processes, protect against disease.

(b) Provide energy, store vitamins, and protect organs.

(c) Build bones, teeth, and support metabolism..

(d) All of the above.

Q15. **Sources of Proteins are:-**

(a) Fish (b) Eggs

(c) Beans (d) All of the above.

Short Answer Questions

Q1. What do you understand by Diet?

Q2. What do you understand by Nutrition?

Q3. What are Macronutrients?

Q4. What is a Food Label?

Q5. What is the nutritional requirement for adolescents?

Long Answer Questions

Q1. What is the function of food components?

Q2. What are the components of a balanced diet for optimal health?

Q3. What is the role of Nutrients for our body?

Q4. What are the types and functions of Macronutrients?

Q5. What do the components ‘serving size’ and ‘calories’ mean on a Food Label and why do they matter?

PHYSICAL AND MENTAL WELLNESS (SD/SW)

CHAPTER PMW IX : YOGA

"Yoga is the practice of quietening the mind."

Maharishi Patanjali



TEACHING INSTRUCTIONS

Period	:	**Two (02)**
Type	:	**Lecture and Practice**
Year	:	**One in 2nd Year and One in 3rd Year SD/SW**
Conducting Officer	:	**Associate NCC Officer**
Training Aids	:	**Script/ Lesson Plan; Presentation on Computer; Charts**

Time Plan

- Introduction : 03 Mins (2nd Year)
- Part I : 07 Mins
- Part II : 30 Mins
- Part III : 30 Mins (3rd Year)
- Part IV : 07 Mins
- Conclusion : 03 Mins



INTRODUCTION

1. Yoga is a practice that encompasses physical, mental, and spiritual elements and has its roots in India, dating back over 5,000 years. The word "yoga" is derived from the Sanskrit term "*YUJ*," which means "to unite" or "to join." The primary goal of yoga is to harmonize the body, mind, and consciousness, promoting overall well-being and helping practitioners realize their full potential. International Yoga Day is celebrated on 21st June every year.

2. Yoga is an ancient practice that improves physical health and enhances mental and emotional well-being. *Maharishi Patanjali*, the father of yoga, initially propounded it. In this chapter, we will explore the importance of yoga for physical fitness, learn about some key yoga postures (*asanas*), and practice breathing exercises (*Pranayama*) that promote overall health and vitality.



LEARNING OBJECTIVES

- Understand the importance of yoga for physical fitness.
- Identify and perform key yoga asanas that improve flexibility, strength and balance.
- Learn pranayama techniques to enhance lung capacity and promote relaxation.
- Develop a basic understanding of how yoga contributes to overall wellness.
- Integrate yoga into their daily routine for physical fitness and mental well-being.

PREVIEW

The lecture will be conducted in the following parts:

(a) Part I : Importance of Yoga

(b) Part II : Major Asanas

(c) Part III : Pranayama

(d) Part IV: Practical Application

INTERESTING FACTS

- Yoga is over 5,000 years old: Yoga originated in ancient India, with the first written records dating back to 3000 BC.
- Yoga was initially a spiritual practice: Yoga was originally a way to connect with the divine and achieve spiritual enlightenment.
- The word "Yoga" means "union": The Sanskrit word "Yoga" means "to unite" or "to join," referring to the union of body, mind and spirit.

PART I : IMPORTANCE OF YOGA

3. **Definition of Yoga.** Yoga is a holistic practice that unites the body and mind through physical postures (asanas), breathing techniques (Pranayama), wisdom, and meditation to foster physical, mental, and spiritual health. It's not just about physical exercise but a complete science encompassing various aspects of life.

4. **Key Elements of Yoga.** Most popularly and commonly, yoga is understood only with one aspect of some physical exercise and postures, while it is a complete Science that includes:-:-

(a) **Lifestyle Practice**. A healthy lifestyle is the fundamental key element of yoga. By adopting healthy habits in daily life, such as mindful eating, regular exercise, proper sleep, stress management, and mental clarity, yoga can be fully integrated into one's lifestyle.

(b) **Physical Posture**. Physical Postures in Yoga are termed as Asana. Asana means Stabilizing the body in a particular posture for some time that enhances muscular strength, proper functioning of targeted organs, and complete physical health.

(c) **Breathing technique**. This is known as Pranayama. The intended breathing with awareness is called Pranayama. It includes Inhalation, Holding the breath in, exhalation, and holding the breath out differently.

(d) **Meditation**. Meditation / *Dhyana* is the relaxation of the mind. It is the only source of mental hygiene.

INTERESTING FACTS

- ***Ashtang* Yoga/ Eight Limbs** of Yoga explains the methods and techniques to practice them and achieve the aim of Yoga- Self-realisation.
  - ***Yama*.** The individual practice of social ethics.
  - ***Niyama*.** Physical & mental hygiene.
  - ***Asana***. Posture in which the body is stabilised for some time
  - ***Pranayama*.** expand vital life force energy through the deliberate control of respiration.
  - ***Pratyahara***. Withdrawing the senses inward.
  - ***Dharana***. Art of effortless concentration.
  - ***Dhyana***. Meditation – the deep relaxation of the mind
  - ***Samadhi*.** The state of equanimity and tranquility.

5. **Benefits of Yoga for Physical Fitness**.

(a) **Flexibility**. Regular yoga practice improves flexibility by stretching and lengthening the muscles.

(b) **Strength**. Specific *asanas* build muscle strength by using body weight to target different muscle groups.

(c) **Balance**. Many yoga poses improve coordination and balance, helping to stabilize the body and prevent injuries.

(d) **Improved Posture**. Yoga promotes proper alignment and posture, reducing the risk of back pain and muscle strain.

(e) **Stress Relief**. Yoga calms the nervous system, reducing stress and promoting relaxation.

(f) **Mind-Body Connection**. Yoga fosters awareness of body movements, making individuals more mindful of their physical health.



PART II : MAJOR ASANAS (INCLUDING PRACTICE)

6. **Asana**. Asana is a Sanskrit word meaning "posture," "seat," or "place." In yoga, *asanas* are the practices performed to develop any part of the body. *Asanas* are the physical body positions or poses of *yoga* that form the foundation of a modern Hatha yoga practice.

7. Preparation before doing yoga is crucial. Make the following preparations before starting yoga:-

(a) **Food**. Empty stomach yoga is more beneficial. One should refrain from doing Asana for at least 2 hours after eating.

(b) **Proper Space**. The place should be clean and free of noise, including outside sounds, for yoga.

(c) **Age Limit**. There is no age limit to do yoga.

(d) **Timings**. Yoga practice is beneficial from 4 am to 7 am in *Brahma Muharat. Brahma Muhurat* is the time before sunrise, considered the most auspicious time for spiritual practices in Indian tradition. It is believed that the mind is most calm and receptive during this time.

(e) **Procedure**. Let's start with *Pawan Mukt Shrinkhala* and then do the Asanas, *Pranayama*, and *Dhyana*.

(f) **Dress**. Loose-fitting Cotton fabric dress is the most suitable for doing asanas.

(g) Bathing. Taking baths with cold water before performing the asana has been considered beneficial.

(h) **Prayer**. One should pray with folded hands while sitting in *Sukhasana* or standing in *Tadasana*.

8. Some of the major *Asanas* are as follows. It is advised that Yoga *Asanas* be performed regularly to develop and maintain good health.

9. **Surya Namaskara**.

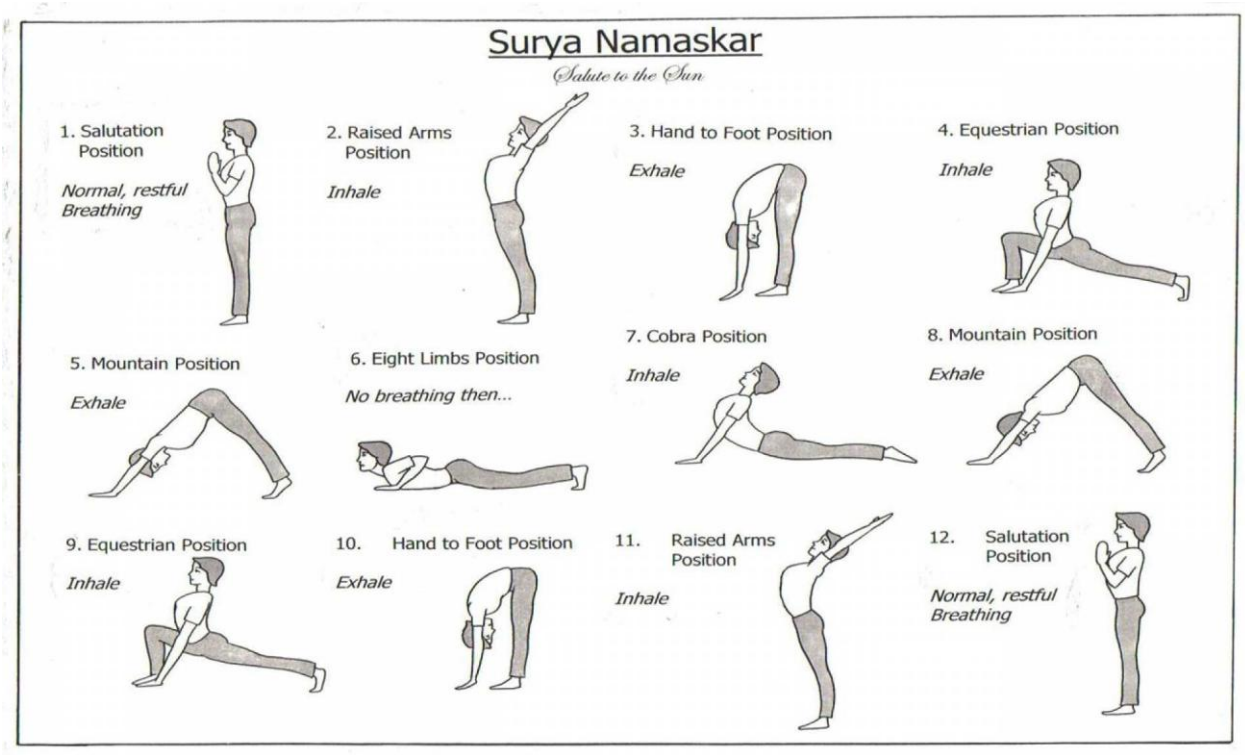
(a) **How to Perform**. The *Surya Namaskara* has 12 positions. A complete cycle is completed in 12 positions, where each position is performed twice.

(b) **Benefits**. This asana provides the following benefits.

(i) Reduces abdomen.

(ii) Increases the flexibility of spinal cord and other joints

(iii) Increases breathing capacity.



10. ***Tadasana*.**

(a) **How to Perform**.



(i) Stand with both feet together, fingers of the hands open and the thighs should be touching each other.

(ii) Distribute the body weight equally between the heels and the toes.

(iii) Raise the hands straight up. Lift the heels. Pull the body upwards.

(b) The **benefits** are as follows:-

(i) A feeling of lightness is experienced in the body.

(ii) The mind is free from stress.

(iii) The stiffness in the leg is relieved

11. ***Trikonasana***.

(a) **How to Perform**.

(i) Stand with your legs apart. Open your arms at shoulder height.



(ii) Bend your waist to the right and bring your right hand down to the ground near the ankle of your right foot.

(iii) Keep your legs straight and your left hand stretched above your head parallel to the ground.

(b) **Benefits**. The **benefits** are as follows:-

(i) The spine and its muscles become strong.

(ii) Fat in the ribs decreases.

(iii) The body's stiffness is alleviated.

(iv) The lungs' capacity to hold air increases.

12. ***Garudasana***.



(a) **How to Perform**.

(i) Lift the right leg and wrap it around the left leg. The right leg should be on the calf of the left leg.

(ii) Now bend the arms and wrap the right arm around the left arm.

(iii) The palms should be together.

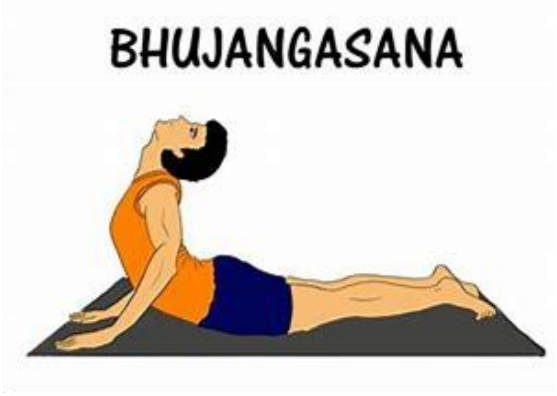
(b) **Benefits**. The benefits are as follows:-

(i) Strengthens the muscles.

(ii) Makes the nervous system function properly.

(iii) Loosens the joints of the legs and increases control.

13. ***Bhujangasana***.



(a) **How to Perform.**

(i) Lie down on your stomach.

(ii) Place your palms beside your shoulders.

(iii) Inhale and raise your head, chest, and abdomen.

(iv) Keep your thighs, hips, and legs touching the floor.

(v) Look upward and hold the pose for four breaths.

(b) **Benefits**. This asana has the following benefits:-

(i) The spinal cord and chest become strong.

(ii) Strengthens the back and stomach muscles.

(iii) Eliminates constipation and increases digestive power.

14. ***Vajrasana***.

(a) **How to Perform**. *Vajrasana* is the only asana that can be practiced immediately after having food.

(i) Kneel on a mat with your knees together. Your feet should be flat on the floor and your toes straight back.

(ii) Gently lower down your hips and sit on your calves. If this is too intense for your knees, place a cushion or folded blanket under your hips for comfort.

(iii) Rest your hands on your knees. Keep your fingers relaxed and straight.

(iv) Engage your core, keep your back straight, and sit upright. Your neck and head should align with your spine. Don't lean forward or backward.

(v) Close your eyes and slow your breath. Hold the position for as long as you feel comfortable.



(b) **Benefits**. This asana has the following benefits:-

(i) Improved digestion.

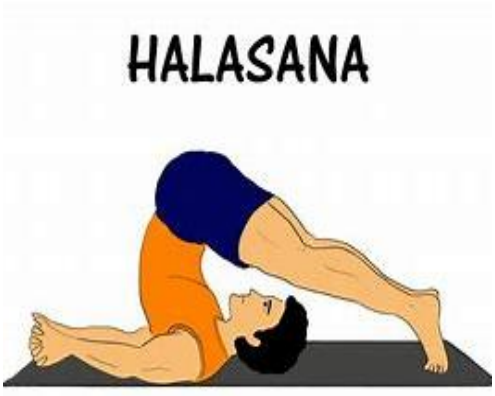
(ii) Strengthening of pelvic muscles.

(iii) Relief from constipation.

(iv) Reduced stress and anxiety.

(v) Improved blood circulation.

15. ***Halasana***.



(a) **How to Perform**.

(i) Lie down in *Shavasana*.

(ii) Bring both feet together while inhaling; lift the feet up While exhaling, bring the feet back behind the head.

(iii) Keep the feet straight and place the palms above the head.

(b) **Benefits**. The benefits are as follows:-

(i) Flexibility in the spine increases, and the shoulders become stronger.

(ii) The functioning capacity of the lungs increases.

(iii) Gas in the stomach and constipation is relieved.

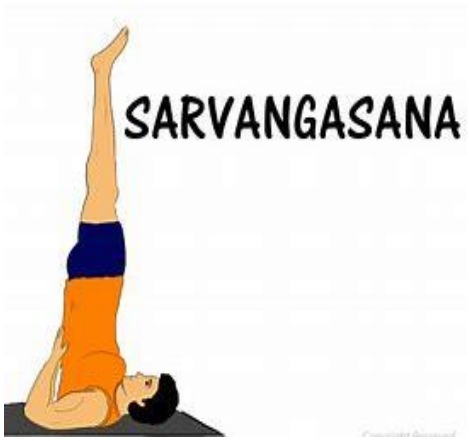
(iv) Digestive power increases.

16. ***Sarvangasana***.

(a) **How to Perform**.

(i) Lie on your back, inhale, and lift your legs, exhale and come into *Halasana*.

(ii) Placing both hands on your back and pointing your legs straight up towards the sky.



(b) **Benefits**. The benefits are as follows:-

(i) Helps to relieve fatigue and brings vitality.

(ii) Removes constipation and enhances digestive power.

(iii) Keeps the throat, nose, and eyes healthy.

(iv) Promotes blood circulation.

17. ***Paschimotanasana***.

(a) **How to Perform**.

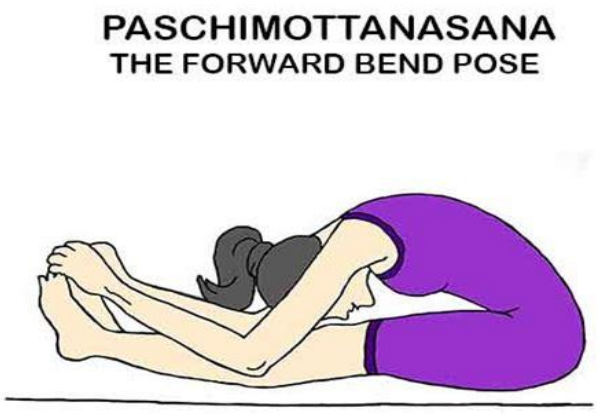
(i) Sit on the ground with your legs straight

(ii) in front of your body.

(iii) Bend your waist forward.

(iv) Hold your toes with your middle finger and

(v) pull them towards you.



(b) **Benefits**. The benefits are as follows:-

(i) This reduces unwanted fat in the stomach.

(ii) This improves the health of the intestines and increases digestive power.

18. ***Ardhmatsyendrasana***.

(a) **How to Perform**.

(i) Sit with both legs extended in front of you. Keep the right leg bent at the knee while placing the sole of the left foot on the ground outside the left knee.

(ii) Bend the left leg at the knee and place its heel against the thigh.



(iii) Twist the torso to the left, wrapping the right arm around the left knee from above to hold the left leg. Rotate the spine to the left and bring the left hand behind.

(iv) Then, turn the head towards the left shoulder.

(b) **Benefits**. The benefits are as follows:-

(i) It makes the spine flexible.

(ii) It eliminates digestive disorders.

19. ***Vrikshasana***.



(a) **How to Perform**.

(i) Stand straight, bend one leg and place it on the other thigh.

(ii) Raise your hands and bring your palms together.

(iii) Keep your gaze forward, straighten your arms and lift your body.

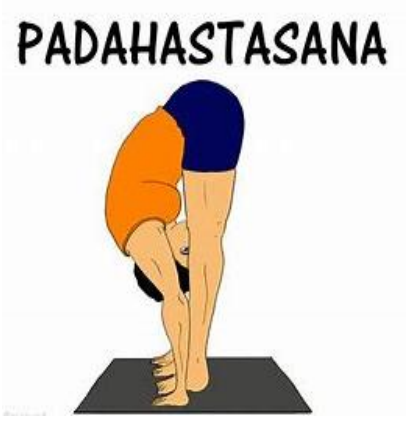
(b) **Benefits**. The benefits are as follows:-

(i) Physical balance is created.

(ii) Relieves flatulence.

(iii) Stomach muscles become stronger.

20. ***Padhastasana***.



(a) **How to Perform**.

(i) Stand straight.

(ii) Exhale, bend forward, and place your hands on the ground.

(iii) Bring your head to your knees.

(b) **Benefits**. The benefits are as follows:-

(i) Hamstring muscle stretched.

(ii) The hip joint becomes loose.

(iii) Reduces belly fat.

21. ***Chakrasana***.

(a) **How to Perform**.



(i) Lie on your back.

(ii) Bend both legs and bring your heels together with your buttocks.

(iii) Place your hands near your head with your fingers pointing towards you and lifting your body.

(b) **Benefits**. The benefits are as follows:-

(i) It brings flexibility and strength to the spine.

(ii) Blood supply to the brain is fulfilled.

(iii) Relief from diseases such as headaches and migraines.

(iv) It removes weakness and stops the trembling of hands and feet.

22. ***Shavasana***.

(a) **How to Perform**.



(i) Lie on your back, bring your arms and legs together, with your palms facing upwards, keep. Make sure that both legs are loose and your entire body is relaxed.

(ii) Keep both eyes closed.



(b) **Benefits**. The benefits are as follows:-

(i) It calms the mind and relieves stress.

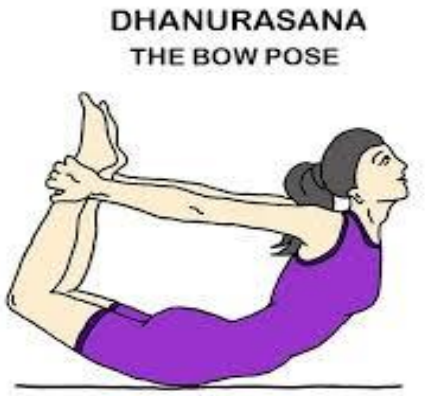
(ii) It helps in achieving yogic sleep.

(iii) It removes physical fatigue.

(iv) It corrects high blood pressure, mental imbalance, and memory power.

23. ***Dhanurasana***.

(a) **How to Perform**.



(i) Lie on your stomach, bend both legs and lift them up from the knees

(ii) Hold the back of the ankles with both hands and pull, bringing the entire body weight onto the stomach.

(b) **Benefits**. The benefits are as follows:-

(i) Tension in the waist is relieved.

(ii) Digestive power increases, and constipation is alleviated.

(iii) The spine becomes more substantial and more flexible.

24. ***Pawanmuktasana***.

(a) **How to Perform**. Lie on your back, bending your legs towards your chest, hold with both hands and place your head between your knees.



(b) **Benefits**. The benefits are as follows:-

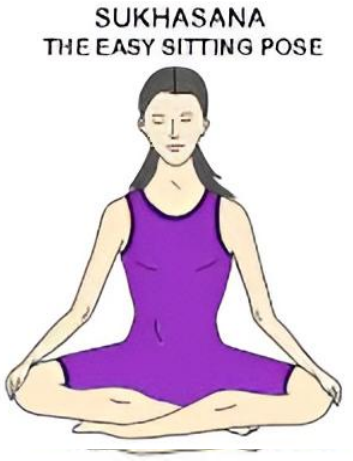
(i) Release of flatulence.

(ii) Digestive power increases

(iii) Intestines gain strength.

25. ***Sukhasana***.

(a) **How to Perform**. Bend your legs and sit in a comfortable position, keep your back straight, and place both hands on your knees, close your eyes, and take long deep breath.



(b) **Benefits**. The benefits are as follows:-

(i) Relieves physical and mental stress.

(ii) It is the best posture for practice and meditation.

(iii) The body gets relaxation.

PART III : PRANAYAMA (INCLUDING PRACTICE)

26. ***Pranayama***. The term is derived from several Sanskrit roots: prana, meaning "vital life force," Yama, meaning "control," and ayama, meaning "extension" or "expansion." The breath is symbolic of prana, and Pranayama is a method of extending and expanding vital life force energy through the deliberate control of respiration.

27. **Pre-Pranayama Preparations**.

(a) Find a quiet and comfortable place to sit.

(b) Sit comfortably with your back straight (e.g., *Padmasana*, *Vajrasana*, or *Sukhasana*).

(c) Close your eyes and relax your body.



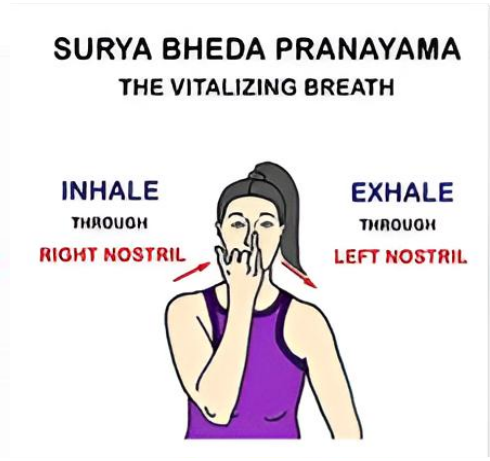
DID YOU KNOW?

The two nostrils harness different types of energy & emotions-

- Right nostril channels the '*Surya Swar'* / *Pingla Nadi*(nerve)
- Left nostril channels the '*Chandra Swar'* / *Ida Nadi*(nerve)

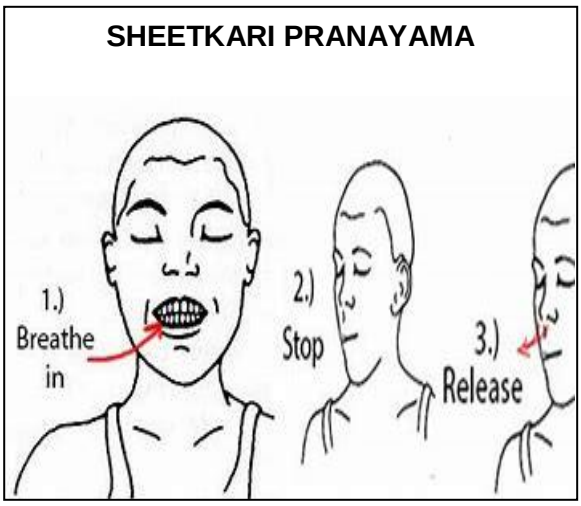
28. **Types of Pranayama**. *Pranayama* can be performed in any meditative posture. The types of *Pranayama* are:-

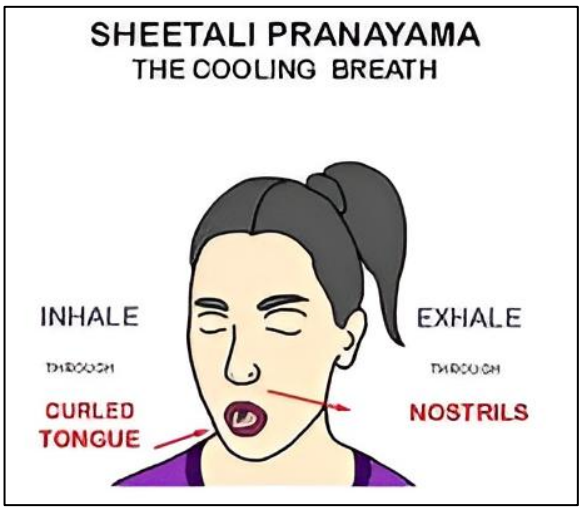
(a) ***Anulom-Vilom* Pranayama**. It is also called Alternate Nostril breathing. Press the right nostril with the right thumb gently to block the correct entrance for breath. Inhale through the left nostril, hold the breath, press the left nostril with middle and ring fingers to block the left exit for breath, exhale through the right nostril, then inhale through the right nostril, hold the breath, and exhale through the left nostril and vice-versa.



(b) ***Surya Bhedi* Pranayama**. Press the left nostril with the help of the ring finger to block the left entrance for breath. Inhale through the right nostril, hold the breath and exhale through the left nostril.

(c) ***Sheetkari* Pranayama**. Place the teeth next to each other, touch the tongue to the lower teeth, inhale through the mouth without holding the breath, and exhale through both nostrils. This is beneficial for good dental health, to control high blood pressure, and to hydrate the body.

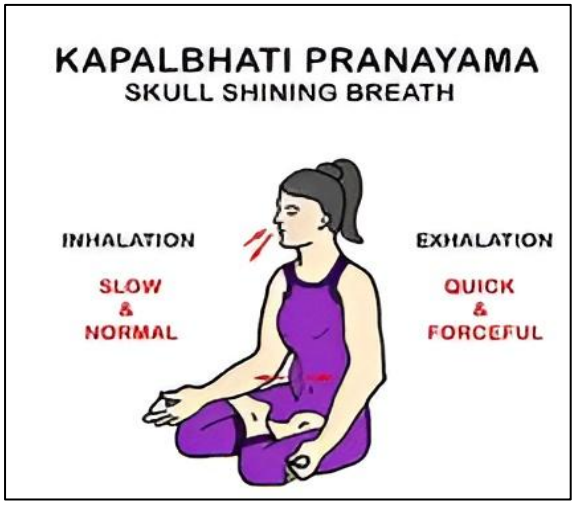


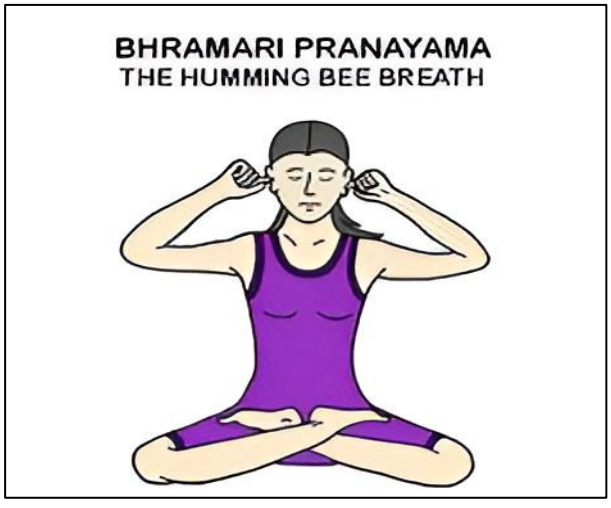


(d) ***Sheetali* Pranayama**. Stick out your tongue, form it like a beak, and breathe in through your mouth. Hold your breath and exhale through both nostrils. It is beneficial to control high blood pressure, hydrate the body, and prevent heat stroke. *Sheetali* Pranayama cools down the body and mind.

(e) ***Kapalbhati* Pranayama**. Inhale slowly and normally. Exhale quickly with force, leaving out any impurities, causing the abdomen to move back and forth. This exercise improves the function of the digestive and excretory systems of the body.

(f) ***Bhramari* Pranayama**. Make a humming sound like a bee while holding your breath through both nostrils and exhaling through both nostrils while making the sound. This practice calms the mind from overthinking and improves concentration.





(g) ***Udgeeta* Pranayama**. Take a long, deep breath from both nostrils and while breathing out, chant 'OM' during the complete exhalation. This is the most essential Pranayama for the practitioner to be in the meditative state.

29. **Benefits of *Pranayama***.

(a) *Pranayama* improves the lung capacity.

(b) It increases stamina by making respiration longer and more profound.

(c) *Pranayama* improves immunity.

(d) It increases the concentration power of the mind.

(e) *Pranayama* cleanses the body from the inside.

(f) It improves anabolism, metabolism, and catabolism of the Human body.

30. **Post-*Pranayama* Relaxation**.

(a) Sit quietly for a few minutes with your eyes closed.

(b) Focus on your breathing and feel calm and relaxed.

(c) Gradually open your eyes and take a few deep breaths before getting up.

PART IV : PRACTICAL APPLICATION

31. Yoga is often celebrated for its physical benefits, yet its philosophical foundation offers profound wisdom applicable to all facets of life. Yoga philosophy encompasses principles that transcend cultural and temporal boundaries, providing a timeless guide for living with purpose, balance, and harmony.



32. **Relevance in Modern Life**. In modern life, yoga helps in the following:-

(a) **Stress Management and Mental Health**.

(i) The principles of mindfulness and non-attachment (*Vairagya*) help individuals navigate stress and anxiety.

(ii) Breathing techniques (*Pranayama*) provide immediate relief and long-term mental stability.

(iii) **Ethical Leadership**.

(iv) Applying *Yamas*, such as non-violence and integrity, can transform workplaces into more compassionate and equitable environments.

(v) Self-discipline fosters accountability and resilience in professional settings.

(b) **Sustainable Living**. Yoga philosophy's emphasis on simplicity aligns with eco-conscious lifestyles, encouraging mindful consumption and environmental stewardship.

HIGHER ORDER THINKING SKILLS (HOTS)

- **Analyze the importance of yoga in maintaining physical and mental wellness. How can yoga help NCC cadets in their daily lives?**
- **Compare and contrast different types of yoga. Highlight the significance of a suitable place for performing Yoga.**
- **Design a simple yoga routine that can help NCC cadets improve these physical attributes.**

CONCLUSION

33. Embracing the timeless wisdom of yoga philosophy offers a transformative approach to modern challenges. Its universal principles provide a framework for ethical living, personal growth, and global harmony. By integrating these teachings into our lives, we enhance our individual well-being and contribute to a more compassionate and balanced world. Yoga is a holistic practice that offers numerous benefits for physical fitness, mental wellness, and emotional resilience. As NCC students, incorporating yoga into your daily routine can significantly enhance physical performance, reduce injuries, and promote overall wellbeing.

SUMMARY

- Yoga is a holistic practice that unites the body, mind, and consciousness to promote well-being and self-realization.
- The founder of yoga is *Maharishi Patanjali*, who combined physical postures (*asanas*), breathing techniques (*Pranayama*), wisdom, and meditation.
- Physical Postures (*Asanas*) stabilize the body and promote bodily health.
- Breathing Techniques (Pranayama) enhance breath control and energy flow.
- Meditation relaxes the mind and promotes mental clarity.
- The benefits of Physical Fitness include improved flexibility, strength, balance, and posture, reduced stress, enhanced mind-body connection, and promotion of overall health.
- Yoga Postures (*Asanas*) are physical postures that target body parts for strength, flexibility, and alignment. Common asanas are *Surya Namaskar* (Sun Salutation), *Taadasana* (Mountain Pose), *Trikonasana* (Triangle Pose), *Bhujangasana* (Cobra Pose), *Vajrasana* (Thunderbolt Pose).
- *Pranayama* (Breathing Techniques) are controlled breathing techniques to improve lung capacity, stamina, and focus. Popular types of pranayama are *Anulom-Vilom* (Alternate Nostril Breathing), *Kapalbhati* (Breath of Fire), *Bhramari* (Bee-Breath).
- Yoga Philosophy focuses on principles like mindfulness and non-attachment and helps manage stress, and improves mental health.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the origin of the word "yoga"?

(a) Sanskrit word "yuj" meaning "to unite"

(b) Hindi word "yoga" meaning "physical exercise"

(c) Tamil word "yoga" meaning "meditation"

(d) Chinese word "yoga" meaning "breathing techniques"

Q2. Which of the following is NOT a benefit of yoga?

(a) Improved flexibility (b) Increased strength

(c) Enhanced cardiovascular endurance (d) Reduced intelligence

Q3. What is the name of the yoga pose that is also known as "tree pose"?

(a) Vrkshasana (b) Virabhadrasana

(c) Trikonasana (d) Bhujangasana

Q4. Who is considered the father of yoga?

(a) Patanjali (b) B.K.S. Iyengar

(c) K. Pattabhi Jois (d) Swami Vivekananda

Q5. What is the name of the ancient Indian text that describes yoga?

(a) Yoga Sutras of Patanjali (b) Bhagavad Gita

(c) Upanishads (d) Vedas

Q6. Which of the following the following benefits are seen by doing Trikonasan?

(a) Spine and its muscles become strong. (b) Fat in the ribs decreases.

(c) The body's stiffness is alleviated (d) All of the above

Q7. The Surya Namaskar has how many positions:-

(a) 10 positions. (b) 11 positions.

(c) 12 positions. (d) 13 positions.

Q8. Which of the following benefits are seen by doing Padhastasan?

(a) Hamstring muscle stretched. (b) The hip joint becomes loose.

(c) Reduces belly fat. (d) All of the above

Q9. Methods to extend and expand vital life force energy through the deliberate control of respiration is called:-

(a) Virabhadrasana. (b) Pranayam.

(c) Salabhasana. (d) Dhanurasana.

Q10. Which of the following yoga postures is beneficial for improving balance and coordination?

(a) Vrikshasana (b) Tree Pose

(c) Eagle Pose (d) All of the above

Q11. What is the name of the Pranayama technique that involves rapid inhalation and exhalation through the nose?

(a) Bhastrika Pranayama (b) Kapalabhati Pranayama

(c) Anuloma Viloma Pranayama (d) Ujjayi Pranayama

Q12. Which of the following pranayama techniques is beneficial for calming the mind and promoting relaxation?

(a) Ujjayi Pranayama (b) Bhramari Pranayama

(c) Anuloma Viloma Pranayama (d) All of the above

Q13. What is the name of the pranayama technique that involves breathing through one nostril and then the other?

(a) Anuloma Viloma Pranayama (b) Kapalabhati Pranayama

(c) Bhastrika Pranayama (d) Ujjayi Pranayama

Q14. Which of the following pranayama techniques is beneficial for improving lung capacity and respiratory endurance?

(a) Bhastrika Pranayama (b) Kapalabhati Pranayama

(c) Anuloma Viloma Pranayama (d) All of the above

Q15. Following is the practical application of Yoga Philosophy:-

(a) Personal growth. (b) Community Building.

(c) Global Perspective. (d) All of the above

Short Answer Questions

Q1. Why is yoga important for physical and mental well-being?

Q2. Name two major asanas and their benefits?

Q3. What is pranayama and how does it help in relaxation?

Q4. How can yoga be incorporated into daily routines?

Q5. Mention one practical application of yoga is stress management?

Long Answer Questions

Q1. Discuss the importance of yoga, focusing on its physical, mental and emotional benefits? Explain the role of major asanas in promoting overall health?

Q2. Explain the concept of pranayama, its techniques and its practical applications in daily life for managing stress and improving focus?

Q3. What are the preparations to be done before doing Yoga?

Q4. What is the relevance of Yoga in Modern Life?

Q5. Give out the practical applications of Yoga.

ANSWER KEY: MULTIPLE CHOICE QUESTIONS (MCQ)

ANSWER KEY TO MCQ : ME

Answer key to MCQ : ME (Chapter - I)

Q1.	(c)	Q2.	(c)	Q3.	(b)	Q4.	(b)
Q5.	(b)	Q6.	(c)	Q7.	(b)	Q8.	(b)
Q9.	(b)	Q10.	(b)	Q11.	(d)	Q12.	(a)
Q13.	(d)	Q14.	(b)	Q15.	(c)		

Answer key to MCQ : ME (Chapter - II)

Q1.	(b)	Q2.	(c)	Q3.	(b)	Q4.	(b)
Q5.	(d)	Q6.	(c)	Q7.	(a)	Q8.	(b)
Q9.	(b)	Q10.	(c)	Q11.	(b)	Q12.	(b)
Q13.	(b)	Q14.	(b)	Q15.	(c)		

Answer key to MCQ : ME (Chapter - III)

Q1.	(b)	Q2.	(c)	Q3.	(b)	Q4.	(b)
Q5.	(b)	Q6.	(c)	Q7.	(b)	Q8.	(b)
Q9.	(c)	Q10.	(b)	Q11.	(c)	Q12.	(b)
Q13.	(a)	Q14.	(b)	Q15.	(c)		

ANSWER KEY TO MCQ: PDLS

Answer key to MCQ : PDLS (Chapter - I)

Q1.	(d)	Q2.	(b)	Q3.	(b)	Q4.	(b)
Q5.	(d)	Q6.	(a)	Q7.	(c)	Q8.	(b)
Q9.	(a)	Q10.	(c)	Q11.	(b)	Q12.	(b)
Q13.	(b)	Q14.	(b)	Q15.	(b)	Q16.	(a)
Q17.	(b)	Q18.	(c)	Q19.	(d)	Q20.	(c)

Answer key to MCQ : PDLS (Chapter - II)

Q1.	(d)	Q2.	(d)	Q3.	(d)	Q4.	(b)
Q5.	(d)	Q6.	(a)	Q7.	(b)	Q8.	(c)
Q9.	(d)	Q10.	(d)	Q11.	(a)	Q12.	(d)
Q13.	(a)	Q14.	(a)	Q15.	(a)		

Answer key to MCQ: PDLS (Chapter - III)

Q1.	(a)	Q2.	(c)	Q3.	(a)	Q4.	(d)

Q5.	(c)	Q6.	(b)	Q7.	(c)	Q8.	(b)
Q9.	(c)	Q10.	(b)	Q11.	(c)	Q12.	(d)
Q13.	(b)	Q14.	(d)	Q15.	(d)		

Answer key to MCQ : PDLS (Chapter - IV)

Q1.	(b)	Q2.	(b)	Q3.	(a)	Q4.	(a)
Q5.	(a)	Q6.	(d)	Q7.	(d)	Q8.	(a)
Q9.	(b)	Q10.	(c)	Q11.	(d)	Q12.	(a)
Q13.	(d)	Q14.	(a)	Q15.	(d)		

Answer key to MCQ : PDLS (Chapter-V)

Q1.	(b)	Q2.	(a)	Q3.	(b)	Q4.	(a)
Q5.	(b)	Q6.	(a)	Q7.	(b)	Q8.	(b)
Q9.	(d)	Q10.	(b)	Q11.	(c)	Q12.	(b)
Q13.	(d)	Q14.	(d)	Q15.	(a)		

ANSWER KEYS TO MCQ : CS

Answer key to MCQ : CS (Chapter - I)

Q1.	(c)	Q2.	(b)	Q3.	(c)	Q4.	(b)
Q5.	(b)	Q6.	(a)	Q7.	(b)	Q8.	(b)
Q9.	(c)	Q10.	(b)	Q11.	(a)	Q12.	(a)
Q13.	(a)	Q14.	(c)	Q15.	(a)		

Answer key to MCQ : CS (Chapter - II)

Q1.	(b)	Q2.	(a)	Q3.	(c)	Q4.	(b)
Q5.	(b)	Q6.	(c)	Q7.	(b)	Q8.	(b)
Q9.	(b)	Q10.	(b)	Q11.	(a)	Q12.	(a)
Q13.	(d)	Q14.	(a)	Q15.	(d)		

Answer key to MCQ : CS (Chapter - III)

Q1.	(b)	Q2.	(c)	Q3.	(b)	Q4.	(b)
Q5.	(c)	Q6.	(a)	Q7.	(c)	Q8.	(a)
Q9.	(c)	Q10.	(b)	Q11.	(c)	Q12.	(b)
Q13.	(b)	Q14.	(c)	Q15.	(b)		

Answer key to MCQ : CS (Chapter - IV)

Q1.	(b)	Q2.	(a)	Q3.	(b)	Q4.	(a)
Q5.	(b)	Q6.	(d)	Q7.	(a)	Q8.	(d)
Q9.	(a)	Q10.	(d)	Q11.	(c)	Q12.	(d)
Q13.	(c)	Q14.	(b)	Q15.	(a)		

Answer key to MCQ: CS (Chapter - V)

Q1.	(c)	Q2.	(b)	Q3.	(b)	Q4.	(a)
Q5.	(a)	Q6.	(c)	Q7.	(b)	Q8.	(a)
Q9.	(b)	Q10.	(a)	Q11.	(c)	Q12.	(c)
Q13.	(b)	Q14.	(a)	Q15.	(d)		

ANSWER KEYS TO MCQ : CC

Answer key to MCQ: CC (Chapter - I)

Q1.	(b)	Q2.	(b)	Q3.	(a)	Q4.	(a)
Q5.	(a)	Q6.	(d)	Q7.	(b)	Q8.	(d)
Q9.	(a)	Q10.	(b)	Q11.	(b)	Q12.	(c)
Q13.	(b)	Q14.	(c)	Q15.	(d)		

Answer key to MCQ: CC (Chapter - II)

Q1.	(c)	Q2.	(b)	Q3.	(d)	Q4.	(b)
Q5.	(d)	Q6.	(c)	Q7.	(c)	Q8.	(c)
Q9.	(b)	Q10.	(d)	Q11.	(c)	Q12.	(c)
Q13.	(b)	Q14.	(c)	Q15.	(b)		

Answer key to MCQ: CC (Chapter - III)

Q1.	(b)	Q2.	(b)	Q3.	(b)	Q4.	(b)
Q5.	(c)	Q6.	(c)	Q7.	(a)	Q8.	(d)
Q9.	(d)	Q10.	(b)	Q11	(d)	Q12.	(b)
Q13.	(b)	Q14.	(b)	Q15.	(c)		

ANSWER KEYS TO MCQ : IC

Answer key to MCQ: IC (Chapter - I)

Q1.	(c)	Q2.	(a)	Q3.	(a)	Q4.	(d)
Q5.	(b)	Q6.	(b)	Q7.	(b)	Q8.	(a)

Q9. (a) Q10. (d) Q11. (b) Q12. (d)

Q13. (d) Q14. (c) Q15. (c)

Answer key to MCQ: IC (Chapter - II)

Q1. (a) Q2. (a) Q3. (c) Q4. (d)

Q5. (c) Q6. (b) Q7. (b) Q8. (b)

Q9. (b) Q10. (a) Q11. (c) Q12. (d)

Q13. (c) Q14. (a) Q15. (b)

ANSWER KEYS TO MCQ : PMW

Answer key to MCQ: PMW (Chapter - I)

Q1. (b) Q2. (c) Q3. (c) Q4. (b)

Q5. (b) Q6. (c) Q7. (b) Q8. (c)

Q9. (a) Q10 (d) Q11. (d) Q12. (a)

Q13. (d) Q14. (a) Q15. (a)

Answer key to MCQ: PMW (Chapter - II)

Q1. (b) Q2. (d) Q3. (c) Q4. (b)

Q5. (b) Q6. (c) Q7. (b) Q8. (c)

Q9. (b) Q10. (a) Q11. (b) Q12. (c)

Q13. (b) Q14. (b) Q15. (c)

Answer key to MCQ: PMW (Chapter - III)

Q1. (b) Q2. (c) Q3. (d) Q4. (c)

Q5. (c) Q6. (b) Q7. (b) Q8. (b)

Q9. (b) Q10. (c) Q11. (b) Q12. (c)

Q13. (c) Q14. (a) Q15. (b)

Answer key to MCQ: PMW (Chapter - IV)

Q1 (b) Q2 (c) Q3 (b) Q4 (d)

Q5 (b) Q6 (c) Q7 (b) Q8 (c)

Q9 (c) Q10 (d) Q11 (a) Q12 (b)

Q13 (c) Q14 (b) Q15 (b)

Answer key to MCQ: PMW (Chapter - IV)

Q1. (b) Q2. (b) Q3. (c) Q4. (b)

Q5.	(b)	Q6.	(b)	Q7.	(b)	Q8.	(b)
Q9.	(d)	Q10.	(b)	Q11.	(a)	Q12.	(a)
Q13.	(c)	Q14.	(b)	Q15.	(c)		

Answer key to MCQ: PMW (Chapter - VI)

Q1.	(c)	Q2.	(c)	Q3.	(b)	Q4.	(c)
Q5.	(b)	Q6.	(b)	Q7.	(b)	Q8.	(c)
Q9.	(c)	Q10.	(c)	Q11.	(b)	Q12.	(c)
Q13.	(d)	Q14.	(b)	Q15.	(c)		

Answer key to MCQ: PMW (Chapter - VII)

Q1.	(c)	Q2.	(c)	Q3.	(d)	Q4	(c)
Q5.	(b)	Q6.	(b)	Q7.	(b)	Q8	(b)
Q9.	(c)	Q10.	(b)	Q11.	(c)	Q12	(c)
Q13.	(c)	Q14.	(c)	Q15	(b)		

Answer key to MCQ: PMW (Chapter - VIII)

Q1.	(a)	Q2.	(b)	Q3.	(a)	Q4.	(b)
Q5.	(d)	Q6.	(d)	Q7.	(a)	Q8.	(b)
Q9.	(c)	Q10.	(d)	Q11.	(c)	Q12.	(b)
Q13.	(d)	Q14.	(c)	Q15.	(d)		

Answer key to MCQ: PMW (Chapter - IX)

Q1.	(a)	Q2.	(d)	Q3.	(a)	Q4.	(a)
Q5.	(a)	Q6.	(d)	Q7.	(c)	Q8.	(d)
Q9.	(b)	Q10.	(d)	Q11.	(a)	Q12.	(d)
Q13.	(a)	Q14.	(d)	Q15.	(d)		



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